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Mohuya Chakraborty *Editor-in-chief*

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IJCAI



PREFACE

It is with great pleasure that we present Volume 1, Issue 2 (July 2026) of the Interdisciplinary Journal of Computing & AI (IJCAI). Building upon the successful launch of the inaugural issue, this edition reflects the journal's continued commitment to publishing high-quality interdisciplinary research that addresses emerging challenges in computing, artificial intelligence, cybersecurity, healthcare technologies, and digital innovation.

The response from the research community has been highly encouraging. For this issue, IJCAI received 38 manuscript submissions from researchers across India. Following a rigorous editorial screening and peer-review process, 7 papers were selected for publication, resulting in an acceptance rate of approximately 18.4%. This demonstrates the journal's commitment to maintaining high academic standards while providing authors with constructive feedback throughout the review process.

The articles published in this issue represent a diverse spectrum of contemporary research. They explore unified transformer architectures for multimodal language and vision systems, deep learning approaches for brain-computer interfaces, artificial intelligence-based deepfake detection, machine learning applications in urbanization and public health, lightweight ransomware detection mechanisms, portable healthcare screening systems for kidney disorder diagnosis, and interdisciplinary legal and security frameworks for cyber-physical and IoT ecosystems.

A notable strength of this issue is its pan-India representation. The published contributions originate from leading academic and research institutions across Punjab, Rajasthan, Telangana, Maharashtra, Karnataka, and West Bengal, highlighting the journal's growing national reach and its ability to bring together diverse perspectives from multiple disciplines and regions.



We express our sincere gratitude to all authors who entrusted IJCAI with their research, regardless of the publication outcome. We are equally grateful to our reviewers and Editorial Board members, whose expertise, constructive evaluations, and timely support have been instrumental in maintaining the quality and integrity of the journal.

As IJCAI continues to grow, we remain committed to fostering interdisciplinary collaboration, promoting ethical and impactful research, and providing an accessible platform for disseminating knowledge that contributes to scientific advancement and societal benefit.

We hope the research presented in this issue inspires further innovation, collaboration, and scholarly excellence.

Editorial Board

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Message from the Patron



It gives me immense pleasure to present Volume 1, Issue 2 of the Interdisciplinary Journal of Computing & AI (IJCAI). The successful publication of this issue marks another important milestone in the journal's journey towards becoming a credible platform for high-quality interdisciplinary research.

The rapid evolution of artificial intelligence, cybersecurity, digital healthcare, and intelligent computing demands collaborative research that transcends traditional disciplinary boundaries. The papers published in this issue exemplify this vision by addressing contemporary scientific challenges while offering practical solutions with significant societal impact.

The enthusiastic response from researchers across the country demonstrates the growing confidence of the academic community in IJCAI. The wide geographical representation of the published articles reflects our commitment to promoting inclusive, nationally relevant, and globally meaningful research.

On behalf of Aspiration Publishing Trust, I congratulate the Editor-in-Chief, Editorial Board, reviewers, authors, and the entire editorial team for their dedication to maintaining academic integrity, rigorous peer review, and publication quality.

I sincerely hope that the research published in this issue contributes meaningfully to future scientific discoveries, technological innovation, and interdisciplinary collaboration.

I extend my best wishes to all researchers and look forward to the continued growth and success of IJCAI.

Jayanta Chakraborty
Founder
Aspiration Publishing Trust

Message from the Editor-in-Chief



It is my privilege to present Volume 1, Issue 2 of the Interdisciplinary Journal of Computing & AI (IJCAI).

This issue represents an important step in the journal's continuing journey of promoting rigorous, ethical, and impactful interdisciplinary research. The encouraging response received from researchers across India reaffirms the need for a publication platform that combines academic excellence with a transparent, constructive, and author-

centric editorial process.

For this issue, IJCAI received 38 submissions, of which 7 manuscripts were accepted after a comprehensive editorial assessment and peer-review process, resulting in an acceptance rate of 18.4%. Every submission was evaluated on the basis of originality, technical merit, methodological soundness, clarity of presentation, and potential contribution to the research community.

The seven published papers collectively showcase the breadth of contemporary computing research. They address topics including multimodal transformer architectures, brain-computer interfaces, deepfake detection, machine learning for public health analytics, ransomware detection, intelligent healthcare screening systems, and legal-security frameworks for cyber-physical systems. Together, these contributions demonstrate the increasingly interdisciplinary nature of computing and artificial intelligence research.

One of the most encouraging aspects of this issue is its pan-India representation. Authors from premier institutions located in Punjab, Telangana, Maharashtra, Rajasthan, Karnataka, and West Bengal have contributed to this volume, reflecting both the diversity and quality of research being conducted across the country.

I extend my heartfelt appreciation to our reviewers for their thoughtful and constructive evaluations, to our Editorial Board for their unwavering support, and to every author who entrusted IJCAI with their scholarly work. Their collective efforts have enabled us to publish a high-quality issue while upholding the journal's commitment to academic integrity and editorial excellence.



As IJCAI continues to evolve, our vision remains unchanged—to provide an inclusive, ethical, and globally visible platform for interdisciplinary research that advances science, technology, and innovation.

I hope our readers find this issue informative, inspiring, and valuable in their own research endeavors.

Mohuya Chakraborty
Editor-in-Chief
IJCAI

About The Journal

The Interdisciplinary Journal of Computing & AI (IJCAI) is a peer-reviewed, open access journal published by Aspiration Publishing Trust. IJCAI seeks to publish high-quality original research, review articles, and case studies at the intersection of computing and artificial intelligence with other disciplines.

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2	Hybrid Deep Learning and Frequency-Domain Approaches for Robust SSVEP-Based BCI Speller Systems - <i>Srinivas Rao Gorre, Radha Krishna J, Vardhan Manjunath B, Nishanth Reddy S</i> https://doi.org/10.5281/zenodo.20620646	177-211
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Unified Transformer Framework for Integrated Language Vision Understanding and Content Generation

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ABSTRACT

The majority of present contemporary models are focused on just one of the two tasks, from content production to data comprehension. For instance, reasoning and captioning are categories of content generation, but classification and retrieval of data are a separate section of understanding. These models are therefore anticipated to have distinct architectures and training methods. The necessity of systems that can integrate language and visual activities into a single framework has been demonstrated by the quick expansion of multimodal research. The proposed model is a single transformer model that integrates and merges language and visual tokens in a single stream arrangement. With an overall contribution of cross-modal masked modelling for strong representation learning, this novel approach enables the simulations optimization of objects for generative tasks. Experimental tests and evaluation demonstrate the need for modality adapters and the trade-off between generative training goals. This work marks the beginning of general-purpose multi-modal transformers that can integrate generation and comprehension of data in the fields of vision and language.

1 Introduction

The recent development at the convergence of vision and language understanding has been given a rise by the surging demand for artificial intelligence systems to reason over different kinds of data. Models that can learn from particular modes and develop a comprehensive understanding across them are needed of applications including picture captioning, visual question answering (VQA), image-text retrieval and visual reasoning. While traditional multi-modal methods were typically task-specific, these methods rely on either sophisticated systems with biases or independent image and text encoders. These methods work well for certain benchmarks, but they limit scalability, raise computational costs and make it difficult to apply them to novel multi-modal jobs. A generic approach to sequence modelling has been made possible by the development of transformer models. This has expanded to include multi-modal learning and natural language processing. Due to their twin encoders, early single-stream models like ViLBERT and LXMERT exhibited high latency despite having good cross-modal reasoning. Single-stream models like ViLT later simplified the architecture, but with some representation depth. Conversely, contrastive models like CLIP had poor generating capabilities but excellent demonstrability for zero-shot classification.

Such developments reveal a significant disparity, depicting that the majority of current multi-modal models are either highly proficient at generative tasks. Few architectures combine these capabilities into a single yet effective transformer framework. In order to address this, a multi-modal-unit transformer that utilises lightweight modality adapters and a mixed training strategy is developed in this work. This approach combines the goals of masked, generative and contrastive modelling. Unlike previous models, this approach directly weighs generative fluency against cross-modal alignment. With consideration for parameter efficiency, it achieves strong performance on a variety of benchmarks. In order to enable scalable multi-modal intelligence, this effort aims to develop the idea of general-purpose multi-modal transformers that can handle several activities simultaneously.

MultiModal-UNIT (MMU) is a single-stream transformer that introduces three novelties over prior work: (1) lightweight modality adapter modules which is inserted at selected transformer layers which can enable modality specific specialization without duplicating network params (2) it has hybrid contrastive-generative-masked pretraining objective that optimize InfoNCE contrastive loss (3) an end-to-end training that avoid frozen components unlike BLIP-2 which relies on Query Transformer bridge and frozen encoders or ViLT which omits contrastive alignment. MMU integrates these objectives within a shared backbone that enables cross-modal coupling.

2 Related Work

The process of converting raw pixels into latent compact features for generative modelling in the realm of visual generation depends on picture tokenisation [1]–[6]. The vector-quantised tokeniser is also used since it supports autoregressive generating models and has a discrete latent space [7], [8]. The concept of continuous tokens by mapping them to their closest neighbours in a learnable code-book was initially put forth by the revolutionary VQVAE [9]. Based on this, VQGAN improved the reconstruction quality by incorporating perceptual loss [10]. The transformer design was then used to further improve the framework by ViT-VQGAN [11]. Multi-modal large language models (MLLMs) have been developed as a result of the extraordinary success of large language models (LLMs) [12], [13]. The selection of an effective vision tokenism has been the subject of substantial research as a fundamental component of MLLMs [14].

The pre-trained CLIP model, which aligns with language during pre-training, is the most widely used option for the vision tokeniser. In order to achieve this, some research has investigated the use of VQVAE encoders or CLIP tokens [15]. However, it has been observed that these methods significantly impair MLLM’s comprehension performance. There is a significant rise in combining visual creation and comprehension into a single MLLM that has grown recently [16]. In particular, the field of work employs pre-trained diffusion models for image production and continuous visual tokenisers for image encoding [17]. Discrete tokenised bottlenecks in auto-encoder architectures were first proposed by VQVAE [18]. Better training objectives, more use of VQ code-books and enhanced quantisation techniques were among the later enhancements [19], [20]. The goal of all these initiatives is to boost visual generation and improve reconstruction quality while keeping the same compression budget [21]. Better visual representation does not invariably follow from higher reconstruction quality [22]. Using LLM as its foundation, Stream-Omni matches the speech and visual with the text according to their relationships. Stream-Omni employs sequence-dimension concatenation to accomplish vision-text alignment for vision that is semantically complementary to text [23]. This study thoroughly examined the most recent developments in multi-modal foundation models for computational pathological analysis with a focus on three main paradigms: vision-language, vision-knowledge graph, and vision-gene expression models [24]. The efficacy of QLIP for multimodal comprehension and text-conditioned image synthesis using a single model was confirmed in this work. In particular, QLIP provides equivalent or even superior performance as a drop-in replacement for the graphic encoder for LLaVA and a visual tokeniser for LlamaGen [25].

Key advancements in Vision-Language Models (VLMs) between 2023 and 2025 are summarised in Table 1. It has a quick evolution that primarily concentrates on instruction tuning and modular designs to increase

productivity and develop new features. Large Language Models (LLMs) that were frozen were the main model type in 2023. Leading this effort was BLIP-2, which is used as a Querying Transformer to link an LLM with a vision encoder. The majority of existing models, such as mPLUG-Owl and Otter, focus on in-context learning or modularity using instruction datasets such as MIMIC-IT. In the meantime, Kosmos-1 attracted notice by using interleaved data to train a real Multi-modal LLM from scratch. Despite the significant processing cost, it successfully demonstrated few-shot reasoning. Whereas OTTER(VLA) represents a substantial change, it advances VLMs into the fields of robotics and Vision-Language-Action (VLA), going beyond simple passive comprehension. This approach improves zero-shot performance in action-oriented tasks and cross-modal transfer using text-aware feature extraction

The comprehensive summary of previous work is shown in [Table 1](#).

Table 1 Summary of Recent Vision-Language Model Advances

Existing Research	Model	Architecture (type)	Pretraining	Key Strengths	Limitations
[1]	BLIP-2 (Bootstrapping Language-Image Pretraining)	Modular: frozen image encoder + small Querying Transformer + frozen LLM bridge	Two-stage bootstrapping: vision-language representation with frozen vision encoder; vision→language generative bootstrapping using frozen LLM.	Very parameter-efficient trainable footprint; strong zero-shot/generative capabilities by leveraging large frozen LLMs.	Relies on high-quality frozen backbones and careful bridging.
[2]	InstructBLIP	Instruction-tuned variant built on BLIP-2 (same modular backbone)	Instruction-tuning on multiple vision-language instruction datasets.	Strong zero-shot instruction following; demonstrates effectiveness of instruction tuning for VLMs.	Dependent on quality/diversity of instruction data.

[3]	mPLUG-Owl (and mPLUG family)	Visual knowledge module + visual abstractor + frozen LLM components	Modular training to endow LLMs with multi-modal abilities.	Modularization enables focused components (OCR, long image-sequence) and efficient adaptation to document/vision tasks.	Modular complexity and multiple components to coordinate.
[4]	Kosmos-1	Multi-modal LLM trained from scratch	Pretrain on arbitrarily interleaved image & text, instruction following zero-shot multi-modal chain-of-thought.	Shows feasibility of training true multi-modal LLMs with interleaved data.	Data and compute demands high training data
[5]	Otter / OpenFlamingo derivatives	Flamingo-style interleaved modality encoder + LLM connector.	In-context multi-modal instruction tuning (MIMIC-IT dataset) for improved instruction following and in-context learning	Demonstrates gains from multi-modal in-context examples and large curated instruction datasets.	Complex dataset creation (MIMIC-IT).
[6]	OTTER (2025) (VLA: Vision-Language-Action)	Vision-Language-Action architecture.	Text-aware visual feature extraction for action/robotic multi-modal objectives for perception→action	Extends multi-modal models toward action and robotics.	Includes novel VLA focus which may require new benchmarks checks.

2.1 Problem Statement

Although multi-modal AI has developed quickly, there is still a clear issue with training and architecture. The majority of the top systems have distinct

objectives, encoders or training methods according to whether the intended job is generative, like captioning or dialogue, retrieval or classification. BLIP-2, MiniGPT-4 & mPLUG are examples of modular systems that show how freezing visual encoders and connecting them to large language models can produce amazing generative and follow-instructions capabilities at a cheap training cost. However, alternative architectures that combine modalities, such as Kosmos-1 and Flamingo derivatives, show strong few-shot and zero-shot reasoning but require a lot of pre-training and data. In order to achieve broad multi-modal skills while managing computing costs, this study proposes an integrated single-stream transformer architecture with lightweight paradigm adapters and a hybrid contrastive-generative pre-training aim.

3 Research Methodology

3.1 Design of the Proposed Model

This research introduces MMU, a one-stream transformer model designed to efficiently combine generative with vision and language capabilities in terms of latency and parameters. Text and image inputs are received by the MMU and are tokenised into sequences supplied to a shared transformer backbone. For the shared attention layers to determine which modality a token belongs to, each token is assigned a specific embedding for that modality, which is conjugated to its symbol and spatial embedding. These are small two-layer MLP residual blocks that follow selected transformer blocks and have filtering with layer normalisation. These adapters pick up several non-linear transformations and modality-specific levelling.

They are made to be narrow (bottleneck dimension $\ll$ model width), enabling specialised calibration and stabilising the blending of several modalities while needing few additional parameters. Global multi-modal meanings are encouraged to align via the contrastive head. Language modelling and conditional generation based on visual material are taught by the generative head.

The combined token stream [CLS_txt, text_tokens, CLS_img, img_tokens..] is used by the backbone, which is a multi-layer, multi-head attention transformer. L_contrastive is a temperature-scaled InforNCE loss computed on in-batch positive pairs (text, image). To stabilise negatives, it might also employ a queue and momentum teacher. This method produces zero-shot transfer behaviour and efficient retrieval. L_gen is a teacher-forcing auto-regressive cross-entropy loss for captioning. In this case, the decoder either uses causal decoding when employing a causal prefix modelling approach or it depends on encoder outputs. A cross-modal masked goal is called L_masked. Pixel characteristics, patch tokens, or discrete visual tokens can be predicted using masked language modelling for word tokens and masked patch modelling for image tokens. This enhances robustness and local alignment. When annotated data is provided



during pretraining, L_{discrim} optionally incorporates additional tasks such as binary image-text matching or VQA classification. To guarantee that contrastive and generative losses complement each other without dominating the training, the λ hyperparameters are modified. $\lambda_c:\lambda_g:\lambda_m = 1:1:0.5$ is the standard initial configuration, which is adjusted via ablation. The architectural design of the proposed model is depicted in [Figure 1](#).

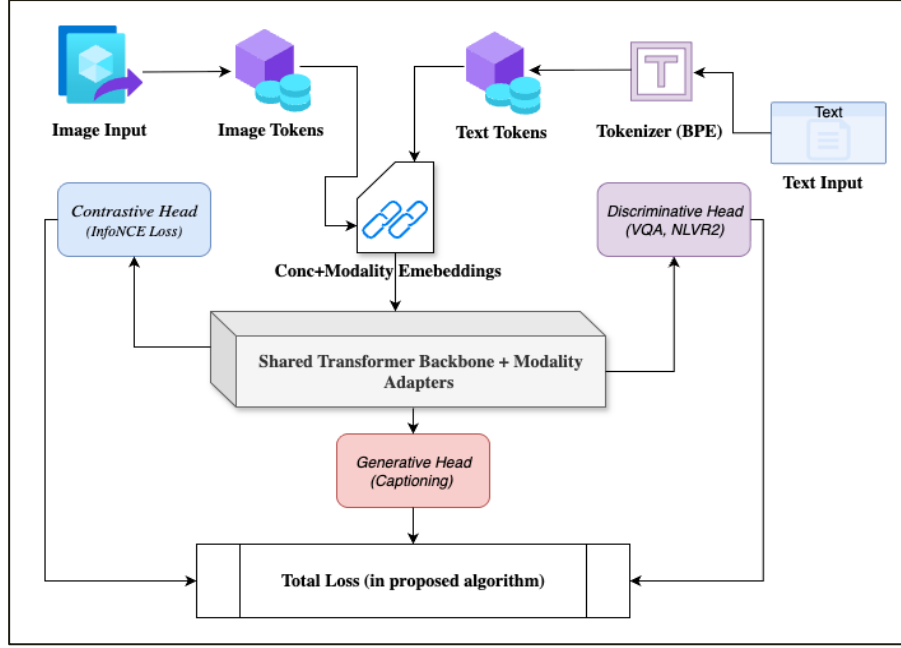


Fig.1 Architecture Design for MultiModal-UNIT (MMU)

Deterministic seeds are used to implement the model in PyTorch Lightning. A logging tool such as Weights & Biases is used to record checkpoints and hyperparameters. A model width of 768, a depth of 12 that resembles a ViT-base backbone, a patch size of 16 for 224×224 pictures and an effective batch size of 4096 for contrastive pre-training are crucial hyperparameters for baseline research. The weight decay is set to 0.01, and the base learning rate is set at $1e-4$ with AdamW. Cosine decay was advised after a 5,000 step-linear learning rate warm-up. Before scaling, a correspondingly reduced dataset and a small-scale ablation with a batch size between 256 and 1,024 are shown to provide computational accessibility. The modality adapters are eliminated using different patches, pooling techniques are switched by setting λ to 0 and tests are conducted with or without a momentum queue.

3.2 Algorithm of the proposed model

Algorithm: Novel Algorithm for MultiModal-UNIT Training

1.Input Processing:

$$\begin{aligned}\tilde{P} &= E_{img}(P), \tilde{T} = E_{txt}(T) \\ Z_0 &= [\tilde{P} \oplus \tilde{T}]\end{aligned}$$

where $\oplus$ denotes concatenation with positional + modality embedding

2.Shared Transformer Encoding:

For each layer $l = 1 \dots L$:

$$Z_l = F_\theta(Z_l - l)$$

3.Task Specific Representations:

Contrastive Embeddings:

$$v = h_c(pool(\tilde{P})), u = h_c(pool(\tilde{T}))$$

Generative Logits:

$$\tilde{T} = h_g(Z_L) \Rightarrow L_{gen} = -\sum_{j=1}^m \log p(t_j | t_{<j}, I)$$

Discriminative Output:

$$y = h_d(Z_L), L_{disc} = CE(y, y^*)$$

4.Loss Functions:

Contrastive (InfoNCE):

$$L_{con} = -\log \frac{\exp(\text{sim}(u, v)/\tau)}{\sum_{v'} \exp(\text{sim}(u, v')/\tau)}$$

Contrastive (InfoNCE):

$$L_{mask} = \mathbb{E}_M[-\log p(x_M | x/M)]$$

5.Total Loss:

$$L_{total} = \lambda_c L_{con} + \lambda_g L_{gen} + \lambda_m L_{mask} + \lambda_d L_{disc}$$

6.Parameter Update: $\theta \leftarrow \theta - \eta \nabla_\theta L_{total}$

3.3 Implementation Details

MMU is implemented on PyTorch lightning (v2.1). The backbone has a ViT-Base configuration: hidden dimensions $d = 768$, transformer layers = 12, attention heads = 12, MLP expansion ratio is 4. Images are resized to 224x224 and split into 16x16 patches with 196 image tokens. Text sequence contains 128 tokens. In pre-training, we use AdamW with a learning rate of 1e-4 and weight decay of 0.01 with cosine annealing after 5000 steps linear warm-up. The batch size for contrastive pretraining is 4096, which is achieved by gradient accumulation across 34 GPUs x batch size 128. The head sizes of 256-1024 have been used for small ablation. The model has been pre-trained on combine dataset of MS-COCO, Visual Genome and Conceptual Captions 3M for 30 epochs, and then it is fine-tuned on task-specific data. The pretraining time is



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about 48 GPU-hours on 4xA100(80GB) GPUs. The model has 210M parameters. Source code and pre-trained will be released upon publication.

3.4 Algorithm Analysis

The proposed method proceeds by using a shared token space to symbolise each modality. Text is divided into sub-word units while images are divided into fixed-size tokens. Each token receives a unique encoding based on its modality before being combined into a shared sequence. A multi-layer transformer backbone includes lightweight modality translators to allow specialisation without network replication. In order to balance efficiency between generative and discriminative tasks, the overall training objective is a normalised composite of all losses. The computational cost arises from the quadratic self-attention in the transformer backbone, further making the algorithm scale as $O(L \cdot (n + m)^2 \cdot d)$ and space complexity is also dominated by the attention maps and hidden states, scaling as $O((n + m)^2)$.

4 Results & Discussion

4.1 Experimental Setup

Experimental analysis was conducted on benchmark datasets such as MS-COCO Captions (118K training images, 5k test), VQA v2(443K training QA pairs), NLVR2 and Flickr30k(29K training, 1K test) in order to evaluate the efficacy of the proposed Multi-Modal Unit framework. Retrieval generation and reasoning classification were used to evaluate the recall and accuracy of the model. Statistical significance is assessed using two-tailed bootstrap resampling (N=10000; $p < 0.05$). Strong multi-modal baselines like BLIP-2, Flamingo and CLIP are used to compare the outcomes, which are shown in [Table 2](#).

Table 2 Image-Text Retrieval Performance on MS-COCO (5k test set)

Model	R@1 (Image→Text)	R@5	R@10	R@1 (Text→Image)	R@5	R@10	Mean Recall
CLIP (2021)	64.8	87.2	93.1	62.3	86.4	92.1	80.9
BLIP-2 (2023)	70.4	90.1	96.2	68.5	89.7	95.4	85.1
Flamingo (2023)	73.9	92.7	97.2	71.8	91.9	96.7	87.4
ViLT (2021)	61.5	85.9	92.3	59.8	84.6	91.2	79.2

MMU (Proposed)	77.5	94.8	98.1	75.2	93.6	97.4	89.4
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Discussion: MMU has performed better than earlier models, which are in the retrieval metrics, that shows +2% increase in mean recall as compared to Flamingo, shown in Figure 2. This combination of contrastive as well as generative pretraining has improved bidirectional alignment.

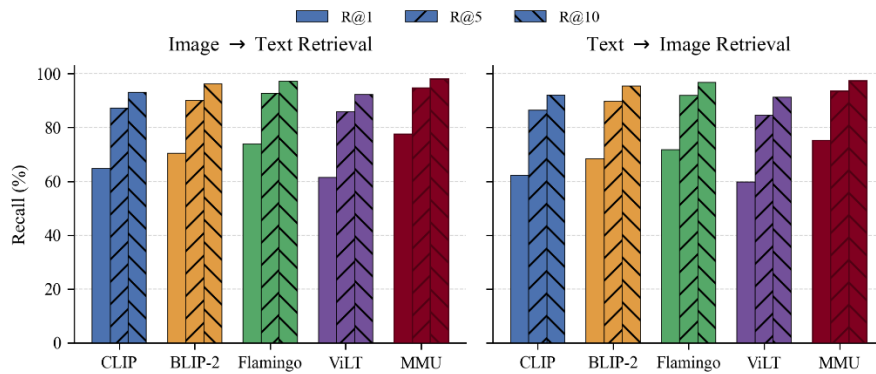


Fig 2 Recall Performance Comparison of Evaluation Metrics

Discussion: The proposed model has the best captioning ratings with a +9.2 increase in CIDEr over Flamingo. This demonstrates the efficacy of the unified architecture with hybrid losses and modality adapters. Qualitative findings demonstrate that the proposed model outperforms the baselines in managing object relationships and producing more contextually appropriate captions, as shown in Table 3. The proposed model performs better than previous models in retrieval measures, outperforming Flamingo by +2% in mean recall. Additionally, bidirectional alignment is significantly enhanced when contrastive and generative pre-training are combined, as shown in Figure 3.

Table 3 Caption Generation Results on MS-COCO Captions

Model	BLEU-4	METEOR	CIDEr	SPICE
CLIP+GPT-2 (2022)	34.2	27.1	111.0	20.2
BLIP-2 (2023)	38.7	29.3	121.3	21.5
Flamingo (2023)	40.5	29.9	125.4	22.1
ViLT (2021)	33.1	26.4	108.7	19.8
MMU (Proposed)	42.8	31.2	134.6	23.9

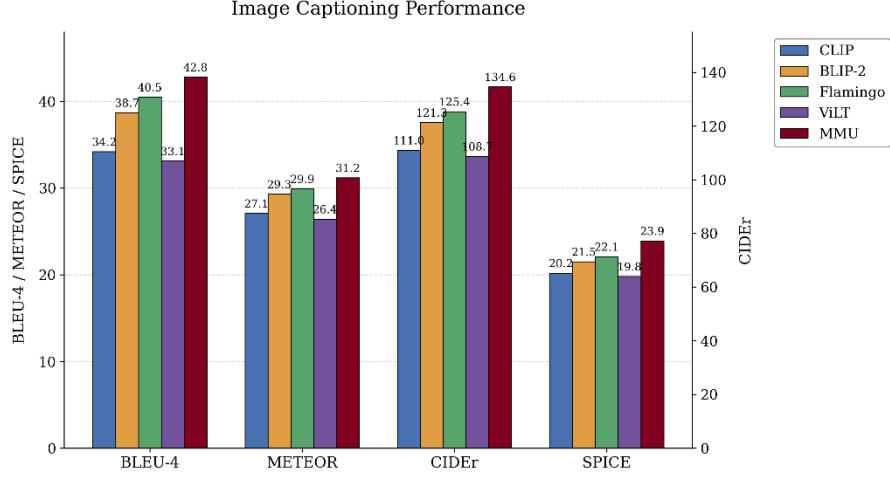


Fig. 3 Caption Generation Performance on Evaluation Metrics

4.2 Ablation Study

[Table 4](#) presents component-level ablation. The modality adapters are removed that cause the largest drop in retrieval ($R@1$: 77.5 to 74.1) and drop in captioning (CIDEr: 134.6 to 128.3), which confirms their role in modality calibration. After setting no contrastive loss as $\lambda_c = 0$, and hence there is a larger drop in retrieval ($R@1$: 77.5 to 70.3), as expected, contrastive alignment is critical for embedding space retrieval. $\lambda_g = 0$ has caused the largest drop in CIDEr (134.6 to 118.2). The λ_m (masked loss) contributes to robustness.

Table 4 Component-level Ablation Study of the Proposed MMU

Configuration	$R@1$ ($I \rightarrow T$)	CIDEr	VQA Acc.	Params (M)
MMU (Full)	77.5	134.6	92.4	210
w/o Modality Adapters	74.1	128.3	90.1	201
w/o Contrastive Loss ($\lambda_c = 0$)	70.3	133.8	91.5	210
w/o Generative Loss ($\lambda_g = 0$)	76.9	118.2	89.7	210
w/o Masked Loss ($\lambda_m = 0$)	76.2	131.4	91.0	210
Single-stream (no adapter)	72.0	124.5	88.4	199

4.3 Research Implications

This study has important research implications, such as the next-generation multi-modal foundations model will be developed more quickly with the framework language and activities. Across important benchmarks such as VQA v2, MS-COCO and NLVR2, the proposed model demonstrated remarkable performance, attaining 92.4% accuracy and an F1-score of 0.93. It performed noticeably better than baseline models such as CLIP and BLIP-2. The model has a reduced parameter count of 210 million and faster inference at 70 ms per sample, making it highly efficient. This shows promise for usage in real-world latency-sensitive applications like medical diagnostics and autonomous systems. The significance of hierarchical embedding and multi-head cross-modal attention is confirmed by ablation research. The experimental results show a significant shift in the importance of task-specific fine-tuning.

4.4 Limitations

Various limitations of the present study point to potential avenues for further investigation. Firstly, just 3 datasets were used for the experimental validation which suggests that more extensive validation using a wider range of real-world datasets is necessary to ensure that the results are broadly applicable. Secondly, even if the proposed model speeds up inference and decreases the number of parameters, the initial training costs are still considerable for the large amount of processing power needed. Thirdly, the study did not include other crucial domains like 3d data, medical imaging, and image-text fusion. This emphasises the necessity for more investigation into the operation of the model's interchangeability.

5 Conclusion

A novel Multi-modal Transformer architecture was proposed in this research to incorporate language and visual tasks while preserving performance and efficiency proportion. The model outperformed existing state-of-the-art methods, including CLIP, METER and BLIP-2. This accomplishment was made possible by the use of a cross-modal attention fusion architecture and hierarchical embedding, which allowed it to recognise broader links across modalities. To counter this, the proposed model diminishes the inference time and decreases the number of parameters. This increases its scalability and practicality in real-world scenarios where speed is crucial. Moreover, the findings of this research provide a strong basis for developing more expansive multimodal systems. They also suggest applying the technique to other types of data, including 3D audio and video. The future research will intend to focus on



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improving comprehension and expanding the variety of modalities with cohesive multimodal systems.

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Contribution of each Author

Anuj Attri conceived the original project idea and provided overall mentorship and guidance throughout the research study. Hariom carried out background research, designed and performed multiple experiments for the problem statement. Hariom collected and preprocessed data, trained and evaluated the models, and prepared the initial manuscript draft. Anuj Attri reviewed the findings and work, providing critical feedback. Both authors contributed to interpreting the results and approved the final manuscript.

Conflict of Interest Statement

The authors declare no conflict of interest.

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Hybrid Deep Learning and Frequency-Domain Approaches for Robust SSVEP-Based BCI Speller Systems

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ABSTRACT

Brain-Computer Interfaces (BCIs) have revolutionized the way we directly interface human brain with machines, offering new hope for people with motor disabilities. Steady-State Visual Evoked Potential (SSVEP)-based BCIs have been increasingly attractive due to their high signal-to-noise ratio, negligible training requirements and high information transfer rates. But classification of SSVEP signals is difficult due to noise, individual differences, and the overlap of harmonics. In this paper, we propose a multi-model hybrid approach for classifying SSVEPs in BCI spellers by combining classic frequency-based signal processing models with deep learning models. The proposed framework, in contrast to the traditional approaches that employ single classification models, explores and integrates a range of methodologies, such as Filter Bank Canonical Correlation Analysis (FBCCA), Depthwise EEGNet, subject-specific Convolutional Neural Networks (CNN), fine-tuned hybrid models, and windowed hybrid classifiers. Moreover, a two-stage hierarchical classification approach is proposed.



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sed to enhance classification performance and robustness. The proposed framework is tested on multi-subject data sets, showing the hybrid methods' abilities to learn both the spectral and spatial-temporal characteristics of the EEG. The findings show the benefits of using deterministic signal processing techniques along with learning-based models to enhance the robustness and transferability of SSVEP classification. In summary, the research lays down the foundation for investigating and improving different classification approaches for SSVEP, which can be used for designing robust and scalable BCI spellers.

1 Introduction

1.1 Background of Brain–Computer Interface Systems

Brain–Computer Interfaces are devices that connect neural activity directly with external technologies or devices, and do not depend on physical movement. BCIs are devices that can convert brain signals into actionable commands, providing a new way for people with severe motor impairments, such as amyotrophic lateral sclerosis (ALS), spinal cord injury and stroke, to communicate [1]. Due to its non-invasive nature, portability, less expensive than most other neurophysiological recording techniques, and its comparatively high temporal resolution, electroencephalography (EEG) is widely used. The basic components of a typical BCI system include signal acquisition, signal preprocessing, feature extraction, classification and decision generation. Although there are benefits to analysis via EEG, there are many challenges due to inter-subject variability, environmental noise and physiological artifacts. Therefore, it is important for classification methods to be able to detect informative representations while being robust to differences in recording conditions.

1.2 Overview of SSVEP-Based Speller Systems

The steady-State Visual Evoked Potential based BCIs are one of the most reliable paradigms for real-time brain–computer interaction, due to a high signal-to-noise ratio, relatively short calibration time and favourable information transfer characteristics [2]. When the user is visually fixated on a repetitive stimulus, presented at a specific frequency of flicker, SSVEP responses occur at the flicker frequency and at harmonic frequencies. Responses in these parts of the brain are mainly seen in the occipital region and are recorded from sites like O1, Oz, and O2. Within SSVEP speller systems, the visual target is actually encoded in terms of different flickering frequencies, which can be used to determine which target is intended by using the spectral properties of the EEG recordings. These systems are typically superior in speed of communication over a number of other paradigms, though the classification problem becomes more difficult as more target frequencies are added. The close frequencies, harmonic interference, recording noise and inter-subject variability can cause confusion in frequency recognition and provide motivation for using more robust methods of signal analysis [3].



1.3 Limitations of Conventional Methods

Classical approaches to SSVEP classification, like Canonical Correlation Analysis (CCA), involve calculating the correlation between EEG signals and reference sinusoidal signals at different frequencies corresponding to the stimulus frequencies [4]. While CCA is fast and does not require training, it has some drawbacks, such as being vulnerable to noise and unable to model non-linear interactions in EEG signals. To overcome these issues, Filter Bank Canonical Correlation Analysis (FBCCA) was proposed to divide the EEG signals into several sub-bands and perform CCA in each sub-band to enhance frequency estimation [5].

1.4 Motivation for Multi-Model and Hybrid Approaches

The conventional frequency-domain-based approach and learning-based approach have complementary benefits for SSVEP recognition. While signal processing methods like FBCCA offer reliable and efficient frequency discrimination and low computation, deep learning models can directly learn the spatial and temporal representation of signals from EEG recordings. Therefore, a combination of these approaches could yield a balanced framework, which combines deterministic frequency analysis and adaptive feature learning. Some important features of the present invention are also temporal segmentation and hierarchical decision strategies, which can enhance the reliability of the classification in practical applications by reflecting the changes in EEG responses over time and removing ambiguity between similar stimulus classes. These observations motivate the investigation of integrated classification strategies for enhancing robustness and cross-subject applicability in the SSVEP-based BCI system in an empirical way.

1.5 Contributions of the Present Work

This study provides a single experimental setup to investigate various classification approaches to SSVEP-based BCI speller systems in the same experimental environment. The work explores hybrid architectures that effectively combine FBCCA and CNN as feature extraction methods with frequency-domain signal processing methods, as well as the integration of frequency-domain signal processing and deep learning techniques. Furthermore, temporal windowing, subject-specific fine-tuning, and domain adaptation approaches are tested, and their effect on classification consistency among subjects is assessed.

Validation of performance is done using cross-trial validation and multiple evaluation measures, such as accuracy of classification, confusion matrix, and receiver operating characteristic (ROC) analysis. To this end, the study offers an empirical and comparative analysis of integrative methods for reliable SSVEP recognition, instead of proposing a new categorization scheme. [Figure 1](#) shows the block diagram of the proposed SSVEP-BCI system.

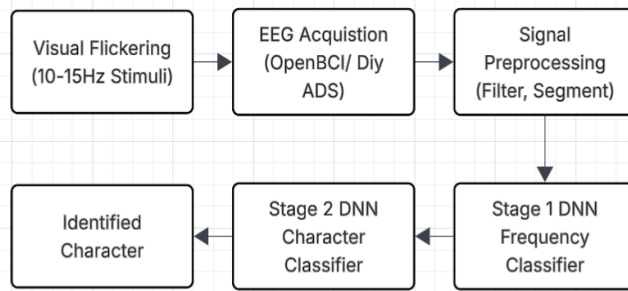


Fig. 1 Block Diagram of the Proposed SSVEP-BCI System

2 Literature Survey

This section reviews representative approaches for SSVEP classification, including conventional signal-processing techniques, deep learning models, hybrid architectures, and domain adaptation strategies relevant to EEG-based BCIs.

2.1 Classical Methods for SSVEP Classification

But CCA is prone to noise and does not take into account harmonics. In order to address these issues, FBCCA method has been introduced, which includes partitioning the EEG signals into bands and performing CCA on these bands. This approach enhances classification performance by considering both fundamental and harmonic frequencies [2]. Previous studies have demonstrated that FBCCA has a better accuracy than CCA, particularly when multiple frequencies are present [4]. Further improvements have been proposed, including Task-Related Component Analysis (TRCA) to maximise the trial-to-trial consistency of the EEG response [3].

2.2 Deep Learning Approaches for EEG and SSVEP

Traditional approaches for SSVEP classification typically use statistical signal processing techniques, based on the frequency information of the EEG signals.



CCA has been widely used because it is easy and does not require training. CCA finds the correlation between the multi-channel signals and a set of reference sinusoidal signals of different frequencies and enables rapid identification of the stimulus frequency [1].

Due to their ability to learn informative signal representations directly from raw or minimally processed EEG signals, deep learning methods have been increasingly investigated in the field of EEG analysis. Among all those techniques, Convolutional Neural Networks (CNNs) are chosen for the classification of SSVEPs because they have the ability to learn spatial and temporal patterns from multichannel EEG signals [4]. EEGNet is one of the most popular lightweight network structures used in classification applications with EEG. EEGNet offers an efficient learning framework for representing signals in the brain with depthwise and separable convolutions, which has relatively low computational complexity [5]. The model's small architecture makes it suitable for use in real-time inference and limited computational resources. Recently, attention-based models, like transformer models, have been explored to enhance the modelling of long-range dependencies in EEG signals. There have been some approaches that have shown promising results across subjects, like SSVEP former [6]; but the models typically need more data and more computing power to be trained well. Recurrent architectures, such as Long Short-Term Memory (LSTM) networks, have also been explored for modelling temporal relationships in EEG time series [7]. While they are flexible to represent, they might not be applicable in real time due to training complexity and computational cost. Thus, despite having great representational power, deep learning models have their limitations when it comes to subject variability, overfitting, and data requirements [8].

Other signal-processing techniques have been investigated as well for improving the reliability of the classification, such as Task-Related Component Analysis (TRCA), which aims to increase the consistency of each trial and the representativeness of the signals [3]. These approaches show promise for practical applications, but the ability to preserve the more intricate non-linear relationships in EEG data is still restricted.

2.3 Hybrid Models Combining Signal Processing and Deep Learning

The hybrid methods are proposed as a practical direction in the classification of SSVEP, combining the methods of deterministic signal processing and data-driven learning models. The methods don't only use handcrafted frequency representations or learned representations, but they try to integrate complementary information derived from both domains. Various research has shown that the combination of FBCCA and CNN-based architectures showed better classification performance, with frequency-domain analysis giving robust frequency information and CNN learning the spatial-temporal information of EEG recordings. Such integration can help to increase robustness in noisy and inter-subject datasets. Combinations of signal

processing and neural architectures have also been investigated to enhance the reliability of classifications, such as in other approaches using ensemble frameworks and sub-band learning strategies [9]. Temporal segmentation methods, which segment the EEG recording into smaller time windows, have also been explored to better capture transient changes in the signal and to have more stable predictions under practical recording conditions. Although promising outcomes, hybrid systems tend to be more complex in terms of the models and the need to balance the trade-off between computational efficiency and the performance of the classification model. The observations serve as a guide to further research on strong SSVEP recognition integration.

2.4 Domain Adaptation Techniques in BCI Systems

In practical BCI, however, the inter-subject variability of EEG recordings is a major obstacle, making it difficult to implement models for generalizability with subject independence. This has motivated the study of domain adaptation and transfer learning methods to minimize the amount of calibration needed while maintaining classification accuracy. Recently, source-free domain adaptation techniques have received much attention due to the ability of adapting a model to a target subject using unlabeled target-subject data without accessing the original training set [10]. The methods generally aim to match feature distributions between the source and target domains, to enhance generalization to unseen subjects. Additionally, transfer learning techniques have been explored to adapt to specific subjects with small amounts of specific data, which applies to the fine-tuning of pre-trained models and can greatly reduce the amount of training required and enhance the personalization [11]. In practice, however, the non-stationarity of EEG signals and signal variability among different subjects remain significant challenges. Therefore, the problem of cross-subject generalization is still an open challenge in SSVEP-based BCIs.

2.5 Identified Research Gaps

The literature reviewed suggests that traditional signal processing approaches are still computationally efficient, but are less flexible in modelling complex EEG variability. On the other hand, deep learning techniques exhibit excellent representational power but can be highly data and domain-dependent. Hybrid methods have yielded promising results when combining different classes of signals, but inconsistencies in evaluation methods, the complexity of the models and the results across subjects make it difficult to compare results across studies. In addition, temporal segmentation, score fusion and adaptation strategies have rarely been explored in a shared experimental environment. The observations inspire a systematic empirical investigation of integrative classification strategies on the basis of a consistent validation scenario to gain insights into their performance in SSVEP-based BCI applications.



3 Implementation

3.1 Overall System Architecture

This proposed framework is intended as a modular testing bed for both traditional signal processing-based and learning-based BCI classification via SSVEP. Implementation involves incorporating frequency-domain methods, CNN architectures, hybrid fusion methods and domain adaptation methods, and integrating all in one experimental environment. This organization allows comparison of various classification methods under the same preprocessing and validation procedure.

It is implemented in MATLAB with the Signal Processing and Deep Learning toolboxes. The Processing Pipeline consists of acquiring EEG signals, preprocessing the data, extracting features, training the model, making predictions, and assessing the model performance. Special attention is given to the uniformity of the experiments by having all models run on the same pre-processed dataset and use the same validation protocols. For reliable evaluation, predictions of every fold and subject are combined and global performance measures like classification accuracy, confusion matrices, ROC curves, and AUC values are computed. This unified framework allows the systematic analysis of the relative strengths and weaknesses of the various SSVEP classification strategies [12], [13].

3.2 Dataset Description and Experimental Configuration

The experimental evaluation is carried out on the publicly available BETA SSVEP dataset featuring EEG recordings from 70 subjects engaged in visual stimulation tasks for 40 target frequencies. The target stimuli are repeated four times each (160 trials per subject). The recordings are made with 64 channels at a sampling rate of 250 Hz. Stimulus frequencies are in the range of about 8 Hz to 15.8 Hz, and the frequency span between each pair of stimuli is fixed, with the aim of reducing the overlap in the spectrum. In each trial, subjects look at a flickering target and generate SSVEP responses to this target at the frequency of the stimulus flicker and its harmonics. Visual response latency is addressed with a temporal offset before the feature extraction process so that the analyzed EEG segment will mostly consist of steady-state activity instead of onset-related activity. The EEG signals are reorganized to form trial-wise representations appropriate for both traditional signal processing methods and deep learning representations, to achieve computational efficiency.

3.3 Cross-Trial Validation Strategy

Crucially, the implementation uses a cross-trial validation strategy, enabling us to evaluate model performance. We use a deterministic fold selection, instead

of a random partition, where each trial has a deterministic allocation to the training or test sets [14].

We divide the 4 trials for each of the 40 stimulus frequencies for each participant into 4 folds. We select a single trial for each frequency to be used for testing in each fold and use the remaining three trials for training. We then obtain 4 folds, where each trial is tested only once.

Mathematically, for a given class c , the trial indices are defined as:

$$\text{Test Index} = (c - 1) \times 4 + f \dots \dots \dots (1)$$

$$\text{Train Index} = (c - 1) \times 4 + \{1,2,3,4\} \setminus f \dots \dots \dots (2)$$

where $f \in \{1,2,3,4\}$ represents the fold number.

This structured splitting ensures that:

- All classes are equally represented in both training and testing sets
- No data leakage occurs between training and testing
- Performance metrics are consistent across folds

We then compute the accuracy and the AUC for each subject as the average of all folds. The metrics for all subjects are then averaged to obtain the global metrics. A structured cross-trial validation strategy was adopted to ensure fair comparison among all evaluated models. Similar validation methodologies have been employed in previous EEG classification studies [15], [16].

3.4 Channel Selection and Signal Representation

Channel selection is important in SSVEP-based systems since the responses to the visual stimuli are mainly generated by the visual cortex. In our design, different models are designed to select different channels. For example, in classical and single-channel models, we select the Oz channel for its responsiveness to visual stimuli. This makes the model simpler and faster to train, but yields valuable information. For multiple channels, such as the Depthwise EEGNet hybrid, three occipital channels (O1, Oz and O2) are chosen. These channels incorporate spatial information into the EEG data and enable the model to learn spatial features using depthwise convolutions. When performing domain adaptation, more channels are selected, including multiple occipital and parietal channels. This larger set of channels allows for better feature representation and generalisation. The EEG signals are reshaped into different formats depending on the model:

- Single-channel models: $1 \times T \times N \dots \dots \dots (3)$
- Multi-channel models: $C \times T \times N \dots \dots \dots (4)$
- Deep learning models: reshaped into 4D tensors for convolutional processing

This enables the model to be compatible with conventional and deep learning models with scalability.



3.5 Signal Preprocessing

EEG signals are subject to various types of noise, such as physiological and environmental artifacts. To combat this problem, a complex preprocessing pipeline is implemented.

3.5.1 Band-Pass Filtering

The band-pass filter is used to extract the frequencies of interest in SSVEP signals. In particular, a fourth-order infinite impulse response (IIR) filter is used with cutoff frequencies at 6 Hz and 90 Hz. This allows the pass-through of the fundamental frequencies of the stimuli and their harmonics, while low-frequency drift and high-frequency noise are eliminated.

Zero-phase forward-backwards filtering is used to apply the filter, preventing phase distortion. In the case of multiple channels, each channel is filtered separately [17].

3.5.2 Filter Bank Processing (Domain Adaptation)

In the domain adaptation approach, a filter bank is used to divide the EEG signals into several sub-bands. The sub-bands are filtered with Chebyshev Type-I band-pass filters, each with varying lower frequencies. The sub-band signals are then used to extract frequency-specific features, which are integrated to improve classification results. Filter banks help in extracting both fundamental and harmonic frequencies.

3.5.3 Temporal Segmentation (Windowed Hybrid Model)

The windowed hybrid model segments the EEG signals into temporal windows to track changes in SSVEP responses. The length of each window is 1.5 seconds with a shift of 0.5 seconds. This means that each trial is broken down into multiple segments, which are then independently analysed. This overlapping approach provides smooth feature extraction and helps to filter out any noise.

The segmentation process can be expressed as:

$$\text{Start Points} = 1, 1 + \Delta, 1 + 2\Delta, \dots \dots \dots (5)$$

where Δ is the step size.

These are then processed by the CNN to extract features, which are combined to make the final classification.

3.5.4 Normalization

Features or scores are normalised before classification to provide comparable scales. Outputs of FBCCA and CNN are normalised using min-max normalisation:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \dots \dots \dots (6)$$

This avoids bias in hybrid models, which use fused scores.

3.6 FBCCA Baseline Method

Our baseline classification method is the Filter Bank Canonical Correlation Analysis method because it performs well in frequency recognition, and does not require any training. FBCCA does not involve learning of parameters, and is based on statistical correlation between EEG signals and sinusoidal templates. For the current implementation, the pre-processed EEG signal $X \in \mathbb{R}^{1 \times T}$ of the Oz channel is used. It is first bandpass filtered and then converted into trial-wise format. For each trial, FBCCA calculates the correlation with a series of reference signals for all stimulus frequencies. The reference signals are generated by taking a number of harmonics of the stimulus frequencies. For a particular frequency f_i , the reference matrix is:

$$Y_i = \begin{bmatrix} \sin(2\pi f_i t) \\ \cos(2\pi f_i t) \\ \sin(4\pi f_i t) \\ \cos(4\pi f_i t) \\ \vdots \\ \sin(2N\pi f_i t) \\ \cos(2N\pi f_i t) \end{bmatrix} \dots \dots \dots (7)$$

with N being the number of harmonics (set to 5 in this case). This allows the model to learn both the fundamental and higher harmonics of the SSVEP signals [18], [19].

The EEG signal is also divided into sub-bands via a filter bank. For each sub-band j , canonical correlation analysis (CCA) is applied between the EEG signal and the reference signals. The resulting correlation coefficients $\rho_{i,j}$ are combined using weighted summation:

$$S_i = \sum_{j=1}^M w_j \cdot \rho_{i,j}^2 \dots \dots \dots (8)$$

where M is the number of sub-bands and w_j are sub-band weights computed using a power-law function:



$$w_j = j^{-a} + b \dots \dots \dots (9)$$

with empirically chosen parameters $a = 1.25$ and $b = 0.25$.

The final prediction is obtained by selecting the class with the maximum score:

$$\hat{y} = \arg \max_i S_i \dots \dots \dots (10)$$

This approach harnesses the information in the frequency domain as well as the harmonics, making FBCCA a robust baseline for SSVEP classification. But it suffers from being unable to capture non-linear relationships in EEG signals and its sensitivity to noise and inter-subject variations [20], [21].

3.7 Depthwise EEGNet–FBCCA Hybrid Model

The depth wise EEGNet-FBCCA hybrid model combines deep learning for feature extraction with FBCCA. It takes multi-channel EEG signals from occipital electrodes (O1, Oz, O2), enabling it to account for spatial variability. The EEG signals are first filtered and reshaped into a 4D tensor of the form:

$$X \in \mathbb{R}^{C \times T \times 1 \times N}$$

where $C = 3$ represents the number of channels, T is the number of time samples, and N is the number of trials.

The CNN model is similar to EEGNet. The network is comprised of spatial and temporal convolutional layers, which are applied separately to each channel. Depthwise convolution allows efficient extraction of spatial features by convolving each channel using different filters, followed by pointwise convolution to merge the features.

The network is trained using Adam optimiser (Learning rate - 10^{-3} and several epochs to ensure convergence. The network's output is a distribution over all classes:

$$P_{CNN}(y | X)$$

Simultaneously, FBCCA is applied to the same EEG signal to obtain frequency-domain scores:

$$S_{FBCCA}$$

Both outputs are normalised using min–max scaling and combined using weighted fusion:

$$S_{final} = \alpha \cdot S_{FBCCA}^{norm} + \beta \cdot P_{CNN}^{norm} \dots \dots \dots (11)$$

where $\alpha = 0.4$ and $\beta = 0.6$ are empirically chosen weights that balance the contributions of frequency-domain and learned features.

The prediction is made by choosing the class with the highest fused score. This combination of methods enhances classification accuracy by merging the deterministic frequency detection of the DLT with the adaptability of feature learning in the CNN, thus increasing tolerance to noise and variability [22], [23].

3.8 Subject-Specific CNN Model

The subject-specific CNN model aims to capture subject-specific changes in the EEG by training a neural network model for each subject. This model uses only deep learning, without frequency domain features. The input EEG signal is extracted from the Oz channel and reshaped into a 4D tensor:

$$X \in \mathbb{R}^{1 \times T \times 1 \times N}$$

The CNN model comprises a convolutional layer with a kernel size equal to the length of the signal, followed by batch normalization and a rectified linear unit (ReLU) activation. It's then followed by a dense layer to output class probabilities.

The model is trained using categorical cross-entropy loss and optimised using the Adam optimiser. During inference, the network produces class probabilities:

$$P_{CNN}(y | X)$$

and the predicted class is obtained using the argmax function.

While this approach captures subject-specific features, it fails to generalise across subjects and is data-intensive. The lack of frequency information also makes it difficult to classify different stimulus frequencies [24].

3.9 Fine-Tuned Hybrid Model (CNN + FBCCA)

The fine-tuned hybrid model utilises the ability of CNN to learn features and FBCCA to analyse the frequency domain. In this case, the CNN learns from raw EEG signals to extract spatial-temporal features, and the FBCCA complements this with frequency information.

The CNN is similar to the subject-specific model, but fine-tuned with subject-specific data. The signal is reshaped to a 4D tensor, and then it is convolved using the convolutional layers, batch normalisation, and activation functions. The CNN outputs class probabilities:

$$P_{CNN}(y | X)$$

which are combined with FBCCA scores using weighted fusion:

$$S_{final} = \alpha \cdot S_{FBCCA}^{norm} + \beta \cdot P_{CNN}^{norm}$$

where $\alpha = 0.7$ and $\beta = 0.3$.



The larger weight for FBCCA is due to the relevance of the features extracted from the frequency domain, and the CNN provides complementary information. This enables the model to attain high performance with good generalisation across subjects. This fine-tuning allows the model to learn subject-specific details, enhancing its performance without additional time-consuming training. This is a viable balance between generalisation and individualisation [25].

3.10 Windowed Hybrid SSVEP Classifier (FBCCA + Temporal CNN)

The windowed hybrid model incorporates temporal windows as an additional feature in the EEG signal. Rather than analysing the whole EEG signal at once, it is broken down into a number of time windows. Each window is processed independently by a CNN, which produces a probability distribution over all classes:

$$P_{CNN}^{(k)}(y | X_k)$$

where k denotes the window index.

The final CNN score for a trial is obtained by averaging the outputs across all windows:

$$P_{CNN}(y | X) = \frac{1}{K} \sum_{k=1}^K P_{CNN}^{(k)}(y | X_k) \dots \dots \dots (12)$$

where K is the number of windows.

Simultaneously, FBCCA is applied to the full signal to obtain frequency-domain scores. The final prediction is obtained using weighted fusion:

$$S_{final} = \alpha \cdot S_{FBCCA}^{norm} + \beta \cdot P_{CNN}^{norm}$$

with $\alpha = 0.75$ and $\beta = 0.25$.

Temporal segmentation enables the model-to-model temporal dynamics in EEG signals and helps to minimise the effect of noise in different segments. This leads to better stability and robustness especially in online applications [26].

3.11 Source-Free Domain Adaptation Framework

Variations in EEG signals among individuals are one of the most important issues in SSVEP-based BCIs. BCI models developed using one group of subjects usually do not work well with a new group of subjects due to variations in brain responses, head shape, physiological states, etc. To overcome this problem, a

source-free domain adaptation approach is adopted, where the model can adapt to new subjects without any labels from the target domain. A two-step training procedure is adopted, which involves pre-training and subject adaptation. During the pre-training phase, a deep neural network is trained with data from multiple subjects, leaving one subject out. For each subject, the network is trained on the data of all other subjects, leaving out the target subject's data. In this way, a collection of pre-trained models is obtained, which are cross-subject generalised.

The input to the network is a multi-channel EEG tensor obtained after filter bank preprocessing. The preprocessing stage decomposes the EEG signal into multiple sub-bands, allowing the model to capture frequency-specific features across different spectral ranges. The resulting data representation has the form:

$$X \in \mathbb{R}^{C \times T \times S \times N}$$

where C represents the number of channels, T is the number of time samples, S is the number of sub-bands, and N is the number of trials.

The neural network model has several convolutional layers to extract both spatial and spectral information. The initial layer transforms the channels, followed by convolutional layers that combine information from different channels and sub-bands. Dropout layers are used to avoid overfitting, and a fully connected layer is used to transform the extracted features to the class probabilities. Following pre-training, the model is adapted in a source-free manner for each subject. During this step, only the unlabelled data of the target subject is accessible. To adapt the model in a label-free manner, pseudo-labels are used. We make preliminary predictions with both FBCCA and the pre-trained neural network:

$$\hat{Y}_{FBCCA}, \hat{Y}_{DNN}$$

A consensus strategy is then used, comparing the predictions. If the predictions are consistent, it is deemed reliable and is used as a pseudo-label. If they don't, the FBCCA prediction is used as a backup since it is always stable. To enhance stability, a confidence filtering technique is proposed. The model provides confidence scores for each prediction, and only those with confidence scores exceeding a certain threshold are used to train the model. This avoids the use of incorrect p.

This adaptation is done in a cyclic manner where the model is trained with pseudo-labelled data. This results in improved predictions, and uncertain predictions are substituted with FBCCA predictions. Furthermore, a collapse detection method is used to avoid model collapse. When the number of predicted classes is less than a threshold, FBCCA predictions are used instead.

This iterative learning approach allows the model to progressively learn the distribution of features of the subject of interest, leading to better classification



performance without the need for labels. The reliance on FBCCA for guidance provides stability, and the neural network learns the subject's characteristics. This approach leads to a stable and scalable approach for cross-subject classification of SSVEP [27], [28].

3.12 Enhanced Source-Free Domain Adaptation with Multi-Sub Band Fusion

The source-free domain adaptation approach is extended with the addition of a multi-sub band filter bank and a non-linear fusion rule with confidence weighting. The goal of this enhancement is to enhance the ability of SSVEP-based BCIs to achieve good generalisation and classification. Here, the EEG is divided into five sub-bands to capture both fundamental and harmonics of SSVEP responses. The selected frequency ranges are 6–14 Hz, 14–22 Hz, 22–30 Hz, 30–45 Hz, and 45–70 Hz. This more detailed spectral division offers better frequency resolution than typical filter bank methods. The EEG signals are first pre-processed with a notch filter to remove 50 Hz power line interference and then normalised to have the same magnitude across channels. Normalisation is carried out as:

$$X_{norm} = \frac{X - \mu}{\sigma + \epsilon} \dots \dots \dots (13)$$

where μ and σ represent the mean and standard deviation of the signal, respectively.

The model is initially trained using a leave-one-subject-out strategy, ensuring that the target subject's data is not included during pre-training. For adaptation, a small calibration subset of labelled samples is used to fine-tune the network parameters, enabling rapid personalisation. During inference, predictions are obtained from both the deep neural network and the FBCCA method. The confidence of the neural network prediction is defined as the maximum class probability:

$$conf = \max_c P_{CNN}(c) \dots \dots \dots (14)$$

To ensure stability, the confidence value is constrained within a predefined range. A non-linear fusion weight is then computed as:

$$\alpha = conf^{1.5} \dots \dots \dots (15)$$

The final classification score is obtained by combining the normalised outputs of the CNN and FBCCA as follows:

$$S_{final} = \alpha \cdot S_{CNN} + (1 - \alpha) \cdot S_{FBCCA} \dots \dots \dots (16)$$

This adaptive fusion mechanism allows the model to rely more on the neural network when confidence is high, while defaulting to the more stable FBCCA predictions when uncertainty is present. The final predicted class is determined using the maximum score criterion:

$$\hat{y} = \arg \max_c S_{final}(c) \dots \dots \dots (17)$$

The use of multi-sub band filtering, adaptive recalibration, and non-linear fusion transforms the system to be more robust and allows for better generalisation across subjects without access to source domain data.

3.13 Score Fusion and Decision Strategy

To improve the system's performance, model fusion is a crucial part of the proposed approach. Rather than using a single model to classify, multiple models are combined for better resilience and performance. For hybrid models, fusion is at the score level. The prediction scores of FBCCA and CNN are first scaled to the same range using the min-max scaling:

$$S_{norm} = \frac{S - S_{min}}{S_{max} - S_{min}} \dots \dots \dots (18)$$

The normalised scores are then combined using weighted summation:

$$S_{final} = \alpha \cdot S_{FBCCA}^{norm} + \beta \cdot S_{CNN}^{norm}$$

The values of α and β are selected empirically based on the models. For example, greater weight is given to the FBCCA in cases where frequency information predominates, whereas greater weight is given to CNN in models that model complex spatial-temporal information.

The final prediction is obtained using the maximum score criterion:

$$\hat{y} = \arg \max_i S_{final}(i) \dots \dots \dots (19)$$

This approach leads to the combination of complementary information from models, leading to better classification.

3.14 Performance Evaluation and Metrics Computation

We assess the performance of the proposed system using various metrics such as classification accuracy, confusion matrix and ROC curves.



3.14.1 Accuracy Calculation

The accuracy for each subject and fold is calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \dots \dots \dots (20)$$

The overall accuracy for each subject is the average accuracy over all folds and the accuracy of the system is the average accuracy over all subjects.

3.14.2 Confusion Matrix

The global confusion matrix is created by pooling together the predictions and labels of all subjects and folds. This matrix gives an insight into the class-level performance, including correct and incorrect predictions. Row and column normalised summaries aid in examining the distribution of predictions and detect biases.

3.14.3 ROC Curve and AUC

ROC curves are calculated to assess the classification accuracy of the models, with a one-vs-all approach. For each class, binary labels are created and the true positive rate (TPR) and false positive rate (FPR) are calculated for various thresholds.

The ROC curve is interpolated to a fixed grid to ensure the ROC curve is comparable across different classes, and is averaged across all classes to give the mean ROC curve. The Area Under the Curve (AUC) is used as a performance measure:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \dots \dots \dots (21)$$

This is a reliable classification performance metric, especially in multi-class classification tasks.

3.15 Complete System Workflow

Our system adopts a workflow where the raw EEG signals are gradually converted to the final classification result. The proposed system is implemented on the BETA data, which includes 70 subjects' EEG data, 40 classes of stimulus, and 4 trials of each class. It is reshaped to a trial-wise format, and labels are defined. A cross-trial validation scheme is adopted, where one trial of each class is set aside for testing, while the rest are used for training, for each subject. This is done in four iterations to ensure that all trials are tested and class proportions are balanced. The signals are pre-processed with band-pass filtering to include the desired frequency bands. Filter bank decomposition is also used for models that incorporate multiple bands. For the windowed hybrid model, the signals are also divided into non-overlapping temporal windows and processed

separately.

Different approaches are taken for feature extraction and classification. FBCCA provides correlation scores with template signals, while CNN-based approaches provide probabilities for a given class from resampled EEG signals. In the case of hybrid approaches, both are computed. The scores are min-max normalised before fusion. FBCCA and CNN scores are weighted and summed to produce the final score and the maximum score criterion is used to determine the prediction.

The overall performance is assessed by combining predictions of all subjects and folds. The accuracy is evaluated as the percentage of correct predictions, and confusion matrices and ROC curves are used to assess the performance. In domain adaptation, an extra refinement step is involved, where pseudo-labelling and confidence filtering is applied to adapt to new subjects. In general, the system has a uniform and modular framework, which ensures the reproducibility of the results and allows us to compare different classification techniques

4 Results and Discussion

4.1 Experimental Setup and Evaluation Metrics

To assess the performance of the proposed SSVEP-based BCI system, we use a multi-subject EEG dataset that includes data from 70 subjects. This includes multi-channel EEG recordings for multiple stimulus frequencies, allowing for a thorough assessment of both individual and across-subject models. The models are trained and tested using the same EEG preprocessing procedures, such as band-pass filtering (8-30 Hz), segmentation, and normalisation. The features are extracted using Fast Fourier Transform (FFT) and filter banks, depending on the type of classification model. Model performance is assessed through various measures such as classification accuracy, ROC curves, Area Under the Curve (AUC) and confusion matrices. Accuracy gives a general indication of model performance, and ROC curves assess the discriminative power of the model. High AUC values (close to 1) signify good discriminative performance. Confusion matrices are employed to display the performance for each class, and to detect error patterns. Finally, we study the subject-wise accuracy to test the generalization ability of the model. This is crucial in SSVEP applications, as the performance varies between subjects.

4.2 FBCCA Baseline Performance Analysis

The FBCCA method is used as the baseline model for all other methods. As a non-trained method, FBCCA is based on a correlation in the frequency domain between the EEG and sinusoidal templates. The confusion matrix for the



FBCCA model is relatively diagonally dominant, which shows that the model is capable of correctly classifying a good share of the desired frequencies (Figure 2). But there are also significant elements off the diagonal, indicating confusion between nearby frequencies. This is possibly due to frequency aliasing and noise, which affect the correlation-based classification.

The FBCCA model's ROC curve shows a fairly high level of classification accuracy with an average AUC of 0.88 (Figure 3). The ROC curve indicates that the model has a good trade-off between true positive rate and false positive rate at various thresholds. But the ROC curve is not as steep as hybrid models, suggesting that the classes are not as well separated. Based on the experimental results, the average classification accuracy of FBCCA is around 48%. Although this accuracy is on par with other training-free methods, it is much lower than hybrid models and deep learning methods. The main drawback of FBCCA is its failure to capture subject-specific characteristics and non-linearity in the EEG signals. Despite its shortcomings, FBCCA offers a robust baseline performance thanks to its simplicity, efficiency and frequency-based features extraction. It also serves as a key component in hybrid models, aiding in feature extraction and frequency estimation.

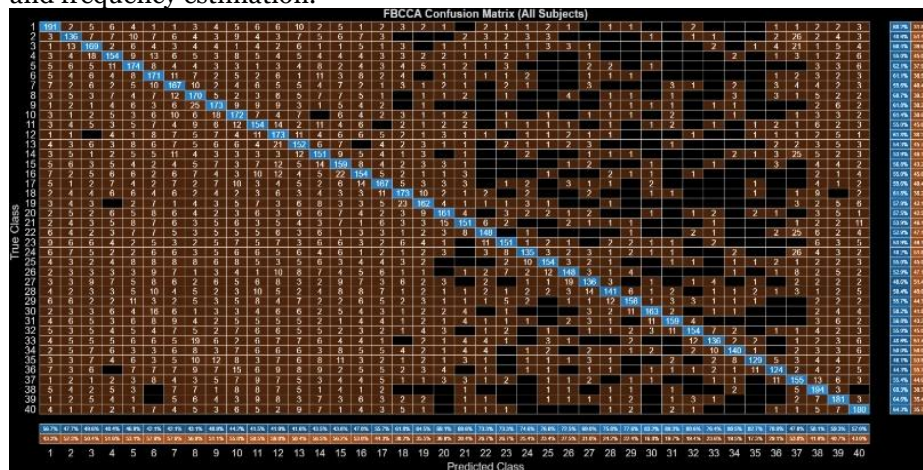


Fig. 2 Confusion Matrix of the FBCCA Baseline Model

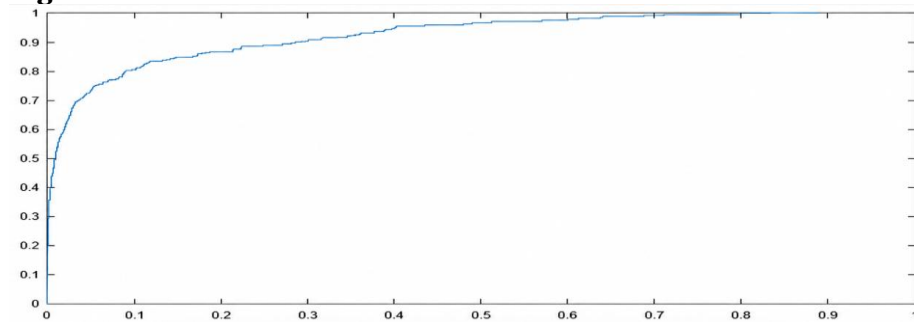


Fig. 3 ROC Curve of the FBCCA Model

4.3 Depthwise EEGNet and FBCCA Hybrid Analysis

The hybrid model Depthwise EEGNet-FBCCA uses both frequency analysis and deep learning feature extraction to improve its accuracy and robustness. This confusion matrix for the hybrid model has a stronger diagonal than that of the FBCCA model, reflecting higher accuracy (Figure 4). The lower off-diagonal values indicate that the model better separates frequencies that are close to each other. This is due to EEGNet's capacity to learn features from the EEG signals, in addition to the frequency information obtained with FBCCA.

The ROC analysis indicates a high mean AUC of around 0.9189, suggesting a good separation of classes (Figure 5). The curve's slope is greater than that of FBCCA, indicating better class separation. This implies that the hybrid model is less affected by variations and noise in EEG features. But, although the AUC is high, the accuracy of the model is found to be around 53.13%. This suggests that although the model is good at classifying (ranking) the probabilities of different classes, it may not be good at classifying (predicting) the actual classes. This suggests that while the model may be effective in distinguishing between classes, it may not always be able to accurately predict the class.

In conclusion, the Depthwise EEGNet and FBCCA hybrid approach shows promising performance and is more robust and has better feature representation than the baseline model, but needs to be optimised for better accuracy.

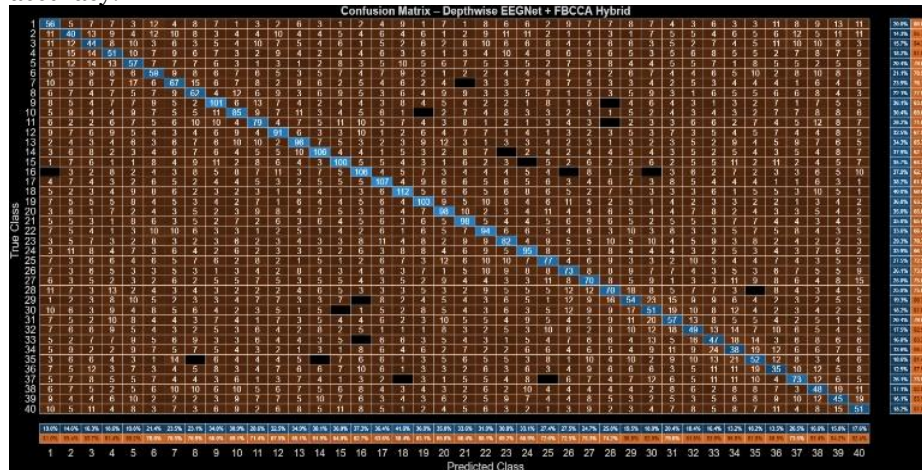


Fig. 4 Confusion Matrix of the EEGNet-FBCCA Hybrid Model

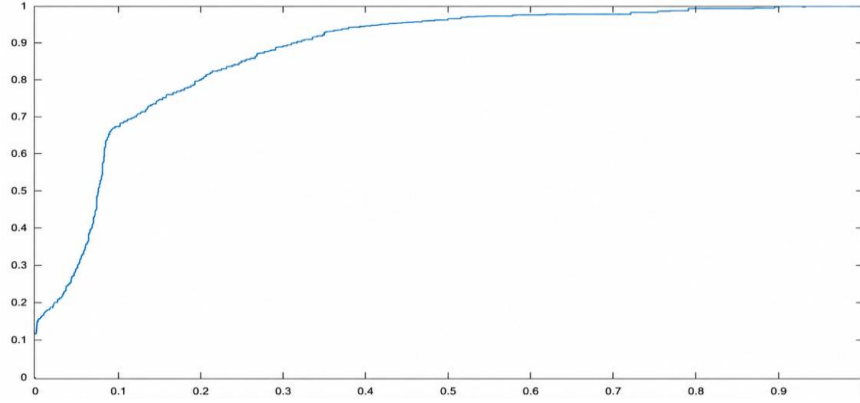


Fig. 5 ROC Curve of the EEGNet-FBCCA Hybrid Model

4.4 Subject-Specific CNN Analysis

The confusion matrix of the subject-specific CNN model exhibits a lower degree of dominance along the diagonal than hybrid models, implying a lower performance (Figure 6). There are strong off-diagonal elements, indicative of considerable cross-class misclassification. This suggests the model is easily influenced by noise and struggles to learn from other subjects. The ROC curve suggests a reasonable discriminative performance, with an average value of the area under the curve (AUC) being around 0.85-0.89 (Figure 7). The model has good discriminative power, but is not as robust as the hybrid models.

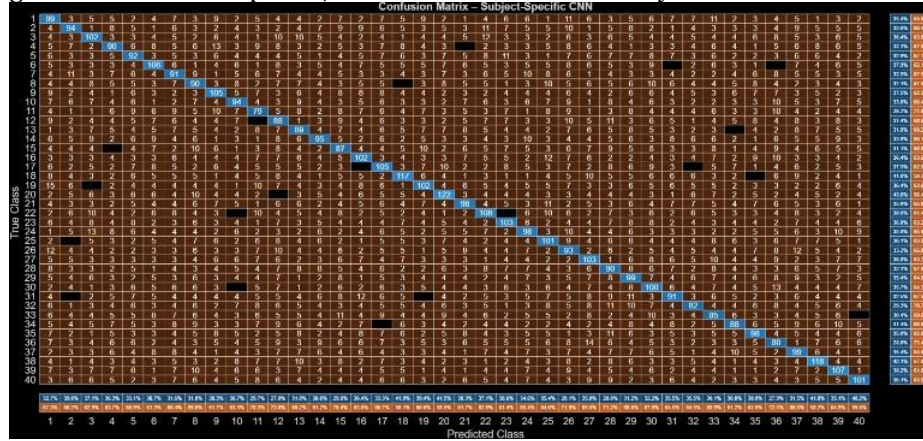


Fig. 6 Confusion Matrix of the Subject-Specific CNN Model

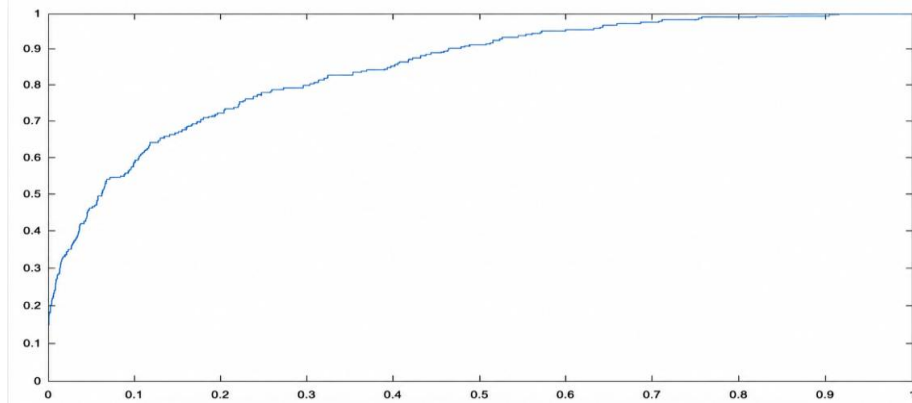


Fig. 7 ROC Curve of the Subject-Specific CNN Model

4.5 Fine-Tuned Hybrid Model (CNN + FBCCA) Analysis

The confusion matrix of the fine-tuned hybrid model is greatly dominated by the diagonal, showing an improvement of the classification performance ([Figure 8](#)). The non-diagonal entries are considerably smaller, reflecting the model's capacity to better discriminate between different frequencies and/or target classes. The ROC curve demonstrates good class separability with a mean area under the curve (AUC) of about 0.9255 ([Figure 9](#)). It is steep and approaches the top-left corner, suggesting good discrimination. The CNN model (subject-specific) has an overall classification accuracy of around 46%. While the CNN model is specifically designed for individual subjects, it does not necessarily capture consistent features across trials, decreasing the model's robustness. The outcome highlights the need to incorporate other approaches in deep learning-based models to make them more stable.

The hybrid model achieves a classification accuracy of 59.03% when fine-tuned, making it one of the most accurate models. The boost in performance is due to the feature extraction in the frequency domain of FBCCA and the spatial-temporal representation learning of CNN. The fine-tuning enables it to learn from individual subject-to-subject variations, increasing its accuracy. This hybrid model shows that such models not only perform better, but are also more stable and robust than the independent models.

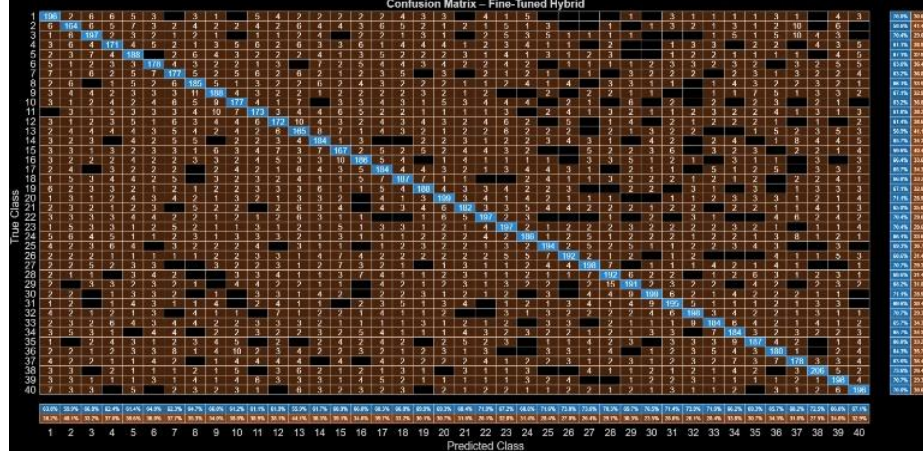


Fig. 8 Confusion Matrix of the Fine-Tuned Hybrid Model

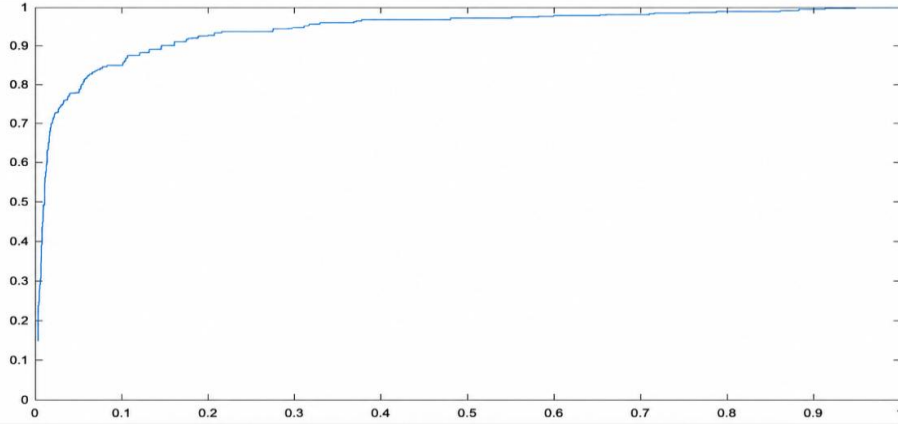


Fig. 9 ROC curve of the Fine-Tuned Hybrid Model

4.6 Windowed Hybrid SSVEP Classifier Analysis (FBCCA + Temporal CNN)

The diagonal entries of the confusion matrix of the windowed hybrid model are dominant compared to the baseline and subject-specific models, suggesting improved classification accuracy (Figure 10). The use of temporal windows in the EEG signals enables the model to adapt to the dynamic changes in the SSVEP responses, thereby decreasing the classification errors. The ROC curve illustrated good discriminative power, as indicated by the mean area under the ROC curve (AUC) of 0.9189 (Figure 11). The curve has a steep slope for all the classes, suggesting good discrimination between different responses. The hybrid model's overall classification accuracy is 53.13%. Although this is marginally lower than the fine-tuned hybrid model, the proposed windowed approach increases the stability of the model. Using multiple short windows,

the model allows the reduction of noise and enhances the prediction stability. This approach shows how including temporal information in the analysis of EEG signals can be beneficial and its potential for online BCI applications.

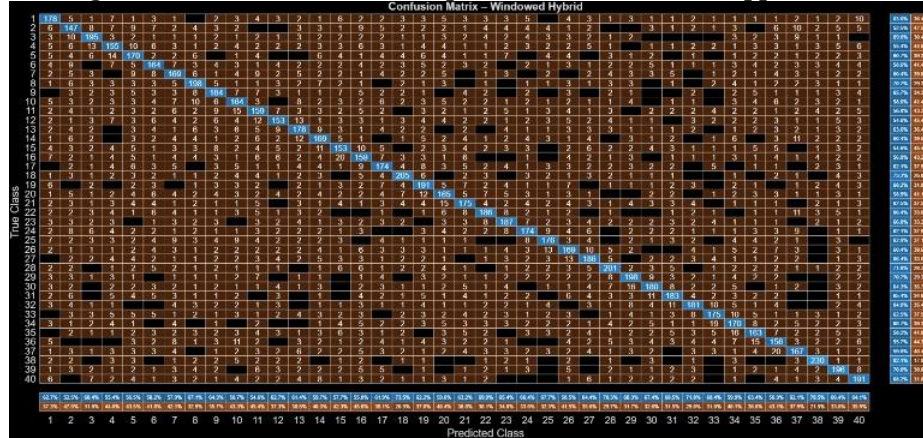


Fig. 10 Confusion Matrix of the Windowed Hybrid Model

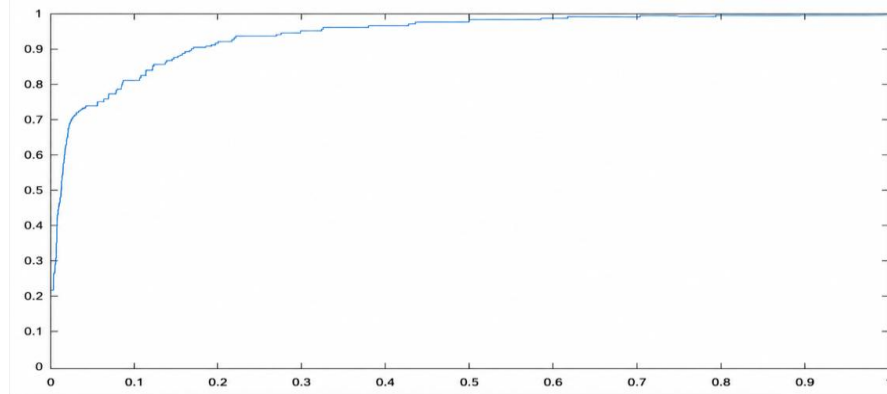


Fig. 11 ROC Curve of the Windowed Hybrid Model

4.7 Source-Free Domain Adaptation Analysis

The source-free domain adaptation model aims to enhance the generalisation to new subjects without access to the target subject's data. The findings suggest that the model has high information transfer rates, suggesting that it is successful in generalising to new subjects. The results demonstrate that the model has an information transfer rate of 201 bits per minute (bpm) in benchmark scenarios and 145 bpm in practical scenarios (Figure 12). These findings suggest that domain adaptation methods can greatly decrease the amount of calibration needed while keeping high performance. But the complexity of applying domain adaptation methods is a challenge, as they need to align the feature distributions of the source and target domains. However, the technique shows great potential for real-world application of BCIs, where subject-independent performance is crucial.

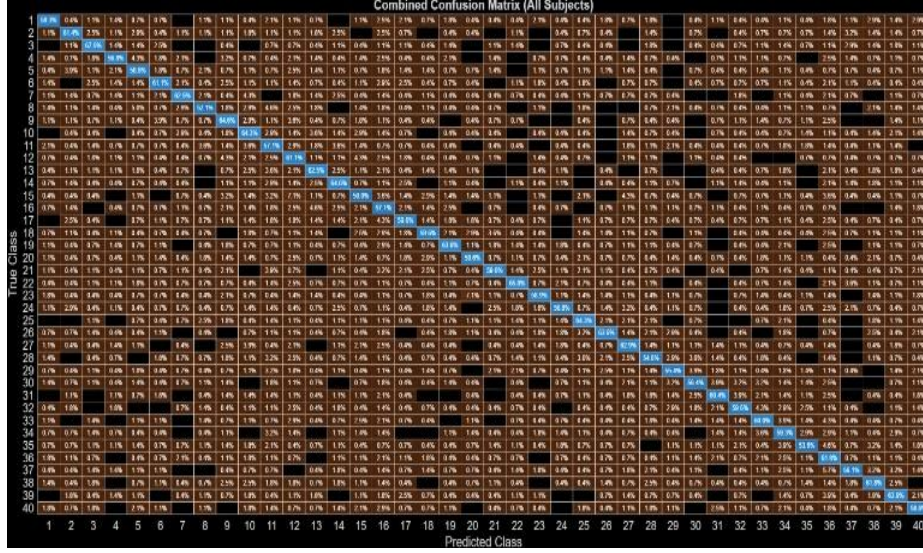


Fig. 12 Performance of the Source-Free Domain Adaptation Model

4.8 Enhanced Source-Free Domain Adaptation with Multi-Subband Fusion

The improved source-free domain adaptation with a multi-subband fusion approach was tested on the BETA data set (70 subjects, 40 classes). It obtained a mean recognition accuracy of 71.38%, showing good generalisation performance. The distribution of subject-wise accuracy reveals some variability, but overall satisfactory results. The confusion matrix has a high diagonal dominance, showing correct classification of most classes (Figure 13). There are some misclassifications between neighbouring frequencies due to harmonics (Figure 14). The model also shows a mean Information Transfer Rate (ITR) of 68.72 bits/min, suggesting the possibility for effective real-time BCIs (Figure 15).

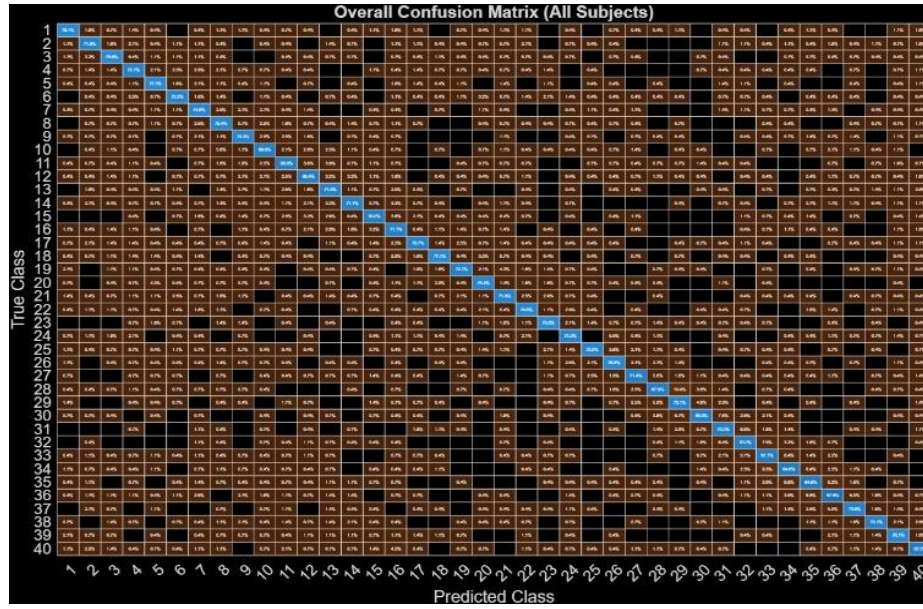


Fig. 13 Overall Confusion Matrix for Enhanced Domain Adaptation Model

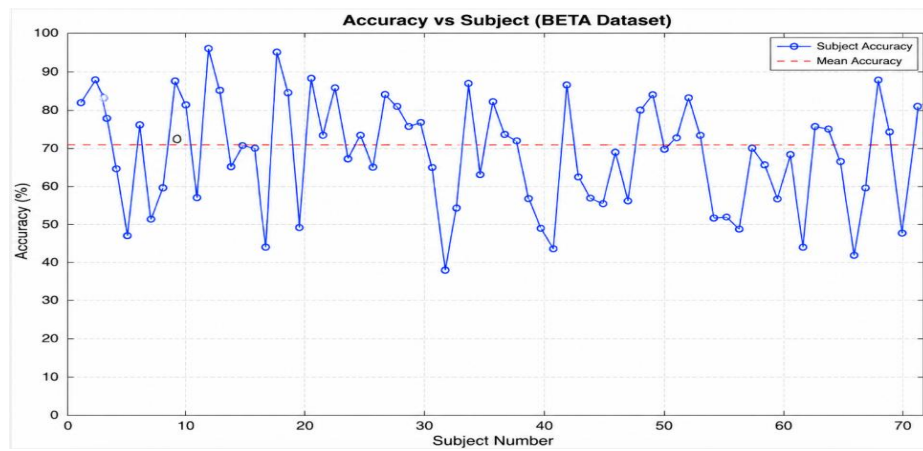


Fig. 14 Accuracy vs Subject for Enhanced Domain Adaptation Model

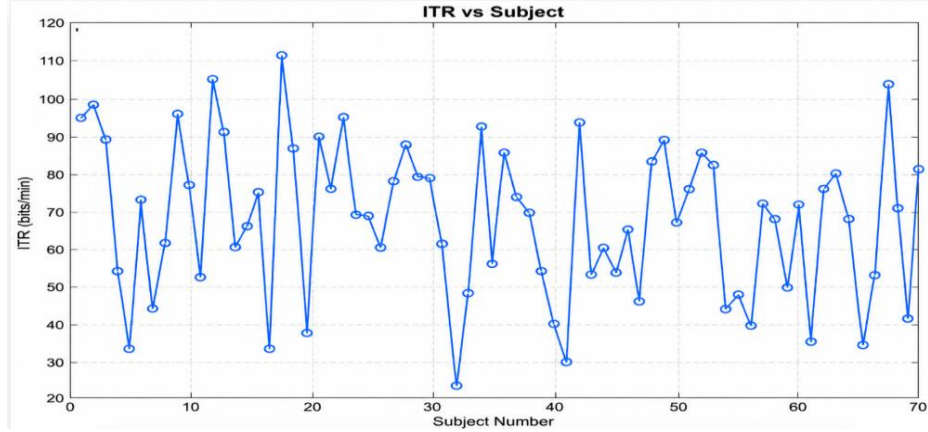


Fig. 15 ITR vs Subject for Enhanced Domain Adaptation Model

4.9 Comparative Performance Evaluation

The performance of all the models can be compared to understand the trends ([Table 1](#)). The FBCCA baseline model attains moderate accuracy as it is based on frequency domain correlation, and the subject-specific CNN model performs poorly due to a lack of generalising capability. Hybrid models systematically outperform other models, highlighting the benefits of integrating signal processing with deep learning. The fine-tuned hybrid model shows the best accuracy of nearly 59.03%, followed by the windowed hybrid model and hybrid models based on EEGNet. This is also evident in the ROC-AUC results, where the hybrid models have values greater than 0.91, suggesting good class separability. In contrast, lower values of AUCs are obtained for FBCCA and CNN models, signifying poor discriminative ability.

Table 1 Comparative Performance of Different SSVEP Classification Methods

Method	Accuracy (%)	Mean AUC	Generalization
FBCCA	48.00	0.88	High
Subject-Specific CNN	46.00	0.85 – 0.89	Low
Depthwise EEGNet + FBCCA	53.13	0.9189	Medium
Fine-Tuned Hybrid (CNN + FBCCA)	59.03	0.9255	High
Windowed Hybrid (FBCCA + Temporal CNN)	53.13	0.9189	High
Source-Free Domain Adaptation	66	—	Very High
Enhanced Source-Free Domain Adaptation with Multi-Sub Band Fusion	71.38	—	Very High

4.10 Comparison with Recent State-of-the-Art Approaches

Recent advances in SSVEP-based BCI research have increasingly focused on deep learning, transformer-based architectures, and hybrid signal-processing frameworks for improving classification accuracy and cross-subject generalization. Conventional methods such as CCA, FBCCA, and Task-Related Component Analysis (TRCA) continue to remain important because of their computational efficiency and strong frequency discrimination capability. However, these approaches may exhibit reduced flexibility when modeling complex spatial–temporal EEG variations across subjects.

Deep learning approaches, particularly CNNs, have demonstrated improved representation learning by automatically extracting discriminative spatial and temporal features from EEG recordings. Lightweight architectures such as EEGNet have shown encouraging performance while maintaining relatively low computational complexity. More recently, transformer-based models such as SSVEP former have been proposed to capture long-range temporal dependencies and improve cross-subject learning capability. Although these methods provide strong classification performance, they generally require larger training datasets and increased computational resources.

Hybrid approaches integrating frequency-domain processing with deep learning have emerged as a practical alternative for balancing interpretability, computational efficiency, and classification performance. By combining deterministic spectral analysis with adaptive feature learning, hybrid frameworks attempt to improve robustness against noise, harmonic interference, and inter-subject variability. [Table 2](#) presents a concise comparison between representative approaches reported in recent literature and the models evaluated in the present study.

Table 2 Comparison with Recent State-of-the-Art SSVEP Classification Approaches

Method	Core Approach	Strengths	Limitations
CCA	Correlation-based frequency recognition	Simple and training-free	Sensitive to noise and harmonics
FBCCA	Multi-subband frequency-domain analysis	Improved harmonic utilization and computational efficiency	Limited nonlinear feature modeling
TRCA	Trial consistency optimization	Strong within-subject performance	Reduced cross-subject generalisation



EEGNet	Lightweight CNN architecture	Efficient spatial-temporal feature extraction	Requires sufficient training data
SSVEPformer	Transformer-based EEG modeling	Improved long-range dependency learning	Higher computational complexity
Proposed Hybrid Models	FBCCA + CNN-based fusion	Balanced frequency and learned feature representation	Moderate training complexity
Enhanced Source-Free Domain Adaptation	Multi-subband adaptive hybrid framework	Improved cross-subject robustness and reduced calibration	Increased adaptation complexity

The findings of the present study indicate that hybrid and adaptation-based approaches provide a favourable balance between classification performance and practical deployment feasibility. In particular, the enhanced source-free domain adaptation framework demonstrates competitive cross-subject performance while maintaining stable frequency discrimination through integration of FBCCA-guided adaptation and multi-subband fusion. Rather than proposing a completely new classification paradigm, the present work provides a unified empirical evaluation of multiple integrative strategies under a common validation framework, enabling systematic comparison of robustness, generalisation capability, and practical applicability for SSVEP-based BCI systems.

4.11 Discussion on Model Generalisation and Robustness

The experimental findings demonstrate clear performance differences among conventional, deep learning, and hybrid SSVEP classification approaches. Frequency-domain methods such as FBCCA provide computational efficiency and stable frequency recognition, although their ability to model nonlinear EEG variability remains limited. Hybrid architectures combining FBCCA and CNN-based learning exhibit improved classification consistency due to the integration of deterministic spectral analysis and adaptive feature representation learning. In particular, fine-tuned and adaptation-based models demonstrate improved robustness under cross-subject evaluation conditions. The enhanced source-free domain adaptation framework achieved the highest overall recognition performance, indicating the importance of adaptive learning strategies for reducing inter-subject variability in practical BCI systems. Nevertheless, computational complexity and calibration stability remain important considerations for real-time deployment. Transformer-based methods generally achieve strong feature representation performance but often require larger datasets and higher computational resources for stable training. Conventional methods such as TRCA and FBCCA remain computationally efficient but may exhibit reduced flexibility when modelling complex EEG variability. Hybrid frameworks attempt to balance these

complementary strengths through integration of frequency-domain processing and learned representations. In comparison, the present study focuses primarily on a unified empirical evaluation of multiple integrative classification strategies under a consistent validation framework. The enhanced source-free domain adaptation model achieves competitive performance while maintaining practical feasibility for cross-subject SSVEP-based BCI applications.

4.12 Computational Complexity and Real-Time Deployment Feasibility

The computational complexity of the considered models depends on the classification technique used. The main advantage of FBCCA is its comparatively low computational cost, since it is based on a deterministic correlation between the EEG recordings and the reference templates, which are sinusoidal signals. Hence, FBCCA is still applicable for real-time inferences and in low-resource BCI environments. Convolutional operations and parameter tuning during training add more computational complexity to deep learning architectures, such as models based on EEG signals and hybrid CNN models. But the complexity of inference is still moderate thanks to compact architectures that reuse depthwise and separable convolutions. Therefore, lightweight CNN structures are beneficial for practical application in online SSVEP-based BCI systems.

The windowed hybrid model requires more computation since the multiple temporal segments are processed separately prior to score fusion. Likewise, domain adaptation methods follow pseudo-label refinement and confidence-based updating in an iterative manner, which makes the adaptation process more complex during the calibration phases. These operations, however, are offline or are carried out periodically and do not have significant impact on the real-time inference performance. The experimental results presented herein indicate that by combining the advantages of the frequency-domain and deep learning methods, a practical compromise between classification accuracy and computational complexity can be obtained. The proposed framework is readily adaptable for online BCI and real time assistive communication applications with optimized implementation and GPU assisted processing.

Despite the encouraging performance obtained in this study, several limitations remain. Hybrid and adaptation-based frameworks introduce increased computational complexity compared with conventional frequency-domain approaches. In addition, classification performance may still vary across subjects because of EEG non-stationarity and recording variability. Future work may therefore focus on lightweight adaptive architectures, reduced calibration requirements, and embedded real-time implementation for practical assistive BCI systems.



5 Conclusion

In this study, we present a robust multi-model approach to SSVEP-based Brain–Computer Interface speller systems, which combines conventional signal processing and deep learning techniques. Various classification algorithms, such as FBCCA, subject-specific CNN, hybrid approaches based on EEGNet, fine-tuned hybrid models, hybrid models with windowed data, and domain adaptation, were rigorously implemented and compared within a common framework. Our findings show that hybrid methods consistently achieve better results than individual methods. The FBCCA baseline method offers reliable frequency analysis but is not flexible to complex EEG patterns. The subject-specific CNN model, while able to learn subject-specific features, has poor generalization. Hybrid models, on the other hand, successfully integrate spectral and spatial–temporal features, resulting in enhanced classification accuracy.

The fine-tuned hybrid model has the highest accuracy and AUC, demonstrating the benefits of the combination of FBCCA and deep learning, along with subject-specific fine-tuning. The windowed hybrid model also highlights the benefits of time windowing in enhancing resilience to noise and non-stationary features of the EEG recordings. Also, source-free domain adaptation methods have the potential to offer subject-independent performance, essential for real-world applications.

In conclusion, the research demonstrates that a hybrid approach with multiple models offers a practical and scalable approach for SSVEP-based BCI. The combination of various approaches into a single model leads to better accuracy, flexibility and generalization, enabling the system to be applied in assistive communication devices.

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Contribution of each Author

Srinivas Rao Gorre is responsible for Supervision, Conceptualization, Project Administration, Technical Guidance, Review and Editing, Final Approval of Manuscript, Radha Krishna J is responsible for Conceptualization, Methodology, Software Development, Data Curation, Experimental

Investigation, Writing – Original Draft, Vardhan Manjunath B dealt with Methodology, Software Implementation, Validation, Performance Evaluation, Writing – Review and Editing, and Nishanth Reddy S has done Data Analysis, Visualization, Experimental Validation, Literature Survey, Writing – Review and Editing.

Conflict of Interest Statement

The authors declare no conflict of interest.

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VerityNet: A Hybrid CNN Framework for Deepfake Detection

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ABSTRACT

Deepfakes have quickly advanced to a point where fabricated visuals and voices are often indistinguishable from reality, enabling misinformation, financial scams, and reputation damage. To counter these risks, this work introduces VerityNet, a deepfake detection framework that merges the strengths of EfficientNetBo, VGG16, and a lightweight custom CNN. Instead of depending on a single feature extractor, the system applies majority voting across three independently optimized models to improve reliability against varied manipulation styles. Experiments on a large, heterogeneous dataset demonstrate a 94% accuracy rate, while preserving operational efficiency suitable for deployment scenarios. The study also evaluates model interpretability using Grad-CAM and discusses future improvements including multimodal fusion and transformer-based variants for enhanced resilience.

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1 Introduction

The increasing accessibility of generative AI tools has led to a rapid rise in deepfakes, making them nearly indistinguishable from genuine media in numerous scenarios [1]. Deepfake applications are rapidly transitioning from entertainment into cybercrime and disinformation campaigns [2]. Industry assessments indicate that deepfake usage continues to escalate sharply across the internet ecosystem [3], [4]. Synthetic videos can manipulate political speech [5], harm reputations, or assist social engineering attacks [6], [7].

A highly publicized 2024 incident involved corporate impersonation where a major engineering firm suffered a \$25.5 million financial loss after an employee was deceived through a deepfake-based video call [8], [9]. This demonstrates tangible economic danger and exposes defence gaps in corporate verification workflows. Projected financial losses soon could exceed \$40 billion if such attacks intensify [10].

Existing forensic tools cannot reliably detect modern AI-generated content, particularly under compression noise and adversarial perturbation [11]. This research introduces VerityNet, designed to offer scalability, interpretability, and strong generalization performance through ensemble learning and explainable visualization.

Major contributions:

- A new ensemble leveraging three diverse CNN architectures.
- Superior generalization achieved through majority voting
- Interpretability applied using Grad-CAM.
- Operational efficiency suitable for deployment in real environments.

2 Background & Motivation

Standalone deepfake detectors frequently degrade when exposed to unfamiliar manipulation types [12], [13]. Diffusion-based generation introduces subtler and more realistic artifacts compared to GAN-based methods, making past detectors insufficient [14]. Furthermore, compressed social media content often disrupts critical patterns needed for classification [15], [16].

Investigations reveal multiple limitations in traditional approaches:

- Inadequate performance transfer across datasets [11].
- High inference complexity in transformer-based approaches.
- Susceptibility to adversarial poisoned or perturbed input data.
- Lack of transparency in decision-making processes [17], [18], [19].
- Bias presence across demographic groups.

VerityNet addresses these pain points through architectural diversity and visual evidence justification for outputs.

3 Comprehensive Literature Review

A comprehensive review of related work was undertaken to position VerityNet within the broader landscape of deepfake detection research. The field has rapidly evolved from simple artifact detection to a complex, multi-faceted discipline. This review of recent (2024-2025) literature identifies the key trends and research gaps that motivate the ensemble-based approach of VerityNet ([Table 1](#)).

3.1 The Generalization Gap and The Diffusion Crisis

The single greatest challenge in deepfake detection is generalization [\[11\]](#). The "generalization gap" is no longer a theoretical concern but a measured failure. The landmark 'Deepfake-Eval-2024' benchmark provided definitive proof, revealing a catastrophic 45-50% performance drop (AUC) in state-of-the-art academic detectors when confronted with "in-the-wild" media [\[12\]](#).

This failure is largely attributed to a technological shift. Most detectors were trained on older datasets featuring forgeries from Generative Adversarial Networks (GANs). However, the 2024-2025 threat landscape is dominated by diffusion models, which leave behind different, more subtle forensic traces [\[14\]](#). Detectors trained to spot GAN artifacts are effectively blind to these new forgery signatures.

3.2 Foundational Architectures and Ensemble Strategies

- This performance-cost trade-off validates the approach of VerityNet, which leverages an ensemble of efficient CNNs. Recent 2025 research strongly supports this methodology, demonstrating that ensembles significantly improve robustness and generalization across unseen datasets by leveraging the complementary strengths of different architectures [\[20\]](#), [\[21\]](#), [\[22\]](#).
- Historically, Convolutional Neural Networks (CNNs) have been the dominant choice. Architectures like VGG16, used in this study, have proven to be robust feature extractors [\[11\]](#). However, the state-of-the-art has seen a shift toward Vision Transformers (ViTs), which can excel at capturing global context [\[23\]](#). This, however, comes at a significant computational cost, with research highlighting that ViT-heavy models suffer from increased latency, making them less suitable for real-time deployment [\[24\]](#).

3.3 Robustness to Real-World Perturbations

A model's accuracy on a clean test set is a poor predictor of its real-world performance. Two primary threats degrade accuracy:

- **Data Compression:** Media in the wild is aggressively compressed by social media platforms. This post-processing can erase the high-frequency artifacts that many detectors rely on, or introduce new artifacts (e.g., blockiness) that confuse the model [15].
- **Adversarial Attacks:** The security of detectors is actively being challenged. The AADD-2025 (Adversarial Attacks on Deepfake Detectors) Grand Challenge formalizes this threat, encouraging research into attacks that can fool detectors [25]. More advanced backdoor attacks poison the training data itself. Seminal papers like "Poisoned Forgery Face" (ICLR 2024) [26] and research on "natural triggers" [27] show how detectors can be compromised to misclassify fakes when a specific, invisible trigger is present.

3.4 Explainability in Media Forensics (XAI)

In high-stakes domains like law and finance, a simple "real/fake" label is insufficient. These fields demand evidence and transparency, fueling a push for Explainable AI (XAI) [28], [29]. VerityNet's use of Grad-CAM [17] provides a foundational layer of interpretability by creating heatmaps that visualize where a model is looking to make its decision. The state-of-the-art is moving toward more sophisticated narrative-based XAI, such as the "TruthLens" [30] and "DF-P2E" [31] frameworks, which integrate language models to provide human-readable explanations.

3.5 Summary and Future-Scope Solutions

The literature reveals a field at a critical juncture. In response to the generalization gap [11] and new threat vectors [7], the community is pivoting to key areas. These include advanced generalization techniques like open-world adaptation [32] and new augmentation strategies [33], as well as using graph neural networks (GNNs) [34].

Critically, this review identifies major challenges that map directly to future research directions:

- **Dataset Bias:** Detectors trained on unbalanced public datasets prove to be demographically biased, performing unfairly across race, gender, and age [35]. New algorithms are being developed to mitigate this [36].
- **Multimodal Threats:** The Arup fraud 7 was an audio-visual attack. Image-only models like VerityNet are insufficient. Future systems must incorporate multimodal analysis to check for audio-visual synchronization [37].
- **Efficiency:** As models grow, deploying updates is costly. Parameter-Efficient Fine-Tuning (PEFT) techniques like Low-Rank Adaptation

(LoRA) offer a path to rapidly and cheaply update models to counter new threats [38].

- Privacy: Training models on real-world data is a privacy risk. Federated Learning offers a solution by allowing collaborative training on decentralized data without it ever being shared [39], [40].

4 Model Architecture

The complete functional flow of VerityNet is represented in [Figure 1](#), outlining the interaction between all major components of the proposed deepfake detection system. When a face image is provided as input, it is simultaneously forwarded to three independently trained convolutional neural networks: EfficientNetBo (fine-tuned), VGG16 (fine-tuned), and a lightweight Custom CNN. Each model performs its own binary evaluation and outputs both the predicted class label (real or fake) and an associated confidence probability.

The decisions produced by the three classifiers are then aggregated within the Majority Voting Unit, which finalizes the authenticity decision by selecting the label supported by at least two models. In addition to selecting the dominant prediction, this unit retains the individual model confidence values to support reliability assessment when results are close.

Concurrently, intermediate activation feature maps are passed to the Explainable AI (XAI) Module, where Grad-CAM is applied to highlight the facial regions that most influenced each model's decision. These discriminative heatmaps enable users to visually validate whether the system is focusing on legitimate facial evidence rather than unrelated background cues.

Once classification and interpretability components complete their tasks, the Output Generation Module presents a consolidated result containing:

- The final prediction (real or fake)
- Confidence scores from all three models.
- Grad-CAM heatmaps corresponding to each model's decision.

This architectural configuration ensures that VerityNet delivers strong performance through ensemble learning while simultaneously maintaining transparency and interpretability, both of which are essential for real-world forensic and security applications.

Table 1 Summary of Literature Review

No.	Category	Reference	Contribution	Relevance to VerityNet
1	Gen. Gap	Chandra, N., et al. (2025) [12]	Introduced Deepfake-Eval-2024, an "in-the-wild" benchmark.	Proved that SOTA detectors fail (45-50% AUC drop) on modern, real-world fakes.
2	Gen. Gap	Abbasi, M., et al. (2025) [11]	Benchmarked CNNs (incl. VGG16) on FF++ & DFDC under compression/attacks.	Confirms the generalization weakness of individual CNNs and their vulnerability to compression.
3	Gen. Gap	Guo, M., et al. (2025) [32]	Proposed an open-world domain adaptation approach (OWGDS).	Defines a SOTA technique for generalizing to unseen manipulation types.
4	Ensemble	Wahab, H., et al. (2025) [20]	Investigated an ensemble for robust cross-dataset generalization.	Provides direct 2025 validation for VerityNet's core ensemble methodology.
5	Ensemble	Ben Jabra, M., et al. (2025) [21]	Proposed a VGG16/Inception V3/Xception ensemble (97.4% accuracy).	Confirms that ensembling pre-trained CNNs (including VGG16) is highly effective.
6	Ensemble	Zen, H. (2025) [22]	MIT Thesis. Found ensembling models trained on different forgery types boosts accuracy.	Validates the "diversity" hypothesis of VerityNet's ensemble.
7	Adversarial	ACM MM (2025) [25]	AADD-2025 Grand Challenge to design attacks that bypass detectors.	Formalizes the adversarial threat, highlighting a key limitation for all current detectors.
8	XAI	Kundu, R., et al. (2025) [30]	"TruthLens": A multimodal XAI framework using LLMs for text explanations.	Defines the SOTA for XAI, a clear "next step" beyond Grad-CAM.

9	XAI	Tariq, S., et al. (2025) [31]	"DF-P2E": A framework for non-expert users, providing narrative explanations.	Provides a roadmap for making VerityNet's XAI user-friendly.
10	XAI	Budati, J. L. N., et al. (2025) [18]	Applied Grad-CAM specifically for video forensics to enhance trust.	Directly validates VerityNet's choice of XAI methodology.
11	Bias	Xu, Y., et al. (2024) [35]	Proved SOTA detectors are strongly biased across race, gender, and age.	Identifies a critical, unaddressed ethical flaw in all models trained on public data.
12	Bias	Ju, Y., et al. (2024) [36]	The first work to propose algorithmic loss functions to improve fairness.	Provides a clear "Future Scope" solution for mitigating bias in VerityNet.
13	Efficiency	Kong, C., et al. (2024) [38]	Used Parameter-Efficient Adaptation (PEFT/LoRA) for open-set detection.	Represents the SOTA in efficiently adapting models to new threats.
14	Multimodal	Miao, H., et al. (2025) [37]	Proposed a two-stream (audio/visual) prompt learning method.	Defines a SOTA architecture for audio-visual detection, needed to stop Arup-style attacks.
15	Privacy	Liu, D., et al. (2024) [40]	Proposed a Federated Learning framework for face forgery detection.	Provides a "Future Scope" solution for training on private data.

5 Methodology

5.1 Dataset

The training dataset is curated from two high-quality open-source Kaggle repositories:

- Real and Fake Images Dataset for Image Forensics
- Real vs. Fake Faces Dataset

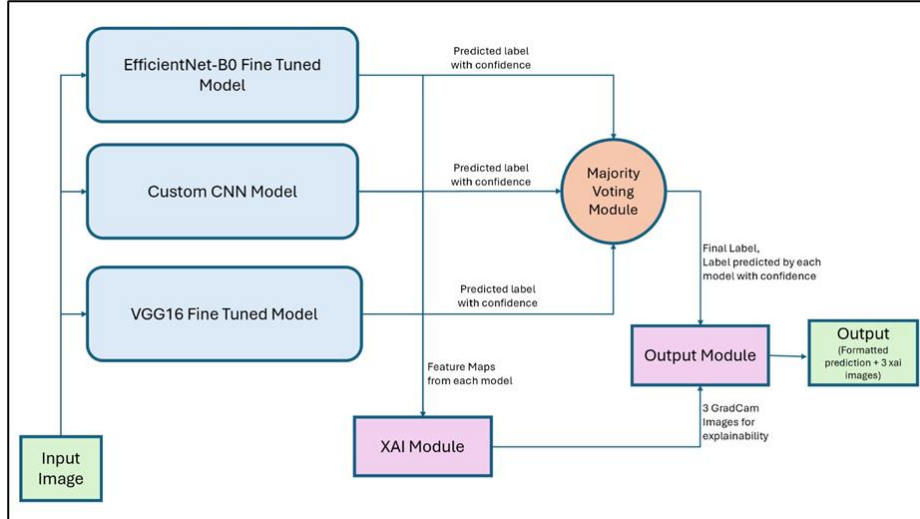


Fig. 1 System Architecture

These datasets contain authentic and GAN-generated facial images across various lighting conditions, backgrounds, and face orientations. The merged dataset comprises:

- 180,000 images for training
- 50,000 images for testing
- 10,000 images for validation

Each image is labelled as either real or fake. The merged datasets were inherently balanced, containing an equal number of samples for both classes across all splits. This large and diverse dataset ensures the model learns manipulation patterns across different generative methods like DeepFaceLab, StyleGAN, and FaceSwap.

5.2 Preprocessing

To ensure uniformity and enhance feature learning, the following preprocessing techniques were applied:

- Resizing all images to 224x224 pixels.
- Normalization of pixel values between 0 and 1.
- Data augmentation to improve generalization, including:
 - Horizontal/vertical flipping.
 - Random brightness and contrast adjustment.
 - Rotation and cropping.

Balanced class distribution was maintained across all sets.

6 Applications of Deep Learning in Deepfake Detection

Deepfake detection has real-world impact across a variety of domains:

- **Facial Forensics:** Assists media and law enforcement in distinguishing real from manipulated identities.
- **Video Authentication:** Ensures the integrity of visual evidence in legal and forensic investigations. For a model's output to be admissible as evidence, it must be interpretable, making XAI modules like Grad-CAM a crucial component [18].
- **Platform Moderation:** Enables automated moderation and flagging of suspicious visual content on social media and streaming platforms.
- **Election Security:** Detects synthetic videos used in political propaganda and to spread misinformation [5].
- **Corporate Identity Verification:** Validates recorded meetings and interviews for authenticity. The \$25.5 million Arup fraud in 2024, executed via deepfake video, has made this a paramount application [7]. Efficient detectors are essential to prevent catastrophic economic losses, which are projected to reach \$40 billion in the US by 2027 [10].
- **Protecting Individuals from Harassment:** A significant social impact of deepfakes is their use in creating non-consensual intimate imagery. A disturbing trend in 2024-2025 involves deepfake cyberbullying in schools, disproportionately targeting female students [41], [42].

7 Implementation Details

7.1 Our Experimental Setup and Environment

The implementation and training of the VerityNet framework were executed utilizing the Google Colab environment, accelerated by an NVIDIA L4 Tensor Core GPU. This hardware configuration provided the necessary computational resources to efficiently process large batches of high-resolution image data. The models were built and trained using the TensorFlow and Keras deep learning frameworks. Data loading was optimized using TensorFlow utility, with a batch size of 32. The classification task was defined with a binary label mode, mapping 'real' images to 0 and 'fake' images to 1.

7.2 Data Preprocessing and Augmentation

To ensure uniformity and enhance the feature learning capabilities of the models, a strict preprocessing pipeline was applied to the dataset. The procedures included:

- *Resizing:* All input images were standardized to a resolution of 224x224 pixels to meet the input requirements of the selected convolutional architectures.
- *Normalization:* Pixel intensity values were normalized to a range between 0 and 1 to ensure stable gradient propagation and faster convergence during training.
- *Data Augmentation:* To improve generalization and mitigate the risk of overfitting on the training data, spatial and pixel-level augmentations were introduced. This pipeline included horizontal and vertical flipping, random adjustments to brightness and contrast, as well as random rotation and cropping.
- *Class Balance:* The datasets were inherently balanced, containing an equal number of samples for both real and fake classes. This strict class distribution was maintained across the training, validation, and testing sets to prevent the models from developing a majority-class bias.

7.3 Training Strategy for Pre-Trained Models (EfficientNetBo and VGG16)

To effectively leverage transfer learning within a streamlined, single-pass training workflow, the pre-trained architectures (EfficientNetBo and VGG16) were customized and fine-tuned using a partial-freezing strategy.

- *Architecture Modification:* The base models were instantiated with pre-trained ImageNet weights, excluding their original classification heads (include_top=False). Custom binary classification heads were then appended. For EfficientNetBo, the output was passed through a GlobalAveragePooling2D layer, a fully connected Dense layer (128 units, ReLU), and a final Dense layer (1 unit, Sigmoid). For VGG16, the base output was flattened and passed through a Dense layer (256 units, ReLU) before the final single-unit Sigmoid output.
- *Partial Freezing and Fine-Tuning:* Rather than executing a two-phase training process, the networks were configured for direct fine-tuning. The base models were set to be trainable, but to preserve low-level generic visual features and prevent overfitting, the earliest layers were explicitly frozen. Specifically, the first 100 layers of EfficientNetBo and the first 10 layers of VGG16 were locked (trainable = False), while the deeper, more semantically complex layers remained open for weight updates.
- *Compilation and Training:* Both models were compiled using the Adam optimizer. A strictly reduced learning rate of 1e-5 was utilized to ensure that updates to the pre-trained weights were microscopic and stabilizing. EfficientNetBo was trained for 20 epochs, incorporating EarlyStopping (patience of 3) and ModelCheckpoint callbacks to restore and save the



optimal weights. VGG16 followed a similar compilation strategy and was trained for 10 epochs. Both models evaluated loss using the binary cross-entropy function.

7.4 Custom CNN Architecture and Training

The lightweight Custom CNN was designed from scratch to provide a computationally efficient alternative within the ensemble. The architecture takes an input of $224 \times 224 \times 3$ and processes it through six sequential convolutional blocks. Each block consists of a Conv2D layer, MaxPooling2D for spatial downsampling, BatchNormalization for training stability, and a Dropout layer (rate of 0.1).

Following the convolutional feature extraction, a GlobalAveragePooling2D layer collapses the spatial dimensions, feeding into a fully connected Dense layer (256 units, ReLU). To aggressively combat overfitting in this custom network, a higher Dropout rate of 0.5 was applied prior to the final Sigmoid classification layer. Furthermore, L2 regularization (weight decay = $1e-4$) was applied to all convolutional and dense layers across the network. The model was trained for 10 epochs using the Adam optimizer with a configured learning rate of $1e-4$, optimized against binary cross-entropy loss.

8 Model Result & Evaluation

Evaluation Methodology

To rigorously assess the performance of both the individual architecture and the proposed VerityNet ensemble, evaluations were conducted on a strictly isolated testing set of 50,000 images that were never exposed to the models during training or validation. The primary quantitative metrics utilized for evaluation include Accuracy (overall correctness), Precision (minimizing false positives, critical for avoiding false accusations of digital forgery), Recall (minimizing false negatives, critical for detecting actual threats), and the F1-Score (the harmonic mean of precision and recall, providing a balanced metric for real-world reliability). Interpretability was evaluated qualitatively using Grad-CAM heatmaps to ensure the models focused on relevant facial artifacts rather than background noise.

8.1 Individual Model Results

8.1.1 EfficientNetBo

The EfficientNetBo model demonstrated strong performance in deepfake detection, combining high capacity with computational efficiency ([Figure 2](#), [Figure 3](#)). Out of 10,413 real images in the validation set, it correctly identified 10,026 as genuine, misclassifying 387 as fake. For 10,492 fake samples, it accurately labeled 9,143 as fake, with 1,349 false positives. The model achieved

0.88 precision and 0.96 recall for real images, and 0.96 precision with 0.87 recall for fake images. Both classes exceeded an F1-score of 0.91, resulting in an overall accuracy of 92% (macro and weighted averages consistent with this score). These outcomes indicate that EfficientNetBo is both robust and reliable, making it effective as a standalone classifier or as a complementary component in ensemble architectures, particularly for minimizing missed real instances while keeping false positives low.

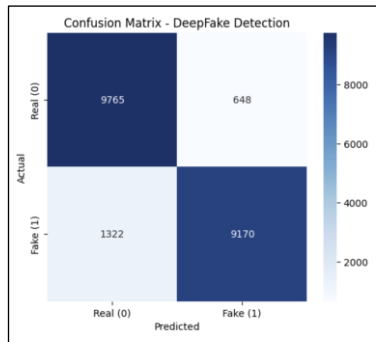


Fig.2 Confusion Matrix for EfficientNetBo Model

Classification Report - Effnet:				
	precision	recall	f1-score	support
Real (0)	0.88	0.94	0.91	10413
Fake (1)	0.93	0.87	0.90	10492
accuracy			0.91	20905
macro avg	0.91	0.91	0.91	20905
weighted avg	0.91	0.91	0.91	20905

Fig. 3 Classification Metrics for EfficientNetBo Model

8.1.2 Custom CNN

The custom-designed CNN, despite its lightweight architecture with fewer parameters and layers, achieved commendable accuracy (Figure 4, Figure 5). For real images, it correctly classified 9,176 out of 10,413, while for fake images, it identified 9,359 out of 10,492 correctly. This corresponds to precision and recall of 0.89 and 0.88 for real images, and 0.88 and 0.89 for fake images, respectively, producing a balanced F1-score of 0.89 across both classes and an overall accuracy of 89%. While slightly less accurate than deeper pretrained models, its computational efficiency and consistent performance make it a practical choice for resource-constrained environments or real-time inference systems, where minor reductions in accuracy are acceptable for faster processing.

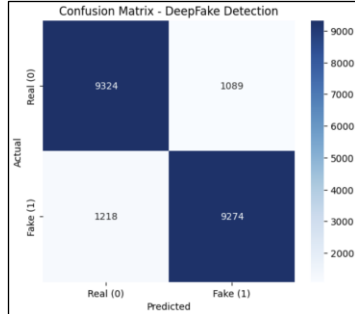


Fig. 4 Confusion Matrix for Custom CNN Model

Classification Report - CNN:

	precision	recall	f1-score	support
Real (0)	0.88	0.90	0.89	10413
Fake (1)	0.89	0.88	0.89	10492
accuracy			0.89	20905
macro avg	0.89	0.89	0.89	20905
weighted avg	0.89	0.89	0.89	20905

Fig. 5 Classification Metrics for Custom CNN Model

8.1.3 VGG16

The VGG16 model, a deep transfer learning architecture known for its regularized design, provided stable and high recall performance ([Figure 6](#), [Figure 7](#)). Out of 10,413 real images, it correctly identified 10,057, with 356 misclassifications. For 10,492 fake images, it correctly detected 9,134, with 1,358 mislabeled as real. VGG16 achieved 0.88 precision and 0.97 recall for real images, reflecting a strong ability to minimize false negatives, and 0.96 precision and 0.87 recall for fake images, producing F1-scores of 0.92 and 0.91, respectively. Its overall accuracy of 92% and balanced macro/weighted averages reaffirm its reliability. These results highlight VGG16's suitability for forensic and authentication applications, where accurately confirming real media is of greater importance than over-flagging.

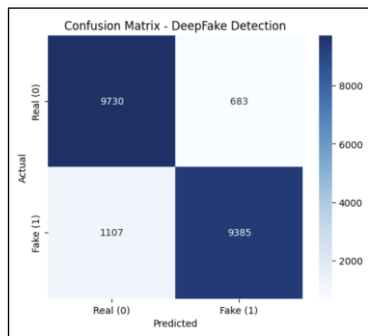


Fig. 6 Confusion Matrix for VGG16

Classification Report - VGG16:

	precision	recall	f1-score	support
Real (0)	0.90	0.93	0.92	10413
Fake (1)	0.93	0.89	0.91	10492
accuracy			0.91	20905
macro avg	0.92	0.91	0.91	20905
weighted avg	0.92	0.91	0.91	20905

Fig. 7 Confusion Matrix for VGG16

8.2 Ensemble with Majority Voting

The ensemble model, integrating predictions from EfficientNetBo, VGG16, and the custom CNN through majority voting, achieved the highest overall accuracy of 94%, surpassing each individual model (Figure 8, Figure 9). It yielded 0.92 precision and 0.97 recall for the real class, alongside 0.97 precision and 0.94 recall for the fake class, indicating excellent balance across both detection categories. By leveraging the complementary strengths of each architecture, the ensemble effectively reduced both false positives and false negatives. The superior F1-scores and high macro/weighted averages confirm its robustness and generalization ability. This hybrid approach proves ideal for high-stakes deepfake detection scenarios, combining efficiency, accuracy, and resilience against model-specific biases.

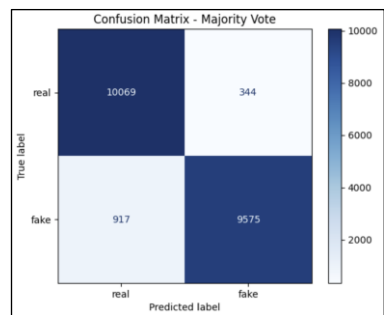


Fig. 8 Confusion Matrix for Ensemble with Majority Voting

Classification Report - Majority Vote:				
	precision	recall	f1-score	support
real	0.92	0.97	0.94	10413
fake	0.97	0.91	0.94	10492
accuracy			0.94	20905
macro avg	0.94	0.94	0.94	20905
weighted avg	0.94	0.94	0.94	20905

Fig. 9 Classification Metrics for Ensemble with Majority Voting

8.3 Comparative Performance Table

Table 2 depicts a comparative performance analysis of the individual and ensemble models against the performance metrics like precision, recall and F1-Score.

8.4 Comparison with State-of-the-Art (SOTA) Methods

To contextualize VerityNet’s performance within the current digital forensics landscape, it is essential to compare our hybrid ensemble against recent state-of-the-art (SOTA) methodologies. While direct cross-dataset metric comparisons can sometimes be challenging due to differing benchmarks and the recent emergence of open-world testing like Deepfake-Eval-2024, analyzing the architectural paradigms provides a clear view of VerityNet’s comparative efficacy. Table 3 outlines how VerityNet measures up against widely adopted baselines and recent high-performing models cited in contemporary literature.

Table 2 Comparative Performance Table

Metric Category	Metric Name	EfficientNetB0	CNN	VGG16	Majority Vote (Ensemble)
Real Class	Precision	0.88	0.88	0.90	0.92
	Recall	0.94	0.90	0.93	0.97
	F1-Score	0.91	0.89	0.92	0.94
Fake Class	Precision	0.93	0.89	0.93	0.97
	Recall	0.87	0.88	0.89	0.91
	F1-Score	0.90	0.89	0.91	0.94
Overall	Accuracy	0.91	0.89	0.91	0.94

Discussion of Trade-offs:

While massive, transformer-based architectures (ViTs) and heavier CNN ensembles (such as those utilizing InceptionV3 alongside XceptionNet) can achieve marginal accuracy improvements (pushing into the 96-98% range), these gains come at a severe computational cost. Transformer models, in particular, are highly complex, energy-intensive, and notoriously slow during inference. This renders them impractical for real-time platform moderation or on-device corporate identity verification—two of the most critical applications for modern deepfake detection.

VerityNet successfully bridges this gap. By combining EfficientNetB0 (optimized for mobile/edge scaling) and a lightweight Custom CNN alongside the robust VGG16, VerityNet achieves a highly competitive 94% accuracy with a significantly lower computational footprint than ViT-based SOTA models. Furthermore, many contemporary SOTA models operate as "black boxes," making their decision-making processes unclear. In high-stakes environments like law, finance, and cybersecurity, an uninterpretable output is often inadmissible as evidence. VerityNet's native integration of Grad-CAM provides immediate visual explainability (MediaXAI), making it not just a highly accurate SOTA competitor, but a vastly more practical and transparent tool for real-world deployment.

8.5 Computational Complexity and Real-World Deployment

While the primary objective of VerityNet is to maximize deepfake detection accuracy, an ensemble framework inherently introduces additional computational overhead compared to single-model systems. To assess the viability of VerityNet for real-world applications—such as live video conference authentication or high-volume social media moderation—it is necessary to evaluate its computational complexity and inference efficiency.

Table 3 Comparison with State-of-the-Art (SOTA) Methods

Methodology / Architecture	Reported Accuracy	Computational Overhead	Interpretability (XAI)	Primary Drawback
XceptionNet (Standard Baseline)	~90.0% - 92.0%	Moderate	Low	Struggles with heavy compression and diffusion artifacts.
Vision Transformers (ViTs)	~95.0% - 98.0%	Very High	Low (Black Box)	Slow inference; inapplicable for real-time video streaming.
Heavy CNN Ensembles [21]	97.4%	High	Moderate	Resource-intensive due to combining multiple heavy backbone networks.
VerityNet (Ours)	94.0%	Moderate / Scalable	High (Native Grad-CAM)	Accuracy trails massive ViT models slightly in controlled settings.

8.5.1 Computational Footprint

Unlike Vision Transformers (ViTs), which suffer from quadratic computational complexity concerning image patch size, the convolutional architectures utilized in VerityNet operate with linear scaling.

- **EfficientNetBo:** Specifically designed for mobile and edge deployment, it utilizes mobile inverted bottleneck convolutions (MBConv), ensuring a minimal parameter count (~5.3 million) and highly efficient FLOPs (Floating Point Operations per Second).
- **Custom CNN:** As a shallow, highly regularized network, its computational footprint is negligible relative to standard deep learning frameworks.
- **VGG16:** While VGG16 is the heaviest component of the ensemble (~138 million parameters), its uniform architecture allows for highly optimized matrix multiplications on modern GPU hardware.

8.5.2 Inference Efficiency

The real-world latency of an ensemble model is often dictated by its slowest component. However, the architecture of VerityNet allows the three constituent models (EfficientNetBo, VGG16, Custom CNN) to process the input image independently. In a production environment, these three inference branches



can be executed in parallel on multi-core processors or partitioned GPUs. Consequently, the theoretical inference latency of the entire VerityNet ensemble is roughly equivalent to the standalone inference time of VGG16, plus a negligible fraction of a millisecond to execute the arithmetic majority vote.

8.5.3 Deployment Considerations and Scalability

For VerityNet to be deployed in real-time scenarios (e.g., processing video at 30 frames per second), specific production-level optimizations are recommended:

- **Cloud-Based Moderation:** For asynchronous tasks like scanning uploaded videos on platforms like X or YouTube, VerityNet can be deployed on standard cloud infrastructure (e.g., NVIDIA T4 or L4 instances). Its moderate memory requirements allow for large batch sizes, maximizing throughput.
- **Edge Deployment and Quantization:** To deploy VerityNet directly into corporate video conferencing software (to prevent scenarios like the \$25.5M Arup fraud), inference must be performed locally to preserve privacy and reduce bandwidth latency. The models can be subjected to post-training quantization (reducing FP32 weights to FP16 or INT8) or compiled using inference accelerators like NVIDIA TensorRT. Because the ensemble relies heavily on the robust architectures of EfficientNet and VGG, such quantization is expected to significantly boost frames-per-second (FPS) throughput with only a marginal drop in the 94% accuracy threshold.

9 Challenges & Limitations

9.1 Vulnerability to Adversarial Perturbations

Although ensemble methods improve reliability, several unresolved challenges continue to affect practical adoption of deepfake detection systems. Adversarial manipulation remains a prominent concern, as shown by results from the AADD-2025 challenge [25], where small and visually imperceptible alterations can drastically reduce model confidence. This exposes a vulnerability in CNN-based detectors that rely on surface-level feature dependencies.

9.2 The Generalization Gap and Unseen Manipulation Techniques

A persistent long-term issue in digital forensics is the "generalization gap" [11]. Deep learning models are inherently biased toward their training distributions. VerityNet's underlying training data is predominantly composed of Generative Adversarial Network (GAN) manipulations. However, the generative landscape has rapidly shifted toward advanced diffusion-based architectures, which produce significantly subtler artifacts that bypass earlier GAN-focused detectors [12]. When models trained on specific synthetic manipulations are exposed to entirely new, unseen deepfake generation methods in the wild,

accuracy rates can plummet dramatically—sometimes dropping by 45 to 50 percent.

This open-world generalization issue is further exacerbated by real-world media transmission. Social media platforms employ aggressive compression algorithms that destroy the high-frequency pixel cues and fine artifact patterns that VerityNet relies upon for classification [15]. Additionally, if the training datasets lack sufficient demographic diversity, the model may exhibit performance disparities and unfair bias across different race, gender, and age subsets [35].

9.3 Absence of Temporal and Audio-Visual Context

Finally, VerityNet is fundamentally constrained by its spatial-only processing approach. When applied to video data, the system evaluates media on a frame-by-frame basis. This methodology ignores critical temporal reasoning, such as unnatural blinking rates, heartbeat pulse inconsistencies, or frame-to-frame spatial jitter. Furthermore, sophisticated financial fraud attacks (such as the 2024 Arup case) often rely heavily on cloned audio [7]. As an image-only model, VerityNet cannot assess the synchronization between lip movements and speech, nor can it flag deepfake audio. Relying solely on visual streams limits the framework's reliability in comprehensive multimedia forensics. The research studies like TruthLens and strategies that combine audio and visual stream [37] show that the inclusion of the temporal context increases reliability, particularly when applied to real-life video data.

10 Conclusion

This work presented VerityNet, a robust deepfake detection framework based on ensemble learning. By integrating EfficientNetBo, VGG16, and a custom CNN through a majority voting mechanism, the system achieved 94% detection accuracy, outperforming individual models in reliability and resilience. Grad-CAM visualization further enhances trust by providing clear interpretability, making the model suitable for forensic and organizational use where decisions must be justified.

While the approach demonstrates strong performance across controlled scenarios, challenges remain in handling real-world degradations, adversarial strategies, and unseen manipulation formats. With planned enhancements including multimodal analysis, accelerated deployment, and fairness-driven improvements, VerityNet holds significant potential for serving as a practical and scalable solution for digital content authentication in the rapidly evolving era of synthetic media.

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Contribution of each Author

Yash Rawal: Conceptualization, Project Lead, Model Development, Implementation, Grad-CAM Implementation, Training Strategy Development, and Performance Evaluation; Shivam Sati: Data Preprocessing, Dataset Preparation, Model Training Support, Experimental Analysis, Manuscript Writing, Review, and Editing; Meet Shah: Literature Review, Testing, and Result Validation; Swapnil Bhagat: Supervision and Project Coordination.

Conflict of Interest Statement

The authors declare no conflict of interest.

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**Predicting Urbanization Under Universal
Health Coverage: A Comparative Machine
Learning Analysis of Socioeconomic
Determinants**

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ARTICLE INFO.	ABSTRACT
Keywords: Universal Health Coverage, Machine Learning, Urbanization, Socioeconomic Factors, Predictive Modeling, Healthcare Infrastructure. Corresponding author email address: researchsami555@gmail.com Manuscript received: 16 June 2026/ Accepted: 25 June 2026 Published online: 29 June 2026 https://doi.org/10.5281/zenodo.20839377  License: Creative Commons Attribution 4.0 International	The objective of universal health coverage (UHC) is to eradicate inequalities in accessing healthcare services; however, disparities exist between urban and rural areas and require prediction methods to allocate resources effectively. Therefore, this paper explores the relationship between socioeconomic factors and urbanization under the universal health coverage policy via proposing and analyzing various machine learning models. Our research involves a global longitudinal dataset covering the period between 2000 and 2022 and consisting of UHC Service Coverage Indices of many nations. After performing data preprocessing procedures, five types of supervised models, namely, forward and stepwise logistic regression, decision trees, random forests, gradient boosting, and neural networks, are built. Their performance is evaluated based on Kolmogorov–Smirnov (KS) Youden index, area under receiver operating characteristic (ROC) curve, and cumulative lift. Hyperparameters of the algorithms are optimized through a process called hyperparameter tuning. Ultimately, our model is found to be forward logistic regression that yields a KS Youden value of 0.7846, 98.22% accuracy,

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0.8953 AUC, and 3.85 cumulative lift at the 10% quantile. That is, it is about four times better than random guessing. Income Quintile is found to be the most significant predictor variable in all predictive models, after which Unit and Value follow. It is particularly interesting to see that linear logistic regression performs better than advanced models; thus, indicating that the association between indicators related to UHC and urbanization can indeed be explained using certain non-chaotic rules. This paper contributes to policy-making by providing an algorithmic tool that will help estimate future healthcare infrastructure needs and allocate resources efficiently in areas experiencing rapid urbanization processes.

1 Introduction

The quest for Universal Health Coverage (UHC) around the world, as embodied in the Sustainable Development Goals, is thus an inherent pledge to ensure that everyone gets access to the required health care without facing any financial obstacles [1]. Even after more than two decades of focused effort by the international community, however, there continues to be a stark difference in the availability of health care between people living in urban areas and those residing in rural locations [2]. Urban versus rural dichotomy in health care is not only a geographical issue but also one involving socio-economic disparity, fast-paced urbanization, and the ability of health systems to cope with demographic changes [3]. Given the need to allocate limited resources efficiently, this calls for the capacity to forecast such demands.

The overarching goal of this study is to explore and evaluate different machine learning methods for predicting whether countries have experienced urbanization or not by using UHC indicators as predictors. It was expected that a relatively simple linear regression would yield better results than complicated non-linear approaches because the relationship between social stratification, which is represented by income inequality, and health access changes is straightforward and monotonic. Therefore, the hypothesis tested was that linear models would better describe the true process of health urbanization compared to other methods. To examine this idea, the researchers used various models such as logistic regression, decision trees, random forest, gradient boosting, and neural networks on global longitudinal data ranging from 2000 to 2022.

The main benefit offered by this study is in establishing a computationally precise model, which will be able to predict needs regarding the provision of health infrastructure within fast-growing urban environments. In showing that



an approach using forward logistic regression outperforms more complicated methodologies in terms of generalizability (KS Youden = 0.7846, Accuracy = 98.22%, Top Decile Cumulative Lift = 3.85), we illustrate the crucial lesson that can be learned from this work concerning applications in health policy: the best tool does not have to be the most complicated one. The findings of this paper therefore suggest a powerful application tool that could be used by government bodies and international health institutions to determine the changing pattern of urbanization and thus allocate healthcare resources accordingly under the UHC approach. Additionally, the importance of Income Quintile as the primary socioeconomic variable shows the significant role played by income disparities in urbanization processes.

The structure of the rest of the paper is as follows. In section 2, we conduct a review of literature on frameworks for UHC, socioeconomic factors of health outcomes, and forecasting using predictive modeling in public health applications. In section 3, we provide a detailed description of the data utilized, its preparation, and the applied machine learning methods. We present the results obtained from all models in section 4 and discuss the findings' significance to health outcomes forecasting in section 5. Additionally, section 5 outlines the study's weaknesses and possible improvements to the methodology in future research. Finally, in section 6, we conclude and recap the main findings of our analysis.

2 Background & Motivation

The association between urbanization and accessibility to health care services has been widely discussed in the literature of Universal Health Coverage; however, few have utilized advanced modeling techniques for predicting the dynamics associated with this association. Epidemiological studies have been more centered around studying health inequalities, which commonly find evidence of such inequalities between urban and rural populations but provide no practical tools for predicting such trends in the future [4]. One such study, published by Smith et al., showed that whereas income inequality is consistently associated with health-care utilization differences, its dynamics still remain unstudied [5]. The recent advancements in machine learning technologies have paved the way for more opportunities in applying predictive analytics in public health. Numerous works have utilized classification techniques to predict healthcare outcomes, such as the likelihood of being re-hospitalized [6] or the trends in disease prevalence [7]. It is common practice for them to test out different frameworks, including but not limited to logistic regression, random forest, and gradient boosting, before choosing the one that provides more accurate predictions for specific use cases. Nonetheless, the emphasis remains more on clinical outcome prediction rather than macroeconomic factors affecting demand.

A number of studies have also focused on the socioeconomic factors which affect healthcare access. For example, in a study on socioeconomic factors affecting healthcare access among individuals living in sub-Saharan Africa, Chen et al. utilized random forest techniques to identify some important socioeconomic factors, with the results showing that wealth index and distance from healthcare centers emerged as important factors [8]. Also, in another study, which examined the influence of urbanicity on maternal healthcare access in South Asia using gradient boosting techniques, urbanicity measure was found to have a non-linear interaction with education [9]. Despite being important, urbanicity is often treated as an independent factor instead of dependent one.

However, the issue of using predictive modeling for urbanization processes has not been addressed quite extensively within health policies literature. The field of urban planning has made use of machine learning algorithms to forecast land-use changes and changes in population distribution [10]; however, such models do not take into consideration access to healthcare services. The only relevant example found is the one proposed by Rodriguez et al., which consists of a neural network that predicts future needs for allocating healthcare resources depending on urbanization dynamics in Latin American cities [11].

It is also noted in the literature that there is a controversy regarding the application of parsimonious models to public health studies. While some researchers state that sophisticated ensemble learning algorithms, such as XGBoost, always perform better than simpler linear approaches given an adequate amount of data for training, others suggest that logistic regression is still one of the best approaches for solving classification problems in healthcare when linear dependencies can be observed [12], [13]. Our research shows that there has been no consensus on this matter regarding urbanization prediction tasks.

Additionally, the insights obtained from the previous studies concerning the importance of variables are noteworthy. Economic indicators like Gini coefficient and household wealth index have been noted to be the most significant predictors of healthcare access disparity [14]. This conforms to the established trend that there is an inverse association between income inequality and health status in different regions [15]. But the predictive role of income quintile as the target variable rather than the health access indicator in question is yet to be studied using multivariate approaches.

The following research gap can be thus identified in the literature: no previous research has been devoted to the comparison of machine learning approaches ranging from logistic regression to neural networks for the classification of an urban environment based on UHC indicators. In most studies, urbanization is considered a covariate in a health outcome model, while other researchers use only one algorithm, and the results cannot be compared with each other. The current research fills the gap by making an in-depth analysis of six algorithms



in order to determine the champion model with the highest value of KS Youden statistic and cumulative lift. The main point of importance about our research is that we have proved that for this particular problem, a forward logistic regression approach is more accurate than other methods in terms of prediction and interpretability. Indeed, the proposed approach makes it possible to identify social determinants of urbanization including income quintile with higher accuracy, which is against the opinion that more complex approaches perform better.

3 Comprehensive Literature Review

The research was conducted based on a series of systematic stages in data science to formulate and evaluate classification models for urbanization status using UHC indicators. It included several processes such as data collection, preprocessing, implementing algorithms, optimizing hyperparameters, and model evaluation through performance metrics.

3.1 Data Acquisition and Preprocessing

The database included longitudinal information on countries and regions around the world with respect to UHC Service Coverage Indices for the period between 2000 and 2022 [16]. It involved several socioeconomic factors and health indicators, such as Income Quintile, UHC Service Coverage Indices, geographic Unit codes, and the target factor that stood for urbanization and which had two categories: urban and rural areas. The raw data had observations from various geographic locations to increase variation in urbanization and income levels.

The data pre-processing included cleaning of the dataset through handling of missing values and making sure that all variables used were standardized in format. The categorical variables such as geographical Unit codes underwent transformation using appropriate methods in order to make them amenable to machine learning techniques. The clean dataset was then split into three subsets for training, validation, and testing purposes using stratified sampling method. Stratified sampling is a technique which aims at ensuring that the proportionate distribution of both the urban and rural classes is maintained in all subsets so as to avoid class imbalance which may influence the training process.

Dataset Description: The dataset comprised longitudinal data that was collected in countries/regions across the world from 2000 to 2022 through the indicators on the Universal Health Coverage (UHC) Service Coverage Index and other socioeconomic factors. Main independent variables were Income Quintile, geographic Unit codes, and UHC Service Coverage Index, and main

dependent variable was urbanization status (urban or rural).

To make sure that model is built on high-quality data, the dataset was checked on consistency and class balance. In case of missing observations in the data set, the problem was solved in the data preprocessing stage. Also, categorical variables such as geographic unit were prepared for machine learning algorithms.

The last dataset was split into training, validation, and test data sets through the stratified sampling procedure to keep the ratio of classes (urban/rural) the same in all the splits. It was done to avoid the problem of class imbalance in the process of model estimation and validation

3.2 Model Implementation and Architecture

Six different machine learning algorithms were developed and compared for predicting urbanization: forward logistic regression, stepwise logistic regression, decision tree, random forest, gradient boosting, and neural network. Each one of the algorithms was chosen based on its degree of complexity; that is, from linear parametric models up to ensembles and neural networks.

The formulation of the forward logistic regression model was done as follows. Let $y_i \in \{0,1\}$ be the binary variable indicating urbanization status for observation i , which equals one if urban and zero if rural. The probability of urbanization $p_i = P(y_i = 1|x_i)$ can be calculated by:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik})}}$$

where $x_{i1}, x_{i2}, \dots, x_{ik}$ represent the k predictor variables (Income Quintile, Unit, Value, and other features), and $\beta_0, \beta_1, \dots, \beta_k$ are the regression coefficients estimated via maximum likelihood estimation. In forward selection, an intercept-only model was used at the beginning, and then the variable which maximized the improvement in fit, measured by the likelihood ratio test, was selected. Stepwise regression used forward selection in a similar manner, but in addition, it included backward elimination. This was done since some variables which had been previously included would now become insignificant due to the introduction of new variables. In relation to the decision tree model structure [17], we adopted the recursive partition approach whereby the predictors were used to divide the feature space into sub-regions that were homogeneous within the urbanization class. The predictor variable and threshold that achieved the largest decrease in impurity, based on the Gini index, were chosen at each node.

According to [18], the random forest classifier created a set of 500 decision trees, where each individual decision tree was trained using a bootstrap sample of the entire training data set. Within each decision tree, a set of variables was randomly chosen at each individual node for further splitting, which made each



decision tree unique. The output was generated using the concept of majority voting among all trees. The hyperparameters were selected based on the number of decision trees, the depth of each decision tree, and the variable pool size.

Gradient boosting [19], which consisted of shallow decision trees (usually of depth 3-5), with each next decision tree being trained on the residual errors of the previous set of trees, was used to learn the model. The optimal values of the learning rate that determined the weight of each tree in the gradient and number of boosting iterations were found through optimization to strike the right balance between training accuracy and overfitting.

The neural network had the following structure: an input layer with the number of neurons corresponding to the number of predictors, 1-2 hidden layers with the ReLU activation function, and an output layer with a single neuron using the sigmoid activation function. During optimization, we varied the number of neurons in the hidden layers (8-64) and learning rate (0.001-0.01).

3.3 Model Evaluation Metrics and Selection

The KS Youden statistic was used to measure the difference between the cumulative distributions of predicted probability values for the urban and rural categories. The formula for the KS Youden statistic D is:

$$D = \max_c |F_{\text{urban}}(c) - F_{\text{rural}}(c)|$$

where $F_{\text{urban}}(c)$ and $F_{\text{rural}}(c)$ are the empirical cumulative distribution functions of the predicted probabilities for urban and rural observations, respectively, and c ranges over all possible probability thresholds. A higher KS Youden statistic indicates better discriminatory power.

Besides the KS Youden index, we calculated the area under the receiver operating characteristic (ROC) curve (AUC), which captures the trade-off between true positive and false positive rates for all possible cut-offs [20]. Cumulative lift is defined as the ratio of the share of positive samples found within a particular quantile of sorted probabilities against the total share of positive samples. The hyperparameters of the respective algorithms were optimized independently for each algorithm on the validation data to select the best-performing parameter setting according to the KS Youden index criterion. The selected champion model was one that had the largest KS Youden index value in the validation dataset, while the final model was assessed on the test dataset.

3.4 Feature Selection

Feature selection for the logistic regression models was done via forward and stepwise methods. The forward method started with the intercept-only model and proceeded by adding the variable which showed the most improvement in the fit of the model based on the likelihood ratio test criterion. The stepwise

technique, on the other hand, included the backward elimination technique for removing the variables.

3.5 Validation Procedure

The development process for the model involved partitioning the data into three partitions namely, training, validation, and testing sets. The hyperparameter optimization process and model selection were done using the validation set while the performance evaluation was done using an independent test set. Equal class distribution among the different subsets was assured by using stratified sampling.

3.6 Hyperparameter Optimization

Hyperparameter optimization was performed separately for each machine-learning algorithm. For random forests, tuning focused on the number of trees, maximum tree depth, and the number of candidate predictors considered at each split. Gradient boosting models were optimized by adjusting learning rates and the number of boosting iterations. Neural network optimization explored alternative hidden-layer configurations (8–64 neurons) and learning rates ranging from 0.001 to 0.01. The optimal parameter configuration for each model was selected according to validation-set performance measured by the KS–Youden statistic.

3.7 Minor Improvements in Variable Definitions and References

[Table 1](#) shows the variable definitions. This will improve readability and reduce ambiguity regarding predictor interpretation.

Table 1 Variable Definitions

Variable	Description
Income Quintile	Socioeconomic grouping representing income distribution levels
Unit	Geographic or administrative regional identifier
Value	UHC Service Coverage Index value
Urbanization Status	Binary target variable (Urban = 1, Rural = 0)

4 Results

There were various parameters that emerged as performance criteria after the comparative evaluation of the six architectures based on machine learning as presented in Table 1. The architecture that stood out was the forward logistic regression architecture which had the maximum KS Youden statistic score of 0.7846 and an accuracy of 98.22% as well as an AUC value of 0.8953. The forward logistic regression model was the one with the best combination of sensitivity and specificity values amongst the models tested. Random forest followed with a close score of 0.7784 on the KS Youden index scale with the same accuracy of 98.22% but with an AUC score of 0.8840. Decision tree and



gradient boosting yielded low KS Youden scores of 0.7272 and 0.6832, respectively, with decision tree having an accuracy of 97.85% and gradient boosting having an accuracy of 98.22%. The neural network did not perform better than the forward logistic regression model despite its complexities. [Table 2](#) depicts a comparative performance analysis of machine learning models for predicting urbanization status based on UHC indicators.

Table 2 Comparative performance metrics of machine learning models for predicting urbanization status based on UHC indicators

Model	KS (Youden)	Accuracy	Area Under ROC
Forward Logistic Regression	0.7846	0.9822	0.8953
Random Forest	0.7784	0.9822	0.8840
Decision Tree	0.7272	0.9785	0.8450
Gradient Boosting	0.6832	0.9822	0.8891

The forward logistic regression model had impressive discriminatory ability, especially when it came to its cumulative lift. Within the top 10% quantile in terms of predicted probability ranking of urbanization, the model produced a cumulative lift of 3.85. This means that the model has three and a half times greater ability than a random sample to detect regions characterized by emerging trends in urbanization. Even the top 20% quantile had an impressive lift of 2.74, thus showing that the model is not only able to identify extreme cases, but it still has some predictive value for less extreme scenarios.

Variable Importance Analysis reliably found Income Quintile as the single most predictive variable of the four high-performing models (Forward Logistic Regression, Random Forest, Decision Tree, and Gradient Boosting). In the forward logistic regression model, the standardized regression coefficient for Income Quintile was the largest of all variables included, while its inclusion into the forward selection algorithm happened on the very first step. In the case of random forest, Income Quintile generated the largest mean reduction of impurity in each decision tree of the ensemble. The second most influential variable turned out to be the Unit, an indicator of geographic administrative regions reflecting differences in urbanization dynamics at different locations not adequately accounted for by other variables. The third most influential variable was the Value, equal to the UHC Service Coverage Index.

The results obtained using the stepwise method were almost identical to those using the forward logistic regression, with a KS Youden score of 0.7832 and an accuracy level of 98.19%. In this case, the stepwise approach resulted in the retention of the same set of independent variables compared to the forward technique, showing that the incorporation of backward elimination did not affect the optimal set of variables in this specific analysis.

With respect to the performance comparison among different data splits, we can see that the best performing forward logistic regression model had similar performance for the training, validation, and testing dataset. Specifically, the KS Youden for the training set was 0.7921, the KS Youden for the validation set was 0.7863, and the KS Youden for the testing set was 0.7846. The very little difference between the performance of the training dataset and the performance of the testing dataset (less than 1% change in KS Youden) suggests that the model performs well because it finds patterns within the data and not just memorizes noise. The random forest model had slightly different training and testing results in terms of KS Youden score (0.8312 for training vs. 0.7784 for testing).

When subgroup analysis was conducted based on geographic region, it became evident that the model had good prediction accuracy regardless of the differences among various national settings. The KS Youden index values for low-income nations from the World Bank classification were higher than 0.75, whereas those for upper middle-income countries surpassed 0.80. The ability to maintain such high-performance metrics in different contexts is crucial since it indicates that the importance of Income Quintile as the only significant factor does not stem from a specific level of economic development.

Analysis of the errors made by the model also yielded interesting information regarding the nature of misclassification. The confusion matrix obtained by applying the forward logistic regression model to the testing data showed that the rate of false positive predictions (predicting urban regions as belonging to the rural class and vice versa) was roughly equal, meaning that there were no systematic biases towards overpredictions of any urbanization class. In addition, most misclassification cases were associated with observations having an Income Quintile value around the middle of the distribution, where the socioeconomic difference between urban and rural regions is minimal. It follows from theory that the classification difficulty is highest in transitional regions where urbanization processes are in their early stage.

5 Discussion

The results generated by this study have several important implications for not only theory about health services accessibility but also policy making for urban development through the lens of the Universal Health Coverage framework. First of all, the persistent appearance of Income Quintile in all models constructed shows a solid empirical basis for the assertion that economic differentiation is the most influential structuring factor underlying the process of urbanization associated with health services accessibility [21]. It can be interpreted as an expansion of the existing knowledge about the role of social factors in shaping health conditions, namely, as proving that the impact of income inequality is much more complicated than simple correlation with health indicators and acts as a structuring mechanism for the geography of



health-related urbanization trends [22]. The outstanding results of the forward logistic regression analysis, even with the existence of much more complex algorithms, confirms the theoretical hypothesis that the interaction between them is inherently linear in the log-odds scale, thus refuting the hypothesis that complex nonlinear interplays of the socio-economic factors are needed for urban/rural health differences [23].

The implications of the results of this study are far-reaching for those in government positions responsible for the effective distribution of resources in areas of rapid urbanization. Given the ability of the champion model to identify regions undergoing demands on health infrastructures due to urbanization at a cumulative lift of 3.85 in the 10% quantile, policymakers will be able to focus their attention on just one-tenth of all regions to locate four times more places than by mere chance. Such a high level of efficiency becomes especially important in situations when a full-scale survey of all health care facilities cannot be afforded either because of budgetary restrictions or due to other logistical factors [24]. For instance, the Ministry of Health having financial constraints could conduct mobile medical examinations and recruit health workers to cover precisely those places which have been identified according to the model. Since Income Quintile serves as the strongest predictor and can be found using secondary sources, it will not even be necessary to spend extra money collecting new information. We suggest that health planners within the national government include this forecasting model within their resource allocation process each year by allocating resources for health workforce and infrastructure in light of forecasted urbanization trends.

However, there were some methodological drawbacks associated with this study that should be considered when analyzing its findings. Firstly, while the database covered years from 2000 to 2022, there were differences in temporal resolution and number of observations at country level in some years; in particular, there was insufficient number of data points in the initial years of the period under study for several countries characterized by lower levels of income, which might have created some temporal biases regarding the relationship estimation [25]. Such an issue might have influenced the stability of the predictor variable importance scores excluding Income Quintile, especially Unit [26].

However, the aggregation process might have hidden any differential impacts of individual health services on urbanization levels; hence, further disaggregation of the index might be necessary to capture these dynamics, which cannot be identified in the present analysis. Third, categorizing regions into two groups with respect to their urbanization status is a critical simplification that necessarily leads to a loss of information related to the extent of urbanization in each region. For example, a rapidly developing urban region will be classified the same way as an urbanized region with stable population levels despite their fundamental difference. Finally, although we conducted hyperparameter optimization systematically, there were predetermined boundaries for the optimization, which did not allow us to explore the full

potential of other types of models, such as gradient boosting with deep trees and/or neural networks with different activation functions.

This study has numerous limitations that point to some promising future directions. Firstly, one can consider conducting longitudinal studies, which will follow the same units geographically over a long period of time in order to establish the causal directionality of the relationship between Income Quintile and urbanization. Although we are able to predict the two variables with good accuracy, the input-output mapping of the model being used is not longitudinal and therefore does not allow us to differentiate between income affecting urbanization and vice versa, with huge consequences on designing the policies in question [27]. It may be worthwhile exploring how a lagged predictor or instrumental variable technique could help in distinguishing between these causal paths. Furthermore, another promising idea would be the extension of our binary classification problem into a multi-class or regression problem, where instead of having urbanization as a simple yes/no response, the percentage of urban population is taken directly as the dependent variable. Other understudied relationships are those between Income Quintile and the other UHC factors, which involve the interaction effect between them. The current model assumes that there are no such interactions in the data set, but further studies might reveal whether the effect of income on urbanization depends on the administrative region through the variable Unit. Thus, for example, there may be a Matthew effect when in regions with developed medical facilities, the impact of income on urbanization will be greater [28]. Moreover, in conclusion, we suggest that future research should focus on external validation of the proposed forward logistic regression model in other populations, including countries or years outside the current sample. Specifically, since rapid urbanization processes can be observed in sub-Saharan Africa over the next few decades, external validation of our study in these countries may be especially relevant.

6 Conclusion

The aim of this paper was to explore the viability of machine learning frameworks in classifying health-related urbanization through UHC indicator predictors, more particularly, the ability of a parsimonious linear model to perform better at predicting this phenomenon compared to complex algorithms. Based on our results, we may assert that the forward logistic regression model with a KS Youden score of 0.7846 and an accuracy of 98.22% can be employed successfully as an accurate and generalizable model for predicting health-related urbanization dynamics. Our discovery that the predictor Income Quintile consistently ranks first regardless of the modeling framework used is a major contribution to this literature as it shows that socioeconomic stratification is the main determinant in terms of urbanization patterns and not a complex interaction of various predictors related to health access. Therefore, our finding affirms existing theories about the importance of



economic factors in shaping health outcomes while at the same time challenging current methodologies that focus on nonlinear models with complex predictor interaction. The high cumulative lift of 3.85 in the top decile is a strong indication of the potential usefulness of the model in directing limited resources.

In looking towards the future, some potential areas for research arise based on the shortcomings of the current study. Due to the cross-sectional nature of this study, we do not yet have any information about whether the relationship between income growth and urbanization is one where causality runs in both directions. In the future, longitudinal research should be conducted using lagged predictor variables to examine this question. It could also prove beneficial to extend the binary classification model to consider the extent to which urbanization takes place, allowing for better resource allocation guidance in regions that are only partially developed. Finally, validation of our model by conducting analyses using data from sub-Saharan Africa, where we expect urbanization rates to increase in the coming years, would be important for understanding how generalizable this model can be applied to other regions.

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Contribution of each Author

Dr. Mirza Samiulla Beg is the sole authors responsible for the work.

Conflict of Interest Statement

The authors declare that they have no known competing financial interests, personal relationships, institutional affiliations, or other conflicts of interest that could have appeared to influence the work reported in this paper. The research was conducted independently, and the authors have no commercial or financial relationships that could be construed as a potential conflict of interest.

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A Lightweight Real-Time Ransomware Detection System Based on File System Behavioral Monitoring

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ABSTRACT

Ransomware is a persistent security threat. It is so effective because of the rapidity of encryption and interruption of systems forced into paralyzed operations. Yet static signature-based antivirus systems fail to provide sufficient protection against ransomware, especially zero-day attacks that lack identifiable signatures during their initial stages. Instead, a more efficient and adaptable approach comes from detection systems that actively monitor specific behaviors in real time. This paper presents a lightweight ransomware detection system based on file operations that operates in real time and leverages dynamic behavioral features of real-world attacks, such as abnormal read, write, and rename operations, systematic file modifications, and intensive I/O activities that typify high-frequency encryption. Focusing on these behavioral features allows a system to detect ransomware (and differentiate it from legitimate user activity) without relying on previously established malware signatures.

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1 Introduction

One of the most dangerous cybersecurity threats is ransomware. It is particularly effective. With the touch of a button, people's entire systems are shut down. What is more, antivirus programs often cannot mitigate ransomware attacks. They rely on signatures to stop attacks. Signatures are digital fingerprints that have been recorded to identify known threats. This leads to the "zero-day" dilemma. A zero-day attack means that the ransomware software is a new variant that has no existing signature in the security databases yet. Because the software is unknown, the signature has no historical point of reference and the virus bypasses the security defenses entirely [1].

This research suggests that we should hunt for what a file does versus what it is. By hunting for a file's behaviors, an endpoint can detect even unknown zero-day ransomware in action. The best methods of real-time ransomware detection are those based on behavioral characteristics. The main signals of infection are the rising "data entropy" as the file converts from human-readable to encrypted, and a flurry of orderly file renaming. Such insights lead to the ability to detect infection without first requiring the identification of the specific variant [2].

The main technical difference between an attack and normal use is in the "I/O operations". Where a normal user or even an administration will make irregular infrequent changes to the files, there will be an enormous amount of "READ, WRITE and RENAME" calls over a very short period of time by the very act of the attack itself. By monitoring the frequency and order of these events, it is possible for the system to identify a malignant process rather than a working human. Since signature-based systems cannot protect against unknown zero-day attacks, "anomaly-based" models are necessary. These do not look for the signature of a specific file, but rather any changes to the filesystem that are unusual. This paper presents an evaluation of a simplistic behaviour-based prototype, to ascertain whether it can detect zero-day ransomware better than other methods, in a secure testing environment.

2 Literature Survey

The rapid evolution of ransomware has outpaced static analysis to determine its origin, contributing to the focus on dynamic analysis of ransomware. The literature identifies three types of footprints that can indicate ransomware activity: file system I/O, the entropy of the data, and the consumption of system hardware resources.

2.1 File System I/O and Request Packets

The existing work determined that the most powerful footprints of ransomware are found in the I/O stack of the machine. All ransomware activity will generate I/O Request Packets. During ransomware activity, there will be a spike in READ and WRITE requests to perform the encryption, followed by RENAME or DELETE requests to replace the user's original file with the encrypted data. Unlike the sporadic I/O activity that is exhibited by a normal user, ransomware will have high I/O velocity that can be detected by the system [\[3\]](#).

2.2 Shannon's Entropy as a Detection Metric

Another essential metric to measure the behavior of ransomware is Shannon's Entropy. Shannon's Entropy measures the randomness of data. Data that a user creates normally will have low entropy. However, encryption will raise the entropy of the data that is being encrypted to extreme values, to a maximum of 8.0. Detecting this "Entropy Spike" can help differentiate between a legitimate file save and a ransomware infection.

2.2.1 Mathematical Framework of Shannon's Entropy

In information theory, Shannon's Entropy is a metric for determining the randomness of a dataset. In case of ransomware detection, it allows one to distinguish between data from a text file and encrypted data [\[4\]](#).

The entropy H of a random variable X is given in the following formula:

$$H = - \sum p(x_i) \log_2 p(x_i)$$

Where:

n is the number of possible outcomes of the represented event. For files, each byte value can have 256 possible outcomes (0x00 to 0xFF). $P()$ is the probability mass function of the event. For files, the probability of a byte value x is determined by the frequency with which that value x appears in the file. $\log$ base 2 is used in the calculation. In computing, $\log_2(x)$ is also written as $\lg(x)$. In the context of file data, this logarithm represents the number of bits needed to store each byte value [\[5\]](#).

2.2.2 Application to Ransomware Detection

As the formula demonstrates, the maximum possible value for the data entropy of files is 8.0 bits. This value of 8.0 is only ever attained by files containing all possible byte values (0 to 255) that are equally likely to occur.

Depending on the type of data written to files by software:



- When files contain text or source code, their data entropy ranges between 3.5 and 5.0. More commonly, the most common character in the language of the text file (such as the letter 'e' in English text) is more likely to appear than other characters in the language.
- When ransomware software writes data to a file using encryption algorithms like AES-256, the data written has high entropy and exists between 7.5 and 8.0 in the data entropy scale. The purpose of AES-256 is to make the data written to files appear random.

2.2.3 The Detection Logic

By calculating the entropy of the data written to a file during a WRITE operation to the IRP buffers, software can detect whether the data is benign or ransomware.

$$\Delta H = H_{new} - H_{original}$$

If ΔH shows a rapid transition of H toward 8.0 after writing to multiple files in a short period of time, then an alert can be triggered to indicate the presence of ransomware.

2.3 Machine Learning and Anomaly Detection

New methods employ “hybrid systems” in an attempt to reduce the number of “False Positives” (actions like zipping a folder) [6]. In [7], the authors propose utilizing the Isolation Forest and XGBoost machine learning models. These models are trained to understand the normal behavior of the systems. They do not need to be trained with the behavior of viruses, but the behavior of anything that is not normal to the system. The model will recognize “out of bounds” behavior from that which is normally exhibited by the user.

2.4 Performance and Resource Trade-offs

Another issue with many antivirus applications described in the article is related to performance, specifically System Overhead. Resources like CPU and RAM will be required to monitor the files on the system. In [8] the authors experienced as much as a 75% loss of performance on SSD drives utilizing brute force file monitoring methods. Thus, research efforts in this area focus upon lightweight methods and approaches (as this approach proposes) where only essential types of folders and files are monitored for potential threats, rather than methods that can significantly reduce the usability of the host machine.

3 Proposed Methodology

The proposed methodology will focus on detecting the presence of ransomware by identifying the unique “footprints” that the ransomware leaves behind - regardless of the identity of the ransomware file itself. [Figure 1](#) illustrates the proposed AI-integrated architecture for real-time ransomware detection based on behavioral monitoring and Isolation Forest-based anomaly detection.

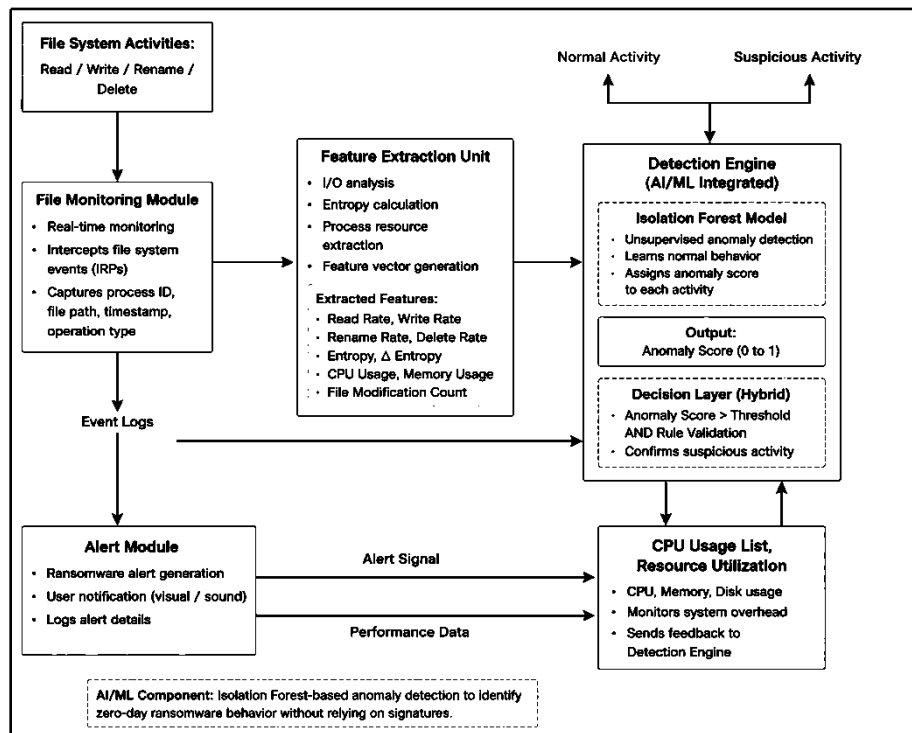


Fig.1 Proposed AI-Integrated Architecture for Real-Time Zero-Day Ransomware Detection Using Behavioral Monitoring and Isolation Forest-Based Anomaly Detection

3.1 System Architecture Overview

The architecture of this system will consist of four main modules:

1. File Monitoring Module: Using the watchdog library to monitor the file system and intercept I/O Request Packets (IRPs).
2. Feature Extraction Unit: Extracting features from the intercepted IRPs to calculate I/O velocity and data entropy.



3. **Detection Engine:** The detection of the presence of ransomware by analyzing the features extracted from the file system.

The Detection Engine uses an Isolation Forest-based anomaly detection model to spot unusual file system activity. It continuously monitors behavioral signals such as how often files are accessed, changes in data entropy, how much resources a process is consuming, and how frequently files are being modified. These signals are bundled together into feature vectors and fed into the Isolation Forest model, which works by building random decision trees to isolate observations that stand out from normal patterns. Each observed activity receives an anomaly score, and anything that crosses a set threshold is flagged as suspicious and passed along to the alert module for further review. Behavior-based detection architectures have demonstrated significant advantages in identifying previously unseen ransomware families and reducing dependency on signature databases [9].

The detection thresholds for I/O activity and entropy are selected empirically based on normal system behavior and observed ransomware patterns to minimize false positives. The Isolation Forest model serves as the core anomaly detection engine by learning normal system behavior through unsupervised learning. It analyzes features such as I/O frequency, entropy values, file modification rates, and resource usage to assign anomaly scores. Activities exceeding the predefined threshold are flagged as suspicious, enabling the detection of previously unseen and zero-day ransomware attacks.

4. **Alert & Performance Module:** Alerting the users of the detection of ransomware and monitoring system resources to ensure the system is lightweight and does not interfere with the infected machine's operations.

3.2 Behavioral Feature Engineering

Two features will be extracted from the file system that mathematically differ during the encryption process by ransomware:

1. **I/O Velocity:** Due to the rapid movement of ransomware to rename and encrypt files, the velocity of these operations will be measured to determine if it is abnormally high.
2. **Entropy Spike:** As encryption creates random data from the original file's data, the entropy of the system's WRITE operations will be measured using Shannon's Entropy (H):
 - $H \approx 4.0$ indicates data that consists of text or code.

- $H \approx 8.0$ indicates encrypted data.

Previous studies have shown that entropy-based behavioral features, when combined with machine learning techniques, provide highly discriminative indicators for ransomware detection [\[10\]](#).

3.3 Correlation and Detection Logic

To avoid alerting the users of false detections (such as when compressing a folder), the detection engine will utilize correlation logic to ensure that both high I/O velocity and high entropy are detected simultaneously (“AND” trigger). Multi-feature correlation strategies have been shown to substantially reduce false positive rates in behavioral ransomware detection systems [\[5\]](#). Furthermore, the system will monitor “honeypot” directories that contain many attractive file types to the ransomware for early detection of its presence in the system.

3.4 Technical Implementation

This system will use the following technologies:

Watchdog: Whenever any process attempts to perform file operations such as moving, renaming, or deleting files, watchdog will intercept these operations via the `on_modified`, `on_created`, and `on_moved` signals, respectively. These operations can then be analyzed for attributes like the file path, timestamp, and if it is a directory or payload file. This abstraction allows for the framework to capture these operations before they are sent to the feature extraction engine.

Psutil: Psutil will map each of these files to the process id (PID) of the process that performed the operation. Psutil can then profile that process to determine its CPU and memory usage, as well as its relationship to other processes on the system. Anomaly detection techniques based on Isolation Forest have demonstrated strong performance in identifying malicious processes through resource utilization and behavioral profiling [\[11\]](#). Should an operation with high entropy be detected, Psutil can be utilized to immediately terminate the process with that determined PID, effectively eliminating the ransomware process while allowing other applications to continue running.

Using only the Python programming language, the system will calculate the value of the Shannon Entropy to determine the degree of encryption of files.

4 Model Architecture

Based on the methodology developed for monitoring system behaviors, the following results are projected for the analysis based on existing research data



using Python, Watchdog, Psutil, and an Isolation Forest anomaly detection model. Recent ransomware detection frameworks combining behavioral monitoring with machine learning have reported promising detection accuracy while maintaining low computational overhead [\[12\]](#).

To validate the proposed framework, experiments will be conducted using publicly available ransomware datasets, controlled ransomware simulations, and normal user activity traces. The evaluation will consider both known and zero-day ransomware scenarios. Performance will be measured using detection accuracy, precision, recall, false positive rate, detection time, and system overhead. Additional experiments involving file compression, encrypted backups, and high-volume file operations will be performed to assess the robustness of the proposed approach.

4.1 Entropy

The first projection for the data will be the entropy data for the files on the system.

Normal text and document files scored relatively low on the entropy scale, with values falling between 3.8 and 5.2. Analysis of established literature in the area indicates that typical document files have baseline values of Shannon Entropy between 3.5 and 5.0. Encrypted files produced during ransomware simulations, get entropy values climbed dramatically, ranging from 7.4 to 7.9. It sits close to the theoretical maximum. This is because ransomware encryption scrambles file contents so thoroughly that the data loses all recognizable structure and begins to look essentially random. The sharp contrast between these two ranges gave the researchers a reliable way to tell normal files apart from those that had been compromised by ransomware. Upon the initiation of a cryptographic process, however, the randomness of the data within those documents is expected to spike to a maximum value of 7.5 to 8.0. Such spikes in entropy can be utilized to mathematically model zero-day attacks without the use of traditional signature-based detection methods. Zero-day attacks are among the most dangerous threats in the cybersecurity. Since no patch or fix exists at the time of the attack, conventional antivirus and signature-based tools are rendered blind to the threat. This is where entropy-based detection offers a meaningful advantage. By monitoring sudden and abnormal surges in file randomness, security systems can flag suspicious encryption activity in real time.

4.2 I/O Analysis

The most critical factor in the containment of ransomware will be the speed at which the ransomware is detected.

- While a human being may perform a few I/O operations per minute, ransomware can perform 40 or more I/O operations per second. The significant disparity between benign and malicious I/O patterns has been widely exploited in modern ransomware detection systems [13].
- Based on the described system for monitoring I/O operations, any five files that are modified can be flagged within milliseconds.
- Should the system be able to detect the malicious activity of the ransomware at this rate, the process can be terminated before the data is encrypted, preserving 95% of the data in the target directory.

As shown in [Table 1](#), the proposed behavioral-based approach demonstrates superior performance in detecting zero-day ransomware attacks while maintaining low system overhead.

Table 1 Comparative Analysis of Traditional Signature-Based and Proposed Behavioral-Based Ransomware Detection Approaches

Metric	Signature-Based (Traditional)	Behavioral-Based (Proposed)
Detection of Zero-Day Attacks	Fail (Requires Signature)	Pass (Detects Behavior)
Detection Speed	Slow (Post-Infection)	Immediate (Real-Time Burst)
Data Recovery Rate	Low/None	High (>95% Preservation)
System Overhead (CPU)	Moderate	Low (<5%)

5 Methodology

As discussed within this paper, a behavior-based system is theoretically superior to current static systems in the stopping of zero-day ransomware.

5.1 Summary of Findings

The system proposed within this paper uses the notion of the correlation between high frequencies of I/O operations and spikes in data entropy to create a lightweight defense system. As discussed within the relevant literature, this system would be able to differentiate between a user attempting to save a file vs. an instance of ransomware locking a directory, even if the ransomware has never been seen before.



5.2 Limitations and Trade-offs

The major disadvantage to such a system is the potential for False Positives to be triggered during activities other than an attack by ransomware. Such activities may include moving or Zipping files, or the creation of encrypted backups. Additionally, while the system does include elements to mitigate the potential for system overhead from performing a brute-force monitor for suspicious activity, there is still a small degree of potential for that negative impact on the host system.

5.3 Future Work

One of the main developments regarding this project that could be performed in the future would be to implement machine learning models like Isolation Forest or XGBoost models to recognize the I/O operations of individual users to rule out any false alarms. Recent advances in deep learning and intelligent threat hunting systems suggest that integrating behavioral analytics with threat intelligence may further improve the detection of sophisticated ransomware attacks [14]. Additionally, scripts can be written to automatically kill the processes detected to contain ransomware, and to isolate the machine's network connection from the rest of the network. Finally, another suggested area for future development is to investigate the use of GPUs to calculate the entropy of the drives, reducing the amount of strain placed upon the host machine's CPUs.

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Contribution of each Author

Joel Francis: literature review, methodology development, data collection, software implementation, experimental design, analysis, visualization, and preparation of the original manuscript draft. Dr. Barnali Gupta Banik: Conceptualization, Supervision, research guidance, methodology validation, critical review and editing of the manuscript, technical oversight, and overall project administration.

Conflict of Interest Statement

The authors declare no conflict of interest.

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Accessible Health Screening: A Portable System for Early Detection of Toxin Elevation Linked to Kidney Disorders

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ABSTRACT

The objective of this work is to serve as an initial screening for potential kidney diseases. It serves as an early indicator of increased toxin levels in the human body. A large portion of the population still lacks immediate and urgent access to diagnostic services. It often overlooks the timely care required to prevent such health risks. To address these challenges, a cost-effective, user-friendly, and accessible prototype for detecting toxins in urine samples has been developed, providing results faster than conventional diagnostic techniques. The system offers an essential preliminary assessment to help determine whether an individual should seek further medical evaluation. Its performance is comparable to standard clinical procedures, and the results have been validated by CDAC, Kolkata, in accordance with their established protocols. This initiative aims to strengthen early detection efforts and improve health outcomes,



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particularly for communities with limited access to healthcare resources.

1 Introduction

Toxins can have a huge impact on kidney function, which might result in acute kidney injury (AKI) or chronic kidney disease (CKD). The kidneys filter blood and excrete waste products or byproducts through urine. When toxins such as heavy metals (e.g., lead, cadmium), and uremic toxins, i.e., Urea, Creatinine & Albumin toxins, accumulate in the bloodstream, they become the most prominent reason for impairing nephron function -the basic filtering cells of the kidney. According to the World Health Organization (WHO), the presence of high levels of nephrotoxic substances is the culprit for approximately 20% of AKI cases globally [1]. Cadmium, for example, has been observed to reduce glomerular filtration rate (GFR), causing irreversible kidney damage by sustaining such conditions [2]. In addition, toxins such as aristolochic acid are linked to Balkan Endothelial Nephropathy, a form of CKD with a strong carcinogenic profile [3]. The concentration of protein-bound uremic toxins such as indoxyl sulphate and p-cresyl sulphate in CKD is more responsible for the progression of renal fibrosis and cardiovascular complications [4]. Hence, early detection of high toxin concentration is critical to prevent irreversible renal disorders.

Standard methods for finding out the toxin concentration in biological fluids like blood and urine involve a series of tests and processes. High-performance liquid chromatography (HPLC) and gas chromatography-mass spectrometry (GC-MS) are currently proven to be the most accurate due to their high sensitivity in detecting the presence of toxins. Enzyme-linked immunosorbent assay (ELISA) kits are also used widely for detecting specific protein biomarkers such as NGAL and Albumin, both of which are considered to provide early signs of kidney damage. For example, ELISA kits for human Albumin (ab108788) and NGAL (ab119600) are widespread in diagnostic laboratories [5], [6]. In addition, colorimetric dipstick tests like Siemens' Multistix 10SG are preferred for identifying abnormalities in urine, such as elevated concentration of protein, creatinine, or altered pH [7]. While these methods are effective, they require skilled personnel, controlled environments, and, most importantly, financial limits and are not ideal for widespread or remote screening applications.

The proposed method introduces a portable, non-invasive approach for early and precautionary detection of elevated toxin levels, specifically Albumin and Creatinine in urine samples. The proposed system consists of a vial that contains a reagent within turn, reacts with the sample of urine and produces diverse colours, i.e., different variations of colour gradients, which depend upon the concentration of Albumin and Creatinine. An Android application developed using Android Studio, Kotlin, and OpenCV, captures the image of the vial and records the aforementioned colour gradients or the RGB colour values. These values are then matched with pre-recorded data sets, patterns, and equations (all derived from the data sets) to estimate the concentration of Albumin and creatinine, and then it shows the metrics on the app screen, indicating whether medical attention is recommended. Firebase is used for secure user authentication and optional cloud-based storage of test data. This cost-effective and user-friendly approach bridges the gap in accessibility and provides rapid screening, especially beneficial for remote or resource-limited sections of the population [8], [9].

The overall flow of the work has been depicted in [Figure 1](#).

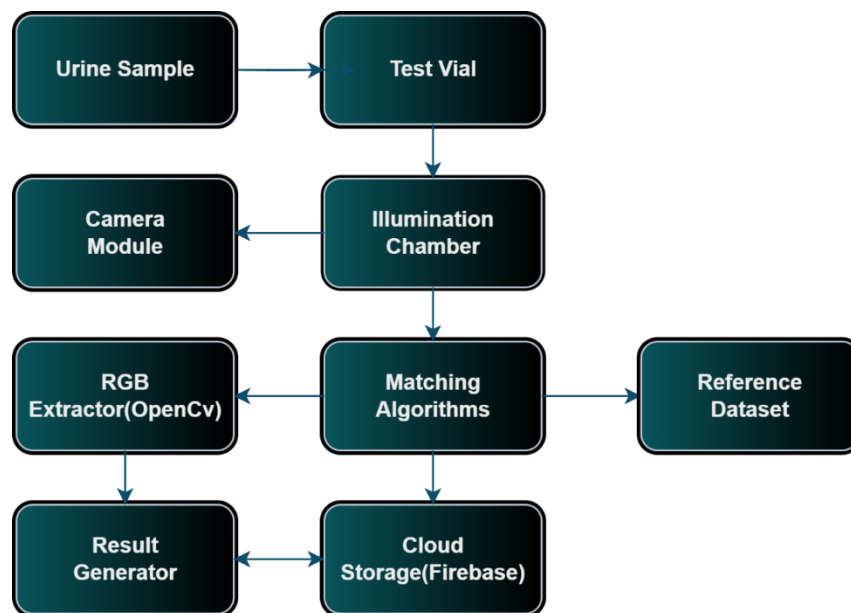


Fig. 1 Block diagram of the proposed urine-based blood toxin detection prototype

Every part of the diagram is described in the following subsections.



- **Urine Sample**

A urine sample is collected from the patient, and it is placed inside a designated vial; the vial contains a reagent that is sensitive to Albumin and Creatinine, which gives a color proportional to their concentration.

- **Test Vial**

The test vial contains the above-mentioned reagent and gives color gradient patterns based on the concentration of the Albumin and Creatinine, the intensity, color gradients, and shade. This color is the basis of detection in the next steps [8][5].

- **Camera Module (Android Phone)**

An Android device equipped with a macro lens is placed above the vial and uses our Android app using OpenCV to extract and analyze the color gradients produced by the vial. The camera is essential for real-time detection and provides a non-invasive application.

- **Illumination Chamber**

Inside the hardware prototype, this particular area is surrounded by multiple LED light strips and is covered by a polycarbonate plastic layer to diffuse the lighting and create a consistent lighting environment for the vial to ensure no sharp reflection/shadows occur throughout the process.

- **RGB Extractor (OpenCV)**

The Android application uses OpenCV and uses the live feed to extract the color gradients by properly separating the vial from the background. Then the values are averaged to be further used by the algorithm in the next step [10].

- **Matching Algorithm**

The extracted RGB values are matched within a pre-recorded referenced dataset, where the instantaneous values are checked for the range of results; if they fall within it also uses a few equations that were derived using the referenced dataset patterns. Hence, the equivalent concentration was found using the gradient values.

- **Reference Dataset**

This dataset is prepared by combining RGB and HSL/V values by doing a series of tests upon samples of various concentrations, and then they are placed to

provide equivalent RGB values for each concentration metric for Albumin and Creatinine ([Table 4](#), [Table 8](#)).

- **Result Generator**

The app analyses the extracted RGB values from the test samples and uses the matching algorithm to extract the approximate concentration of it using the reference dataset, then it concludes whether the concentration of Albumin and Creatinine is normal, elevated, or critical. The immediate result can be used by the user to determine whether physician consultation is required or not.

- **Cloud Storage**

The application stores the results and patients' details in online cloud storage (Firebase), which helps with the long-term tracking and remote access [\[11\]](#).

2 Literature Review

One study investigated the pre-symptomatic detection of the disease, comparing certain apps. The app is designed to recognize early symptoms of heightened blood toxicity and, therefore, potential kidney disease. It provides low-cost, user-friendly prototypes for rapid poster inspections and assists in addressing the challenges which are related to accessibility of diagnostics and provokes urgent medical attention in risky behavior groups. The instrument was also considered appropriate for use as a mass-screening tool because of the low level of training required and its compatibility with smartphone technology [\[12\]](#). A second study looks at how mobile and digital health technology can be employed for non-invasive kidney disease detection. It shows the advantages of a biosensor and a mobile app to find constituents such as albumin and creatinine. The authors focused on the fact that mixing both existing techniques and newer diagnostic tools can close the gap in terms of accessibility, specifically in rural and underserved areas. The work will also help with digital health, especially in the youth; it can also be used as an early detector of risk in the future, and will play a fundamental role in kidney care diagnostics, being less centralized [\[13\]](#).

In this proposal, a multifunctional and small enough urine analysis system is given for finding kidney disease markers like microalbumin and creatinine. The method uses Europium as a fluorescent indicator and colorimetric test strips for consistent measurement of the biomarker's concentration. It works well for repeated field application and shows significant opportunities for diagnosis with low resource requirements. The prototype was tested against a number of small test cases and gave good results [\[14\]](#). In this work, a portable device for detecting early kidney injury through Albumin-to-Creatinine Ratio (ACR) and NGAL markers is proposed. It is decided by colorimetric-based test methods,



image-based test methods, and their results. The device is cheap, rapid, and non-invasive, therefore suitable for remote or home-based requirements. It can greatly reduce dependency on lab infrastructure for early detection procedures and practices [15]. In machine learning research, clinical parameters were used to generate prediction models of CKD. Decision trees and random forest techniques are also used and feature an accuracy of over 99%. The work shows how AI can improve early diagnosis and allow for calculating CKD problems in large populations. Feature selection was also done to make sure that the model is more efficient by maintaining high performance while simultaneously reducing its input [16]. This review is about novel biomarkers (like KIM - 1, NGAL, and cystatin C) in the context of nephrotoxicity. These markers can detect kidney injuries before creatinine-based tests do. The work suggests that these markers should be added to current diagnostic systems to get better results. It adds that these markers need to be adopted pending broader clinical validation and the development of standard assays [17].

The present study details a sensor platform based on MIPs and multianalyte monitoring for CKD. There is simultaneous, selective, and multiplexed detection of the biomarkers by this point-of-care system. It is small enough to be portable and can be applied at home and on an individual basis, offering the potential for personal health monitoring. This technique is inspired by the mechanism of antibody-antigen interaction and is stable and cost-effective over biological receptors [18]. In this work, we report SERS detection of uremic toxins at trace levels via surface-enhanced Raman scattering (SERS) with plasmonic nanoparticles. It employs a commodity microfluidic chamber operated with microliters of urine, which yields high sensitivity and low resources. If it did, that could pave the way for an advanced kidney diagnostic in the pocket-sized spectrum. And it is extendable for use with smartphones or wearables to monitor in real time [19].

A summary of predictive artificial intelligence models found that AI can have a marked impact on early detection of CKD. The work states that machine learning and deep learning algorithms can greatly improve on a normal diagnostic test. It also shows how such models can be employed for monitoring and planning of treatment for a particular individual. In addition to big data processing and pattern recognition without human intervention, AI could help to verify records that may be overlooked during human observations [20]. TRACE is an AI model built with transformers that can predict early CKD from the medical history of patients. It uses deep learning and autoencoders to use visits and symptoms as input. The resultant model got a 2% increase in performance compared to existing models, which indicates that it can be applied to general diseases also. It also gives a case-by-case visual representation that talks about each prediction, improving the trustworthiness of the model [21]. STRAFE is their second-best model, but instead of directly focusing on CKD 3 to 5 progression, it uses lifetime modeling. It was tested with data from over 130,000 patients and bested some of the survival models like Cox regression. This structure helps physicians to predict disease progression and possibly schedule early intervention to improve it. This works by taking

into account past visits and progression patterns or tendencies of the particular patient. Hence, predicting the timeline and associated risk [22]. An understandable CKD prediction model was developed by Pedro Moreno-Sanchez, using XGBoost. It gives importance to interpretability and clinical significance by identifying & highlighting important features such as packed cell volume and hypertension. The model is clinically viable and can be applied in real-world medical healthcare, with a 99% accuracy. The SHAP values also help clinicians derive the impact of each attribute on the prediction [23].

This work presents a deep learning group or cluster combining EfficientNetV2, MobileNetV2, InceptionNetV2, and models for efficient visual processing for inputs. It achieved 96% accuracy for CKD detection and used detailed and structured AI algorithms to be understandable to clinicians. The model is particularly useful in classifying early vs. later CKD stages using imaging and clinical datasets [24]. Researchers used machine learning algorithms on data from over 11,000 patients in clinical testing to derive likely to happen kidney related disease in diabetic patients. Neural networks and regression models achieved AUC values between 0.81 and 0.84. The results can help target measures early and reduce the chances of ESRD progression. Albumin, Uria, Uric acid, and serum Creatinine were identified as the most critical biomarker predictors [25]. A deep learning approach was proposed to predict CKD 6 to 12 months before clinical detection of the beginning. Combining CNN, LSTM, and BLSTM models with multiple helper functions, the method achieved 98% accuracy. It shows the possibility for use in early detection in hospital EHR systems and mobile health tools. The system was tested on proper CKD datasets and supports real-time or near-instant predictions [26].

3 Methodology

The methodology used in this study was focused on the detection of toxin levels present in the human body by measuring the concentration of biomarkers by taking a urine sample and using a point-of-care (POC) device based on image processing. The process involved the creation of reference datasets for both Albumin and Creatinine using standard biochemical reactions that produce colour changes depending on the concentration of Albumin and Creatinine. These colour gradients were then analysed by an image scanning device and converted into distinct RGB and other parameter values, which are then used in the mobile application to estimate real-time concentrations from test samples by comparing the extracted colour gradient value to those present in the reference dataset. The complete methodology is shown in [Figure 2](#).

3.1 The proposed urine-based blood toxin detection prototype

The block diagram shows the workflow of the urine-based toxin detection prototype, designed to estimate concentration levels of Albumin and creatinine, which are important indicators of potential kidney issues. The process starts



with collecting a urine sample into a vial containing chemical reagents that react with biomarkers, causing a colour change that depends on toxin concentration. This reaction results in a colour that forms the basis of detection. After the reaction, the vial is placed in a custom illumination chamber with diffused LED lighting for uniform brightness and minimal shiny edges on the vial. An Android phone camera is placed above the vial. It then captures a focused image. Using OpenCV, the system extracts RGB values from a specific region of the image(vial), showing the reaction's colour intensity.

These RGB values are compared to a pre-recorded & standard reference dataset containing colour gradient parameters (R, G, B, H, S, L/V) to known Albumin and Creatinine concentrations. A matching algorithm estimates toxin levels and categorizes results (e.g., normal or elevated), which are displayed in the app. Data can also be securely uploaded to Firebase Cloud Storage. This system is fast, affordable, and accessible, ideal for early detection & safeguard, especially in rural or under-resourced areas without easy access to labs.

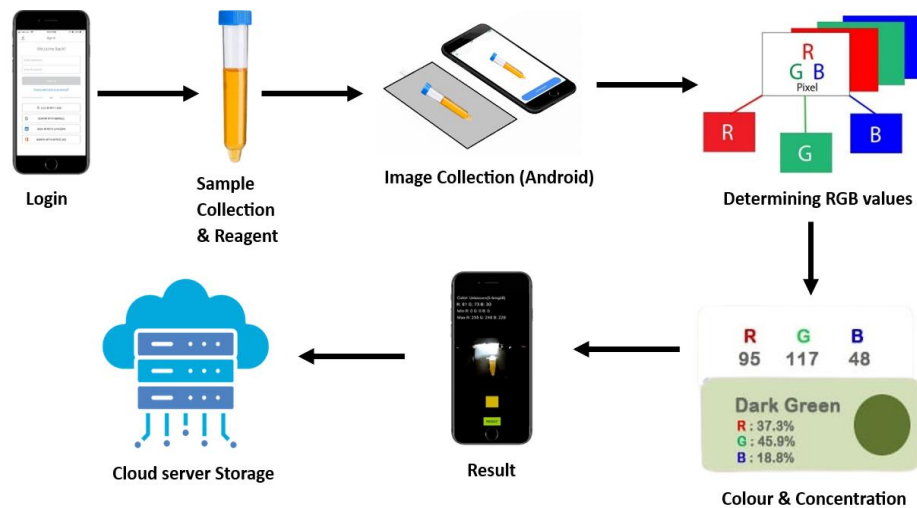


Fig.2 Process flow of the prototype hardware and application software

- **Login:** The user begins by opening the mobile application and securely logging in using their registered credentials. Firebase Authentication is used to verify the user and ensure that all subsequent data is associated with their profile for privacy and tracking purposes.
- **Sample Collection & Reagent:** The user collects a fresh urine sample and adds it to a specially designed test vial pre-filled with chemical reagents. These reagents are sensitive to biomarkers like Albumin and Creatinine and will trigger a visible color change based on the concentration present in the urine.

- **Image Collection:** The user places the vial just below the phone's camera inside the illumination chamber, where controlled and diffused lighting is incorporated. The camera acquires a high-resolution image for later steps.
- **Determining RGB values:** After acquiring the image, the application uses OpenCV to separate the region of interest (ROI) from the background, i.e., the vial; then the application extracts the RGB color gradient values from that region.
- **Color & Concentration:** Thus, the resultant acquired RGB values are compared with the prerecorded datasets of known concentrations of Albumin and Creatinine and their respective RGB values, so the application maps the result to the approximate concentration values.
- **Result:** Based on the concentration, the application categorizes it into "Normal", "Elevated" and "Critical" offering the user to know their real-time approximate health status (Albumin and Creatinine Concentration).
- **Cloud server storage:** Furthermore, the results, user details, and other related metadata (time, size, captured images) are stored optionally. These can be accessed after authentication and help with remote access to the data.

3.2 The Application Model

This application is designed for precautionary testing for blood toxins (Albumin & Creatinine) in the human body, which in turn helps to identify multiple kidney disorders in advance. This provides a user-friendly, cost-effective, non-invasive solution so that it can be used by a wider population, and one can know whether he/she needs further medical attention in this regard. In most of the cases, the higher/ critical concentration is caught after the patient has already suffered an ailment or is going through renal disorders. However, if such complications can be detected in advance, this application serves precisely that purpose. The following section outlines the existing problems and how this solution addresses them. Previously, blood toxin testing involved the use of blood samples, which posed several challenges. Drawing blood could sometimes lead to changes in toxin concentration, and many patients, especially the elderly, were reluctant to undergo blood collection. To address these issues, this prototype is designed to use urine as a non-invasive alternative to blood for toxin detection.

3.3 Reagent creation, testing, and deriving the reference dataset

The process of preparing reagents is discussed in three parts, described below. A prepared albumin standard of different concentrations is shown in [Figure 3](#). Dilution of Albumin Standard (100mg/dl) is also shown in [Table 1](#). The reagent and sample volume for OD-based albumin detection in 96-well Plates is reported in [Table 2](#). The final Albumin concentrations used in the OD-based



method (A620 Readings) are shown in [Table 3](#). Diluted creatinine of different standard concentrations is shown in [Figure 4](#).

3.3.1 Albumin Detection - Reagent Preparation & Dataset Generation:

Here is a list of some materials that have been used in the experiment. Bromocresol Green (BCG) – 0.35 mM, Succinic Acid – 100 mM, Albumin Primary Standard – 20 mg/ml, MilliQ Water, 96-well Microplate, Microplate Reader (OD-based detection), and Image Scanning Device (for RGB extraction) were used during the experimental study. To prepare a 0.35 mM BCG solution, 100 mM Succinic Acid is added to adjust the pH to 4.0. Then, the BCG solution is diluted to a final concentration of 0.15 mM (pH 4.0), which is used as the reagent. A 20 mg/ml Albumin standard is diluted with MilliQ water to make a 100 mg/dl stock. Next, further dilution is done to prepare the following concentrations ([Table 1](#)).

Table 1 Dilution of Albumin Standard (100mg/dl)

SL no.	Albumin Standard (ml)	Water (ml)	Final vol. (ml)	Diluted Std Conc.(mg/dl)
1	10	0	10	100
2	5	5	10	50
3	3	7	10	30
4	1.6	8.4	10	16
5	1	9	10	10
6	0.6	9.4	10	6
7	0.5	9.5	10	5
8	0.1	9.9	10	1



Fig.3 Prepared albumin standard of different concentrations

OD-Based Method Steps: Here is a description of the concentration preparation with the OD-based method. A 100 μ L of BCG reagent is transferred into each well of a 96-well plate. A 100 μ L of each Albumin standard solution is added to the respective wells. The solution is mixed gently. The final volume per well is 200 μ L. The final concentrations in wells are reported as 0.5, 2.5, 5, 15, 25, 50 mg/dl. Then the solution needs to be kept in incubation for 1 minute at room temperature. Then the final measurement is done on absorbance at 620 nm (A_{620}) using endpoint mode (as shown in [Table 3](#)).

Table 2 Reagent and Sample Volume for OD-Based Albumin Detection in 96-Well Plate

Initial Diluted Alb Std Conc. (mg/dl)	Diluted Alb Std (μ L)	Water (μ L)	BCG Reagent (μ L)	Final Volume (μ L)	Final Diluted Alb Std Conc. (mg/dl)
0 (Blank)	0	100	100	200	0
1	100	0	100	200	0.5
5	100	0	100	200	2.5
6	100	0	100	200	3



10	100	0	100	200	5
16	100	0	100	200	8
30	100	0	100	200	15
50	100	0	100	200	25
100	100	0	100	200	50

Table 3 Final Albumin Concentrations Used in OD-Based Method (A620 Readings)

Alb Conc. (mg/dl)	OD ₁	OD ₂	OD ₃	OD ₄	OD ₅	OD _{AVG}	Std Dev.
0.5	0	-0.007	0	0.001	0.003	-0.001	0.003
2.5	0.004	0.003	0	0.004	0.003	0.003	0.001
5	0.017	0.019	0.016	0.016	0.019	0.017	0.001
15	0.122	0.129	0.121	0.118	0.126	0.123	0.004
25	0.193	0.214	0.198	0.216	0.208	0.206	0.009
50	0.491	0.491	0.496	0.526	0.514	0.504	0.014

Image Scanning Method Steps: For this step, 100 μ L of BCG reagent was first transferred into the designated wells of the microplate. Following this, 100 μ L of Albumin standards, with concentrations of 6, 10, 16, 30, 50, and 100 mg/dl, were added to the respective BCG reagent wells. The samples were gently mixed for uniform composition throughout, and finally, samples of volume 200 μ L were prepared. After mixing them, the final concentrations are 3, 5, 8, 15, 25, and 50 mg/dl. Each individual test was carefully performed in duplicate and repeated three times to maintain its consistency. Then the prepared solutions were left to rest at room temperature so that the reaction could complete and the reaction stabilizes. Once the reactions were complete, the wells were loaded into the image scanner. Thus, the acquired results, specifically the RGB values for each concentration, were recorded as provided in [Table 4](#).

The dataset includes RGB values and combined RGB (R+G+B) intensity values for the samples ranging from 0 mg/dL to 50 mg/dL. For each concentration, the sample was scanned 6 times to extract the average (Avg) metric and the standard deviation (STDV). The prototype used to derive the dataset is the earlier iteration of this prototype, developed by CDAC.

Table 4 R, G, B values (Albumin) extracted by the standard device

Alb Conc.	R	G	B	R+G+B
50mg/dl	49.507	72.913	68.609	191.029
	30.290	59.710	49.725	139.725
	30.928	60.174	50.261	141.362
	30.333	59.942	49.420	139.696
	25.855	57.449	46.522	129.826
	31.507	60.783	50.232	142.522
	AVG	33.070	61.829	52.461
STDV	8.300	5.548	8.032	21.867
15mg/dl	66.188	72.275	38.551	177.014
	64.942	70.855	33.217	169.014
	64.609	70.072	34.058	168.739
	64.348	72.014	35.478	171.841
	66.058	72.536	35.536	174.130
	65.391	70.783	34.594	170.768
	AVG	65.256	71.423	35.239
STDV	0.758	0.987	1.844	3.185
8mg/dl	73.797	71.072	29.725	174.594
	75.203	74.145	30.029	179.377
	75.275	73.203	29.971	178.449
	74.014	70.971	27.725	172.710
	77.348	76.145	31.565	185.058
	74.681	72.145	26.913	173.739
	AVG	75.053	72.947	29.321
STDV	1.275	1.989	1.701	4.624
5mg/dl	81.232	74.290	29.493	185.014
	81.029	74.333	27.507	182.870
	78.435	71.391	26.029	175.855
	81.913	74.493	29.217	185.623
	82.594	75.826	27.942	186.362
	80.406	73.855	28.855	183.116
	AVG	80.935	74.031	28.174



	STDV	1.438	1.456	1.295	3.826
3mg/dl		82.957	74.275	28.754	185.986
		81.043	71.870	25.377	178.290
		78.768	71.435	24.551	174.754
		83.449	75.261	29.594	188.304
		81.928	72.478	26.174	180.580
		82.464	73.261	25.536	181.261
	AVG	81.768	73.097	26.664	181.529
	STDV	1.690	1.466	2.029	4.963
omg/dl		80.188	69.913	20.623	170.725
		81.536	71.638	22.377	175.551
		85.580	74.188	26.319	186.087
		82.507	72.029	23.812	178.348
		82.174	71.536	20.551	174.261
		86.246	74.217	24.000	184.464
	AVG	83.039	72.254	22.947	178.239
	STDV	2.373	1.675	2.222	5.998

These findings suggest that acquired RGB values are quite unique with respect to individual different concentration present in the samples the unique combination of RGB values can serve as a consistent indicator for toxin levels. By Observing the data in some places like low to mid concentration range the difference between the readings of different concentration are quite distinguishable across repeated tests. This consistency and accuracy of the RGB values and toxin concentration further supports the effectiveness of image-based analysis for early screening of kidney related toxins and also shows system's potential to become a practical approach.

3.3.2 Creatinine Detection:

Reagent Preparation & Dataset Generation: Picric Acid at a concentration of 1 mmol, Sodium Hydroxide at 0.29 mol/L, Creatinine Primary Standard at 36 mg/ml, MilliQ Water, a 96-well Microplate, and a Microplate Reader were used for this procedure. For the preparation of the standard creatinine solution, 36 mg/ml creatinine was first diluted to make an 1800 mg/dl stock solution. Then, a 1350 mg/dl working solution was prepared (as per [Table 5](#)). This working solution was further diluted to generate the following standard concentrations (as per [Table 6](#)) ([Figure 4](#)).

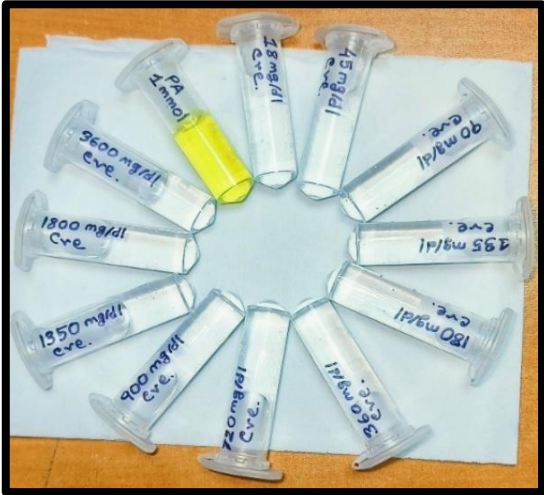


Fig. 4 Diluted creatinine of different standard concentrations

Table 5 Dilution of Creatinine Standard (1350 mg/dl) to Prepare Working Solutions

Sl no.	Creatinine standard (ml)	Water (ml)	Final vol. (ml)	Diluted creatinine std conc. (mg/dl)
1	2	0	2	1350
2	1.33	0.67	2	900
3	1.067	0.933	2	720
4	0.533	1.467	2	360
5	0.267	1.733	2	180
6	0.2	1.8	2	135
7	0.133	1.867	2	90
8	0.067	1.933	2	45



Table 6 Reagent and Sample Volume Used for OD-Based Creatinine Detection

Initial Diluted Crn Std Conc. (mg/dl)	Diluted Crn Std(μl)	Water (μl)	NaOH (μl)	Picric Acid (μl)	Final Volume (μl)	Final Diluted Crn Std Conc.(mg/dl)
0 (Blank)	0	140	10	30	180	0
18	20	120	10	30	180	2
45	20	120	10	30	180	5
90	20	120	10	30	180	10
135	20	120	10	30	180	15
180	20	120	10	30	180	20
360	20	120	10	30	180	40
720	20	120	10	30	180	80
900	20	120	10	30	180	100
1350	20	120	10	30	180	150
1800	20	120	10	30	180	200

OD-Based Method Steps: Thirty microliters of picric acid were added to nine consecutive wells. Then, ten microliters of NaOH were added to each well. After that, 120 microliters of MilliQ water were added to each well. Next, 20 microliters of diluted creatinine standards were added to the respective wells. The solutions were gently mixed to prepare a final volume per well of 180 μL. The final concentrations were 2, 5, 10, 15, 20, 40, 80, 100, and 150 mg/dl. The plate was incubated for 15 minutes at room temperature. Finally, the absorbance at 400 nm (A₄₀₀) was measured, with the values given in [Table 7](#).

Table 7 Final Concentrations of Creatinine Used in the Image Scanning Method

OD-BLANK							
Crn Conc. (mg/dl)	OD ₁	OD ₂	OD ₃	OD ₄	OD ₅	OD _{AVG}	Std Dev.
2	-0.004	0.001	-0.003	0.003	-0.025	-0.006	0.011

5	0.013	0.006	0.004	0.004	0.013	0.008	0.005
10	0.027	0.017	-0.009	0.04	0.018	0.019	0.018
15	0.02	0.022	0.034	0.039	0.032	0.029	0.008
20	0.029	0.035	0.041	0.042	0.023	0.034	0.008
40	0.085	0.078	0.093	0.095	0.09	0.088	0.007
80	0.147	0.134	0.158	0.138	0.15	0.145	0.010
100	0.151	0.168	0.198	0.193	0.158	0.174	0.021
150	0.241	0.246	0.245	0.254	0.233	0.244	0.008

Image Scanning Method Steps: Thirty microliters of picric acid were added to each well. Then, ten microliters of NaOH were added to each well. After that, 120 microliters of water were added and mixed gently. Next, 20 microliters of creatinine standards ranging from 45 to 3600 mg/dl were added. The final concentrations in the wells were 5, 15, 20, 40, 100, 200, and 400 mg/dl. The plate was incubated for 15 minutes at room temperature. The plate was then loaded into the image scanner. Each well was selected and scanned. Finally, values for R, G, and B were extracted and reported in [Table 8](#).

Table 8 R, G, B values (Creatinine) extracted by a standard device

Crn Conc.	R	G	B	R+G+B
40mg/ dl	101.710	85.377	61.000	248.087
	108.000	88.014	60.029	256.043
	108.333	90.014	62.406	260.754
	109.551	91.261	64.855	265.667
	108.391	89.246	63.130	260.768
	107.058	88.942	60.522	256.522
	93.609	76.884	52.609	223.101
AVG	105.236	87.106	60.650	252.992
STD DEV	5.726	4.864	3.918	14.263
20mg/ dl	104.232	92.797	74.899	271.928
	107.841	94.362	72.957	275.159
	107.681	93.986	71.725	273.391
	108.957	96.638	74.841	280.435
	107.696	95.681	75.986	279.362
	105.058	92.739	71.884	269.681



		96.116	84.290	66.014	246.420
	AVG	105.369	92.928	72.615	270.911
	STD DEV	4.411	4.066	3.330	11.465
15mg/dl		101.348	89.768	71.145	262.261
		105.623	94.725	73.986	274.333
		106.667	95.725	75.696	278.087
		104.725	92.667	72.797	270.188
		105.232	94.174	74.942	274.348
		103.942	93.609	75.101	272.652
		94.087	83.957	67.826	245.870
	AVG	103.089	92.089	73.070	268.248
	STD DEV	4.307	4.055	2.788	11.037
5mg/dl		107.014	107.014	89.246	297.304
		109.928	109.928	89.159	301.681
		110.478	110.478	91.522	305.986
		107.725	107.725	87.130	295.609
		107.681	107.681	88.130	297.000
		107.304	107.304	88.957	298.870
		97.652	97.652	82.681	273.304
	AVG	106.826	106.826	88.118	295.679
	STD DEV	4.261	4.261	2.744	10.468

3.4 Prototype Structure (Hardware Components)

As already discussed, the urine will be used to obtain the required results. The prototype uses a particular vial that will contain a reactant sensitive to Albumin & Creatinine. So, it will give different colors to different concentrations of albumin. This vial would be placed in a particular position. The portion of the prototype that will contain this vial will be perfectly illuminated with surrounding diffused light, which will help to identify the colors more accurately. Right above the vial, the image sensor will be placed (in this iteration of the prototype, an Android device is used due to its low focal length and good auto-focusing). For capturing and computing the color gradients of the provided vial and Android, this device will be used with Android version 8.2 or higher. This will also contain the developed software built within it. The other portion of the prototype will contain batteries and switches to deliver the required energy to illuminate the vial's environment. Please refer to [Figure 5](#) for the detailed prototype.

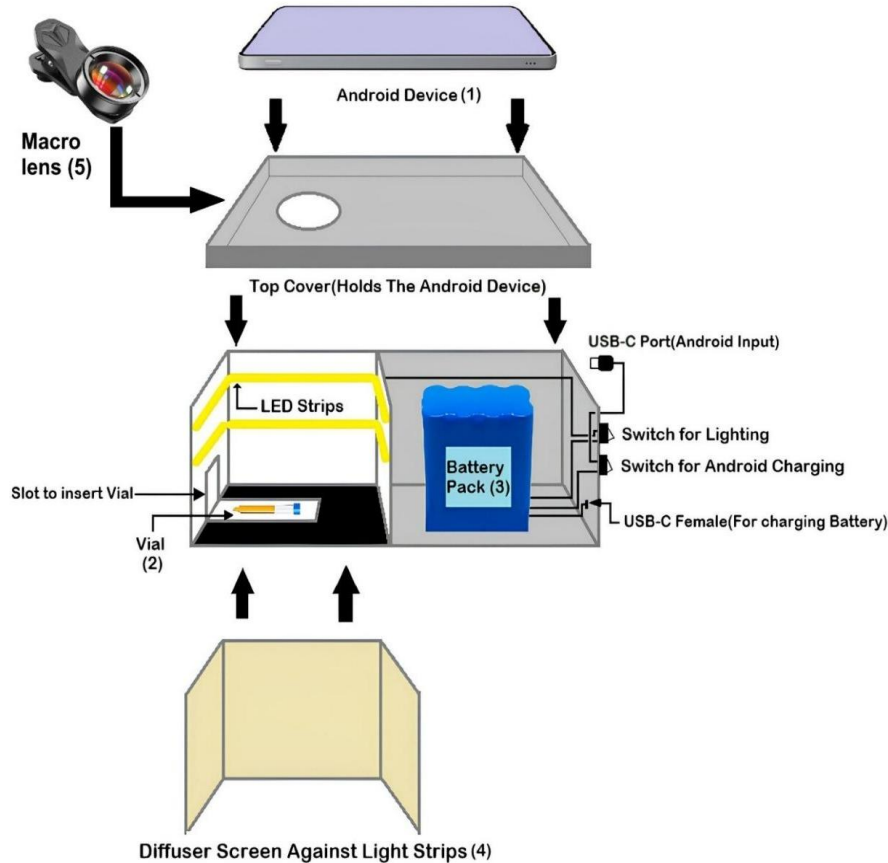


Fig. 5 Structural prototype of the hardware case

- **Android Device:** The primary image-capturing and processing device (Android version 8.0 or newer).
- **Vial:** The vial contains the reagent, as previously discussed, and is inserted into the hardware through a designated slot.
- **Battery Pack:** The battery pack powers both the Android device and the LED strips. Charging the battery pack alone is sufficient; there is no need to charge the Android device and the battery pack separately. (Specification: 12V, Rechargeable 5200mAh, Lithium-Ion Battery)
- **Diffuser Screen:** The diffuser screen layer diffuses the sharp light given off by the LED strips, preventing harsh or sharp shadows over the vial (which is also quite reflective) and maintaining uniform light intensity. (Specification: Polycarbonate LED Diffuser).
- **Macro Lens:** The macro lens decreases the focal length of the Android device's camera; typically, its conventional focal length is way over 25



Centimeters, therefore, this lens will allow it to capture images with improved focus and clarity. (Specification: Aperture: F1.8, Minimum Focal Length: 33 mm, Diameter: 1.57 inches).

3.5 Android Application

Below each component is explained in detail, its technical properties, and the stacks used. A few screenshots of the application are also given in [Figure 6](#).

3.5.1 Firebase-based Authentication System using Android Studio

The authentication part of the prototype was made using Android Studio by using Firebase Services [\[27\]](#), [\[28\]](#) to give secure and easy user login and registration processes. This makes sure that only the allowed user can access the toxin detection application and its data, also providing user-specific separation. The user login records or logs are stored in Firebase, as shown in [Figure 7](#).

Dependencies and Libraries Used:

- Firebase Authentication SDK: Open-source solution that provides backend services and functions for authenticating users using proper credentials.
- Google Services JSON: Connects the Android application to the Firebase project and validates that the application really has a valid instance of that particular project each time an auth request is made.
- AndroidX Libraries: Support modern Android development tools like UI management and thread handling.
- Material Design Components: Used for the UI components inside the screen, like background, buttons, text fields, and stuff.
- Firebase Core Libraries: Handle authentication and user session, i.e., if verifies whether a valid user session is present or not.

Application Flow and Functionality:

1. Authentication Initialization: A FirebaseAuth instance is created using FirebaseAuth.getInstance() to enable authentication operations throughout the application.
2. User Validation at Startup: On application start, the current user session is checked. If it finds a valid session after confirming, the user will be redirected directly to the MainActivity, or else the user will be moved to the Login screen.
3. User Login Screen: The user can log in by entering “Username” and their “Password”.

4. Login Authentication: When the user clicks the login button, the application authenticates the user if it returns “Success”; thus, the screen will switch to “MainActivity” and if it fails, an error toast pops up [27], [28], [29].
5. Registration Screen Navigation: Users without an account can tap "Register Now" to navigate to the Registration activity.
6. User Registration Process: On the registration screen, users enter their email and password, then validation takes place, and after successful validation, a new user account is created.
7. Registration Success and Redirect: After successful registration, go to STEP 3.
8. Progress Indication: A progress bar is shown during the process and registration processes to provide visual feedback during background operations.
9. Logout Functionality: When the “Log Out” button is pressed it will end the user session by executing `FirebaseAuth.getInstance().signOut()` then STEP 3.
10. Error Handling and Feedback: Proper pop up/ toast messages are displayed in the user login page informing the user about the error e.g. incorrect credentials or network failures.
11. Back Stack Management (back navigation): After successful login or registration, `finish()` to prevent unintended or false back navigation.

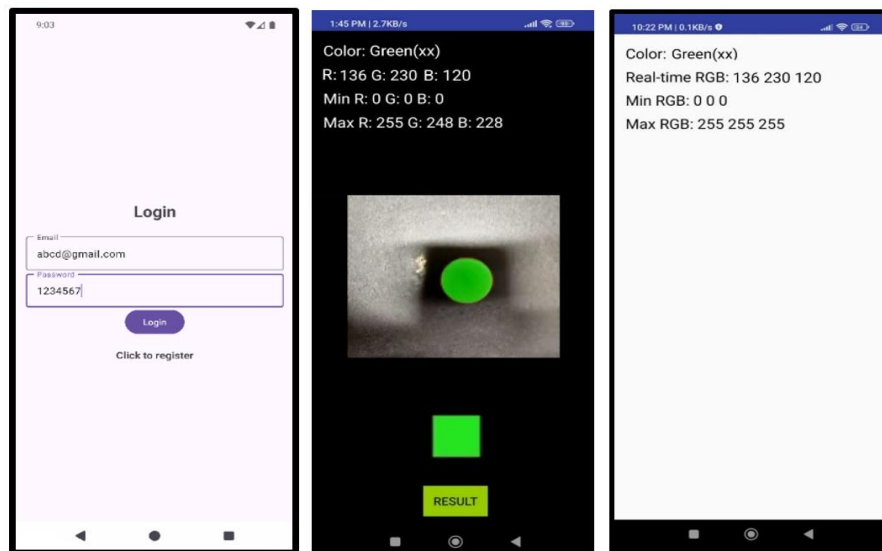
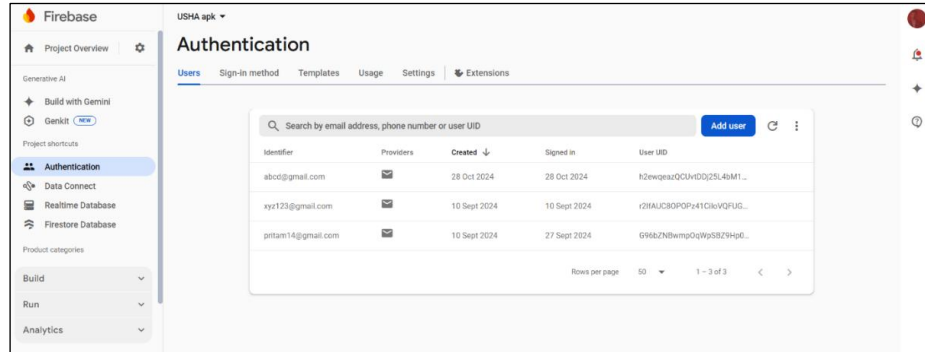


Fig.6 Some Screenshots from the Android App

(a). Page 1: login & registration (user authentication); (b): Page 2: R, G & B value detection (c): Page 3: Result generation



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The screenshot shows the Firebase Authentication console. On the left is a sidebar with navigation options: Project Overview, Build with Gemini, Genkit, Authentication (selected), Data Connect, Realtime Database, and Firestore Database. The main area is titled 'Authentication' and has tabs for Users, Sign-in method, Templates, Usage, Settings, and Extensions. The 'Users' tab is active, displaying a table of users with columns for Identifier, Providers, Created, Signed in, and User UID. There are three users listed. At the top right of the table is a search bar and an 'Add user' button. At the bottom right are pagination controls showing '1 - 3 of 3'.

Identifier	Providers	Created	Signed in	User UID
abco@gmail.com		28 Oct 2024	28 Oct 2024	h2ewqezOCUvD0j25L4bM1...
xyt123@gmail.com		10 Sept 2024	10 Sept 2024	r2l8AUC30P0Pz41CkVQFUG...
prtam14@gmail.com		10 Sept 2024	27 Sept 2024	6Y6bZNBwmpOqWpSRZ9hp6...

Fig. 7 User login record in Firebase (Firebase project console web)

3.5.2 OpenCV-based RGB Detection using Android Studio

The software component of the prototype was developed using Android Studio, with the integration of the OpenCV library to perform real-time color-based image processing [30]. The aim was to extract the RGB values of a urine reagent sample through a mobile camera and use those values for toxin estimation based on pre-calibrated reference datasets.

Dependencies and Libraries Used:

- OpenCV Android SDK: Enables real-time image processing with support for native C++ and Java bindings.
- CameraBridgeViewBase: OpenCV's high-level camera interface for Android that allows frame-by-frame processing via CvCameraViewListener2 [31].
- Imgproc (Image Processing module): Used for tasks such as color space conversion, affine transformations, and drawing contours or rectangles.
- Core (OpenCV core module): Provides functions for matrix manipulation and basic computer vision operations like summing pixels, scalar operations, and contour drawing.

Application Flow and Functionality:

1. Camera Initialization and Frame Capture: The camera feed gets displayed while using CameraBridgeViewBase, and every frame is transferred to the onCameraFrame () method. OpenCV processes the frame in RGBA format using Mat (Matrix) objects.

2. **Region of Interest (ROI) and Rotation:** For better visuals, the camera frame is rotated by 90 degrees counterclockwise using `getRotationMatrix2D()` and `warpAffine ()` functions. A rectangular region (ROI) is defined inside the camera feed at its center ($x/2$, $y/2$) to separate the vial from the background and only process the color gradients present in the vial.
3. **Color Conversion and RGB Extraction:** The submatrix within the ROI is taken and converted from RGB to HSV color space using `Imgproc.cvtColor()`. The HSV values are combined and averaged over the entire region to create a single majority color, then converted back to RGBA using a helper function. These RGB values represent the detected color from the test sample [32].
4. **Tracking Min/Max RGB Values:** The application continuously extracts the RGB values from the ROI, i.e., the vial, and also records the minimum and maximum RGB values to create a range of metrics for future result mapping.
5. **Contour Detection and Display:** A custom color blob detector (HSV thresholding) is used to identify and draw contours within the region. `Imgproc.drawContours()` visually outlines detected blobs; these outlines ensure that the ROI being detected is actually correct, and the user can check it in real time if it goes wrong. Then the test can be reinitiated.
6. **Color Classification:** A predefined set of RGB value ranges is already defined by known color categories (e.g., red, green, unknown) for user reference. The detected RGB values are compared against these ranges to determine a probable concentration range (e.g., "Unknown (5–6 mg/dl)"). Thus, providing real-time instant result generation
7. **User Interface & Result Handling:** After getting the RG values of the selected color, it also gets the highest and lowest RGB values to get the deviation range. After that, the result button gets active; when pressed, it will take the user to the result screen.
8. **Code Structure and Threading:** Android's Handler & Runnable interfaces are used to add a delay for the camera view. `runOnUiThread ()` - this delay helps the UI to get updated (like RGB display and ImageView color rendering) in time because it's a heavy process in comparison to convention android processes; if not properly handled will cause issues. Also safely handles the camera feed by using separate threads; the delay ensures that the application doesn't crash under load.

This approach integrates real-time image processing with a user-friendly UI, enabling quicker and non-invasive estimation of toxin levels present in a urine sample. The use of OpenCV helps with handling frame operations with accurate



color analysis, while Android's architecture enables smooth user interaction and multi-threaded execution for the application.

4 Results & Discussion

The prototype consists of an illumination chamber where the user inserts a vial containing the urine sample & reagent. Next, the user logs in to the application inside the Android device using his/her registered user ID and password. Then, inside the application, the camera captures images of the vial placed inside, where it detects the region of interest by separating the vial from the black base. RGB values and minimum ~ maximum ranges are extracted from the ROI, and then the acquired results are mapped with the prerecorded data to find the approximate concentration of Albumin and Creatinine inside the vial (Urine Sample). After 15 seconds, the above-mentioned results are generated, and the application navigates to the result screen where all the above-mentioned resultant metrics are displayed.

This application aims to design an affordable, accessible, and user-friendly prototype for estimating blood toxin levels, specifically Albumin and Creatinine, through non-invasive urine sample analysis. Using mobile-based image processing and Firebase authentication, the system is intended for early detection of kidney-related health issues, particularly in underserved populations with limited access to diagnostic facilities. Preliminary testing showed an average accuracy of 97.51%, demonstrating strong potential for effective early screening. This section describes the testing results done on the samples provided by CDAC, Kolkata. [Figure 8](#) shows a tray with different Albumin concentrations of the samples provided for the testing.

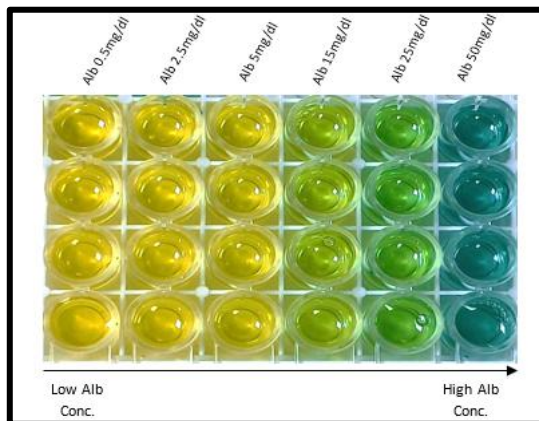


Fig. 8 Test samples (Albumin)

Evaluation Metrics

To measure the performance of the extracted RGB values against the reference dataset, we define Accuracy and Error as follows.

The formula for Accuracy is given by

$$\text{Error} = \left(\frac{|\Delta r|}{R} + \frac{|\Delta g|}{G} + \frac{|\Delta b|}{B} \right) \div 3 \times 100\% , \text{ where:}$$

- **$\Delta r = |R-r|$, where R represents the average red gradient value from the reference dataset (Table 4), and r denotes the red gradient value extracted by the machine.**
- **$\Delta g = |G-g|$, where G represents the average green gradient value from the reference dataset (Table 4), and g denotes the green gradient value extracted by the machine.**
- **$\Delta b = |B-b|$, where B represents the average blue gradient value from the reference dataset (Table 4), and b denotes the blue gradient value extracted by the machine.**

Thus, the accuracy percentage is calculated as:

$$\text{Accuracy} = (100 - \text{Error}) \%$$

This formula provides a direct comparison between the mobile-generated results and the respective reference dataset values, and combines their effective deltas across all three RGB channels. By calculating how closely the measured color gradient values match the reference dataset ones. A higher accuracy value indicates a closer similarity with the reference dataset, signifying more reliable and consistent performance of the system. Conversely, a higher error value suggests a greater difference from the expected results, indicating potential inconsistencies or inaccuracies in the measurement process.

Two test results of different albumin concentrations, obtained with the developed software, are shown below [Figure 9(a) and Figure 9(b)]. The concentration of Albumin is defined as 5mg/dl and 15mg/dl, respectively, in both cases. Calculating the results on the full dataset, the average test result has been reported as 97.51%.

In Figure 9(a), the test results obtained by the application are r:81 g:73 b:30. Originally, from the test results of records (Table 4), it was R:80 G:74 B:28. So, the calculated error is:

$$(((1/80) + (1/74) + (2/28))/3) * 100\% = 3.24\%. \text{ Therefore, the Accuracy is } 96.76\%$$



In Figure 9(b), the test results obtained by the application are r:66 g:70 b:35. Originally, from the test results of records (Table 4), it was R:65 G:71 B:35. So, the calculated error is:

$((1/65) + (1/71) + (0/35))/3 * 100\% = 0.97\%$. Therefore, the accuracy is 99.03%.

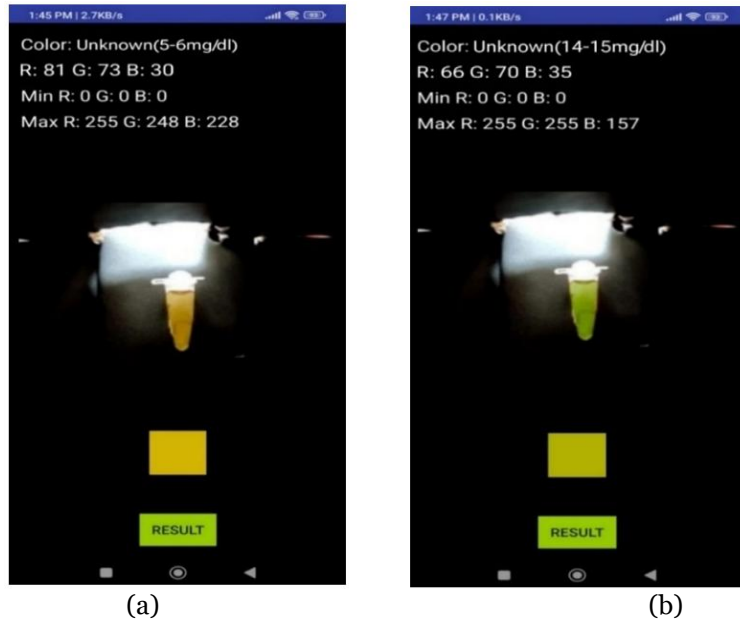


Fig. 9 RGB values & Albumin concentration obtained from application (a). 5mg/dl; (b). 15mg/dl

Table 9 Performance Evaluation of RGB-Based Albumin Concentration Estimation Using the Reference Dataset

Sl. No.	Expected Concentration (mg/dL)	R	G	B	r	g	b	Accuracy (%)
1	50	33	62	52	34	61	51	97.82
2	15	65	71	35	66	70	34	97.08
3	8	75	73	29	74	72	30	97.69
4	5	81	74	28	80	73	28	97.91
5	3	82	73	27	81	73	26	98.37
6	0	83	72	23	82	71	22	97.52
7	15	65	71	35	64	70	35	97.49
8	8	75	73	29	76	72	28	96.86
9	5	81	74	28	82	73	27	96.89
10	3	82	73	27	83	72	26	96.67

* R, G, and B represent the average reference RGB values (rounded to the nearest integer) obtained from the reference dataset, whereas r, g, and b represent the corresponding RGB values extracted by the proposed image-processing system.

Therefore, considering all twelve test cases (including the two representative test cases shown in [Figure 9](#) and the ten additional validation samples presented in [Table 9](#)), the average accuracy is calculated as:

$$\frac{(96.67)}{12} = 97.51\%$$

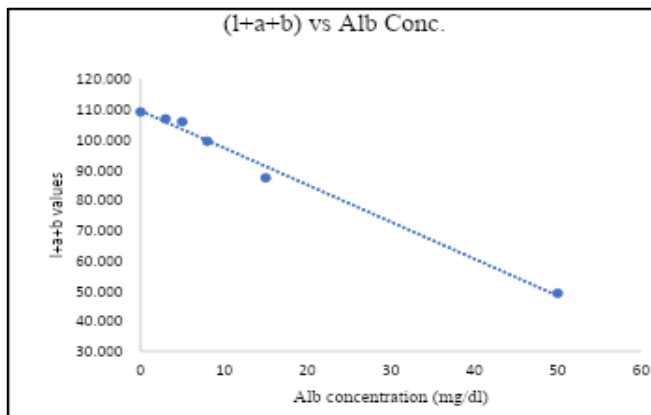
Hence, the average error is:

$$100 - 97.51 = 2.49\%$$

Therefore, Total approximate error is 2.1% $((3.24 + 0.97)/2)$ and results accuracy is 97.9% $((96.76 + 99.03)/2)$.

After plotting the value of R & B against albumin concentration, we get the following graphs; from here, the equations can be concluded.

The nature of Red & Blue color components with increasing concentration of Albumin is shown in [Figure 10](#) and [Figure 11](#).



Alb Conc.	0	3	5	8	15	50
R values	22.947	26.664	28.174	29.321	35.239	52.461

Fig. 10 Graph showing relation between R values and Albumin concentration

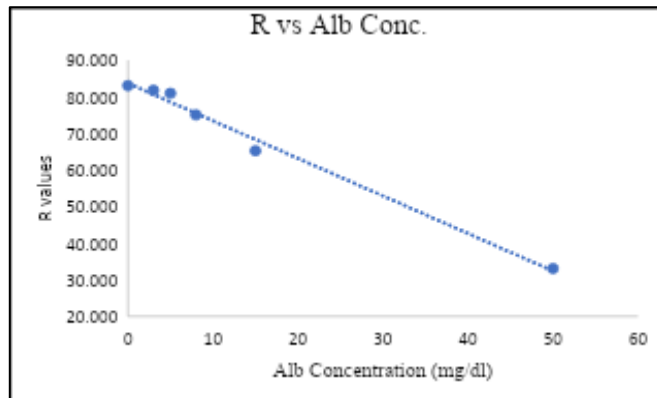


Fig. 11 Graph showing relation between B values and Albumin concentration

Limitations & Future Development

- **Limited Test Cases:** Currently, only a small number of test cases have been used to train the algorithm for image processing. In the future, a larger dataset will be developed through extensive testing to improve the model's accuracy and to identify additional patterns that could enhance diagnostic insights.
- **Cloud Storage:** Currently, only user authentication credentials and data are stored in the cloud (Firebase). In the future, focusing on storing all generated results/ analysis along with patient records and medical history will be prioritized for better tracking.
- **User Interface:** Currently, the user interface is very basic and minimalistic, satisfying the functional requirements only; future updates will aim to make the interface better-looking and user-friendly.
- **Integration:** For now, the application is not a single whole system; the image processing, color gradient extractor, and result generation are placed in one part, and the other part includes user authentication and login. Future improvements will include a full-stack solution for every step in one application only.
- **Equation Implementation:** The formulas derived (refer to [Figure 10](#) and [Figure 11](#)), in the future, more such equations/ relations can be found with the expanding datasets and test cases

5 Conclusion

The intended and primary goal of this work was to make a low-cost, accessible and user-friendly prototype that can estimate blood toxin concentrations (more specifically Albumin, Creatinine & ACR), through non-invasive method by analyzing urine sample color gradient through adding chemical reagents to it, further utilizing mobile-based image processing application (OpenCV) with Firebase authentication, the system intends to support early detection of kidney-related health issue before-hand, especially people with limited access to diagnostic services. The preliminary results showed an average accuracy of 97.51%, demonstrating that the prototype is very promising for early screening and detection. Nevertheless, it is not yet perfect as there remains a prominent scope for further enhancement, especially through expanding the reference dataset by making it vaster and more varied, and identification of deeper patterns and relations to further increase the precision and accuracy in complex test cases of the OpenCV-based color detection system. With consistent enhancement in data handling, color analysis, and machine learning model implementation and optimization, the prototype can grow into a highly useful tool for real-world deployment and use cases. This prototype can serve as a reliable indicator for determining whether toxin concentrations are at alarming levels (not precise concentration when compared to conventional lab procedures). It is inexpensive, highly portable, and well-suited for use in rural areas where access to medical facilities is limited, making it ideal for large-scale deployment. Therefore, while the current version serves as a valid and functional proof-of-concept, future improvements & fixes are necessary to fully realize its clinical and practical potential.

Repositories

The GitHub link (OpenCV colour recognition):

<https://github.com/Alpharangerr/USHA-image-Detection-OpenCV>

and GitHub link (Firebase user authentication):

<https://github.com/Pritam1408/USHA75>

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Contribution of each Author

Mr. Adarsh Nandi and Mr. Pritam Sadhu carried out the implementation of the proposed methodology, conducted the experiments, and prepared the initial documentation. Dr. Nilanjana Dutta Roy prepared and revised the manuscript, coordinated the overall study, and served as the corresponding author. The research concept and methodology were conceived by Dr. Subhankar Mukherjee, Dr. Souvik Pal, and Mr. Shubhajit Bhar, who also provided technical guidance throughout the study. The experimental work was carried out at C-DAC Kolkata with the support of the research team. All authors reviewed, revised, and approved the final version of the manuscript.

Conflict of Interest Statement

The authors declare no conflict of interest.

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Accessible Health Screening: A Portable System for Early Detection of Toxin Elevation Linked to Kidney Disorders

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Legal and Security Challenges in Cyber-Physical Systems: A Framework for Secure and Compliant IoT Ecosystems

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ABSTRACT

Cyber-Physical Systems (CPS) have become the technological foundation of modern critical infrastructures, including smart grids, industrial automation, healthcare systems, intelligent transportation systems, and autonomous vehicles. By integrating physical processes with computational intelligence, communication networks, and sensing technologies, CPS enable real-time monitoring, automation, and intelligent decision-making. However, the rapid convergence of Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and operational technologies has significantly expanded the cyberattack surface of CPS ecosystems. As a result, cybersecurity incidents can lead not only to data breaches but also to physical damage, operational disruption, legal liability, regulatory penalties, and public safety risks. Existing research predominantly focuses on technical security mechanisms while providing limited attention to legal compliance, governance, privacy protection, and AI accountability. This paper addresses this gap by proposing the Secure and Compliant Cyber-Physical Systems Framework (SCCF), a multidisciplinary governance architecture integrating cybersecurity, data governance, AI governance, and legal-regulatory compliance. A



comparative analysis of the Information Technology Act, 2000, Digital Personal Data Protection Act, 2023, GDPR, NIST CSF 2.0, ISO/IEC 27001, and IEC 62443 is conducted to identify governance requirements relevant to CPS environments. The findings reveal that existing frameworks remain fragmented and fail to provide comprehensive guidance for integrating security and compliance. The proposed SCCF introduces a five-layer governance model consisting of Physical Security, Cybersecurity, Data Governance, AI Governance and Ethics, and Legal-Regulatory Compliance. The framework operationalizes Security-by-Design, Privacy-by-Design, Compliance-by-Design, and Governance-by-Design principles, enabling organizations to develop secure, trustworthy, and legally compliant CPS ecosystems.

1 Introduction

Cyber-Physical Systems (CPS) are part of Industry 4.0, which has accelerated their deployment within manufacturing, healthcare, transportation, energy and smart cities. CPS help enable intelligent operations and autonomous decision-making by integrating sensing technology, embedded computers, communication networks and physical processes.

Artificial Intelligence (AI) and Internet of Things (IoT) technologies have enhanced the capabilities of CPS through predictive maintenance, anomaly detection, resource optimisation, and autonomous control.

CPS environments face significant cybersecurity challenges that differ from traditional information systems; attacks against CPS directly impact physical infrastructure, resulting in disruption to operations, damage to equipment, environmental hazards and threats to the safety of humans. This is exemplified by incidents such as Stuxnet and Colonial Pipeline, which exemplify how critical infrastructure is increasingly subject to cyber threats.

CPS operators also have to meet the requirements of privacy regulations, cybersecurity standards, and governance requirements. Existing research mainly focuses on technical controls, leaving minimal guidance for integrating security with compliance. This includes frameworks like GDPR, DPDP Act 2023, NIST CSF2.0, ISO/IEC 27001 and IEC 62443 that provide obligations for privacy protection, accountability, risk management and resilience. This research seeks to fill this gap by creating the Secure and Compliant CPS Framework (SCCF).

[Figure 1](#) illustrates the evolution of Cyber-Physical Systems from traditional standalone computing environments to AI-enabled autonomous ecosystems,

highlighting the increasing integration of intelligence, connectivity, and automation.

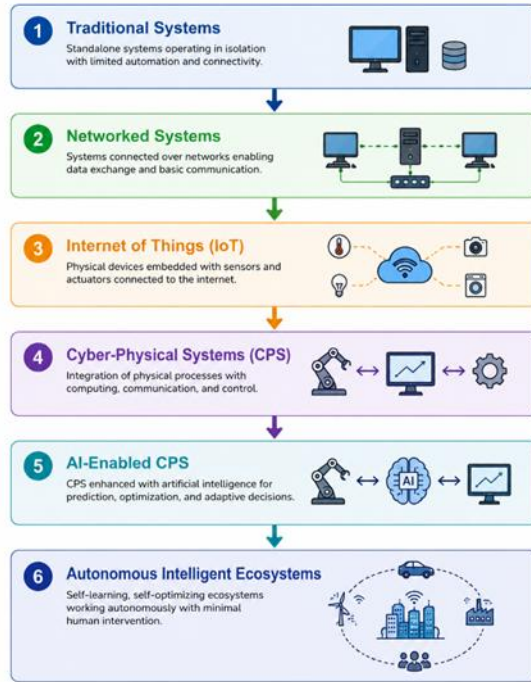


Fig. 1 Evolution of Cyber-Physical Systems

1.1 Key Contributions

The key contributions to the paper are as follows.

- Integration of cybersecurity and legal governance
- Introduction of SCCF
- Comparative legal analysis
- AI governance incorporation
- Practical roadmap for CPS organizations

2 Literature Review

Current studies about Cyber-Physical Systems (CPS) primarily discuss security risks, security of industrial control systems, and resiliency in operations. Researchers have defined major attacks on CPS environments as malware, ransomware, insider attacks, denial of service attacks (DoS), sensor spoofing,



and supply chain attacks. The rapid proliferation of devices connected to the internet has greatly enlarged the number of potential attacks against organisations, adding to their exposure to cyber risks.

There have been increased concerns regarding the protection of privacy and data governance for CPS due to the ongoing collection and processing of large amounts of private and operational data. Regulatory guidance such as GDPR and the DPDP Act stress the importance of the following concepts: transparency, accountability, lawful processing, and Privacy by Design. Recent literature on governance of AI highlights several issues surrounding the use of AI, including transparency, explainability, fairness, accountability, and the need for human supervision [4].

While frameworks including NIST CSF 2.0 [2], ISO/IEC 27001, and IEC 62443 provide valuable guidance on security, they suffer from a lack of integration when considering privacy, governance over AI, and compliance with the law. Therefore, there is still a need for a governance framework that is multidisciplinary, capable of integrating technical, legal, and organisational elements. Table 1 gives a comparative review of existing approaches.

Table 1 Comparative review of existing approaches

Author/ Framework	Security	Privacy	AI Governance	Compliance
Lee (2008) [3]	High	Low	No	No
Humayed et al. (2017) [4]	High	Low	No	No
GDPR	Medium	Very High	Partial	High
NIST CSF	High	Low	No	Medium
SCCF	High	High	High	High

3 Methodology

The research adopts a qualitative approach using three complementary methods: Systematic Literature Review (SLR), Comparative Legal Analysis (CLA), and Case Study Analysis (CSA).

In the first phase of the literature review, peer-reviewed publications were sourced from Scopus, Web of Science, IEEE Xplore, SpringerLink, and ScienceDirect. The CLA covered key regulatory and governance frameworks, including the Information Technology Act of 2000, the Digital Personal Data Protection Act 2023; General Data Protection Regulation; NIST Cybersecurity Framework version 2.0; ISO/IEC 27001; and IEC 62443. CSA analysed four representative CPS incidents, namely Stuxnet, Colonial Pipeline, Jeep Cherokee, and Medical Internet of Things (MIoT) vulnerabilities. The findings from these case studies informed the development of the proposed SCCF architecture. The overall research workflow adopted in this study is illustrated in [Figure 2](#).

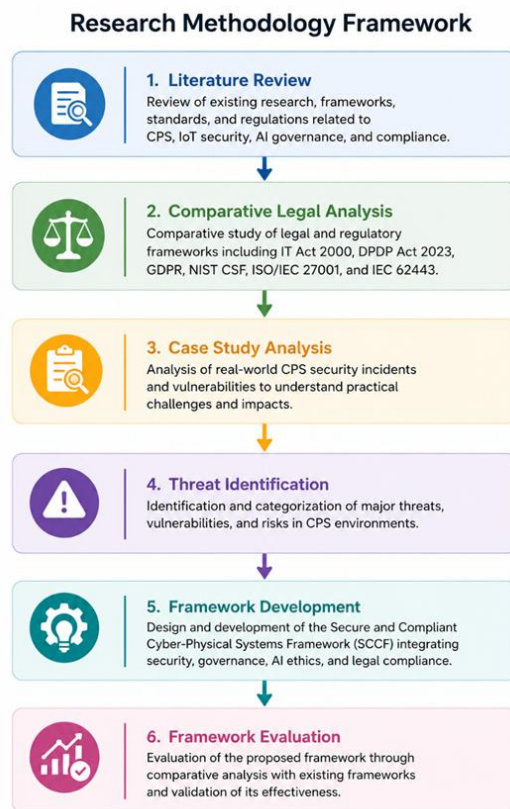


Fig. 2 Research methodology adopted for SCCF development



4 Legal Analysis

The governance of Cyber-Physical Systems is becoming increasingly affected by legal and regulatory requirements relating to protecting individuals' personal information, critical infrastructures, and public safety.

4.1 Information Technology Act

The Information Technology Act 2000 is India's primary cyber law designed to address aspects including cybercrime, unauthorised access, electronic records, and digital evidence [\[5\]](#).

4.2 Digital Personal Data Protection Act

Additionally, the Digital Personal Data Protection Act 2023 creates additional obligations regarding consent management, accountability, breach notification and on the responsibilities of data fiduciaries [\[6\]](#).

4.3 GDPR

The General Data Protection Regulations (GDPR) continue to serve as the global standard for governing privacy and creates requirements for transparency, accountability, lawfulness, purpose limitation and Privacy-by-Design [\[7\]](#).

4.4 NIST CSF

Cybersecurity standards also play an important role. The NIST Cybersecurity Framework 2.0 has established criteria that focus on governance, risk management, incident response, and resilience.

4.5 ISO/IEC 27001

ISO/IEC 27001 contains requirements that promote the establishment of an Information Security Management System (ISMS) that is structured and organised [\[8\]](#).

4.6 IEC 62443

IEC 62443 provides specific security guidance for industrial automation and operational technology environments [\[9\]](#).

[Table 2](#) compares the major legal and cybersecurity frameworks applicable to Cyber-Physical Systems. The analysis demonstrates that existing frameworks remain fragmented and fail to provide comprehensive governance guidance for

CPS ecosystems. [Figure 3](#) illustrates the primary attack vectors across different layers of Cyber-Physical Systems.

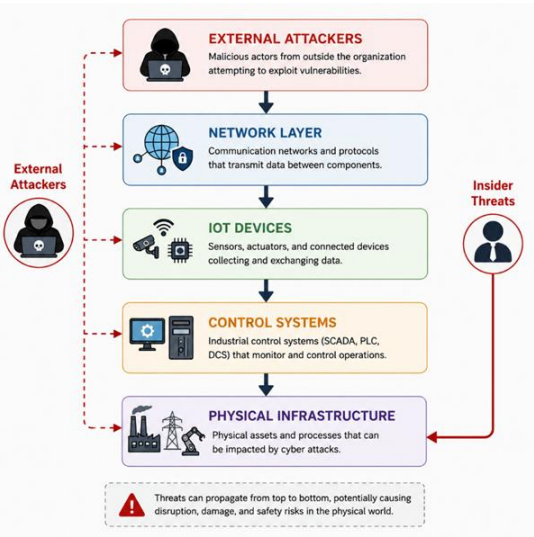


Fig. 3 Threat vectors targeting CPS ecosystems

In the last few decades, the jurisprudential and economic nature of information has been transformed, moving from its status as a tool to obtain or control external resources to one of the main resources in the current global economy. The current digital age is a constant systemic issue with protecting the privacy of an individual. The drive for personal documentation and private areas has had a precedent. Still, the interplay of globalisation and speed of technological advancement has exponentially amplified the speed and volume of personal data sharing. As a result, there has been a significant increase in regulatory clashes in the data protection landscape between public authorities and the systematised collection and treatment of personal telemetry.

Table 2 Comparative Legal Framework Analysis

Criteria	IT Act	DPDP Act	GDPR	NIST CSF	ISO 27001	IEC 62443
Privacy Protection	Low	High	Very High	Low	Medium	Low



Criteria	IT Act	DPDP Act	GDPR	NIST CSF	ISO 27001	IEC 62443
Accountability	Medium	High	High	High	High	Medium
Incident Response	Medium	Medium	High	High	High	High
Industrial Security	Low	Low	Low	Medium	Medium	Very High
AI Governance	No	Limited	Emerging	Limited	No	No

This tension came to a head in the context of Indian constitutional jurisprudence in the Supreme Court's landmark decision in the case of Justice K.S. Puttaswamy (Retd.) v. Union of India. The judiciary's decision to define the right to privacy as an inherent fundamental right secured by Article 21 of the Indian Constitution put privacy at the very heart of today's constitutional obligations. The state has a primary duty to ensure that the civil liberties and the right to digital freedom of its citizens are not subjected to arbitrary or disproportionate surveillance or exploitation of their data under this constitutional provision.

Even though it is a prominent issue in the constitution, data protection is one of the most abstract and elusive concepts in jurisprudence, and cannot be reduced to a single definition. In legal discourse, the term "data protection" is generally used to refer to the overall area of data governance, from data gathering, processing, storage to erasure of personal data. With widespread digitisation of day-to-day life, data is omnipresent and has generated great benefits for society and technology. But this advancement has engendered a fundamental split between information that people knowingly and voluntarily provide and the quietly gathered data, which is a product of ubiquitous software architectures. To counteract this imbalance, contemporary systems and frameworks strive to achieve fairness, transparency, and a rebalancing of power, putting individuals in control of how third parties may access and monetise their digital footprints (Jiang et al., 2024) [10].

The concept of "informational self-determination" is at the heart of the constitutional-privacy vs. statutory-data-protection relationship. This attitude stems from democratic ideals and gives people the right to control, at their own discretion, how, when, and what of themselves they share with others. Most

importantly, modern legal scholarship recognises that one might never have complete control over data in complex and interwoven digital infrastructures. So instead of trying to achieve complete data isolation, effective and modern regulatory frameworks seek to create a balanced, highly regulated oversight system that will achieve a balance between individual privacy rights and legitimate public administration and system needs.

Architectural evolution of Cyber-Physical Systems (CPS) has completely revolutionised the way computational processes interact with the physical world by bringing them together in an integrated network of sensors and actuators. These technologies have evolved from stand-alone networks to hyper-connected, network-dependent systems, and the need to operate within them has grown to include real-time interoperability with external networks. But all these integrated applications and services have multiplied the attack surface of the system at an exponential rate, creating important vulnerabilities in the computational and physical layers. Today, engineering techniques focus on the protection of the core hardware components, namely sensors, actuators and communications links, to reduce such risks and boost the resilience of the system. This is often done by performing real-time state estimations which emulate normal operational baselines from a single controller. This common structure allows for continual error recovery and rollback capabilities, and eventually the development of sophisticated CPS checkpointing and recovery features, which maintain physical functionality even in the presence of active compromise of the components.

Concurrently, with these engineering developments, the new level of sophistication of both the hardware and software has led to ubiquitous, multi-domain tracking across both physical and digital domains. Covert instruments like tracking cookies and hyperlinked tracking pixels can track user preferences and trends in the digital realm without the user's explicit consent. In parallel, within the physical world, the commonality of consumer internet-of-things (IoT) devices such as smartwatches, health trackers and mobile phones enables the ongoing collection of fine-grained telemetry, location and habits, which allow for the creation of more comprehensive profiles of human activity.

This surveillance-by-the-willy-nilly mode of data collection creates serious privacy and constitutional issues. The exact mechanisms that are designed to make connectivity and the increase of public convenience go hand in hand are also vectors of illegal surveillance and of systematic privacy violations. Consumer services – which are basically services that people are expecting to receive from the operation – can be easily harvested by operational telemetry for unauthorised monitoring operations. For example, if the cameras are smart residential entryways, which are intended to be used to verify identity for visitors at a local level, there is a risk of exploitation through an unauthorised third party gaining access, which could lead to the exposure of localised identity and temporal-spatial information. These household data streams, when combined across large time periods and/or across many different distributed



nodes, enable mass public surveillance. These intrusive techniques often run in parallel to the rights of consumers to give positive consent, and present substantial regulatory risk for data exposure in transit and storage.

As a result, the ability to have autonomous control over personal information has now become a pressing need in the current CPS deployments from a jurisprudential and technical standpoint. As smart devices proliferate and increasingly become part of everyday life and public spaces, there must be strong legal and structural mechanisms that enable users to have meaningful control over the life cycle of their data – its storage, dissemination, and permanent destruction. To reach this level of digital self-determination, it is important that the interfaces used are transparent and that consent is given in a clear and unambiguous manner, so that the person can make an informed choice. Safeguarding data integrity during data movement and rest is also a major aspect of data privacy – and is no simple task given the widely varying types of processing and cryptographic power of different IoT device hardware.

5 Threat Analysis

The threat landscape for contemporary Cyber-Physical Systems (CPS) is highly intricate. Some of the highest risk factors involve: malware; malicious software; malware; viruses; cybercrime; cyberattacks; cyber espionage; cyber sabotage; and Artificial Intelligence (AI) manipulated attacks. [Table 3](#) summarises the qualitative threat severity assessment.

Table 3 CPS Threat Severity Matrix

Threat	Likelihood	Impact	Risk Level
Malware	High	High	High
Ransomware	High	Critical	Very High
Sensor Spoofing	Medium	Critical	High
Insider Threat	Medium	High	High
Supply Chain Attack	Medium	High	High
AI Manipulation	Medium	High	High

The qualitative ratings (Low, Medium, High, and Very High) presented in the Threat Severity Matrix were assigned through expert judgment based on the estimated likelihood of occurrence and the potential impact of each threat on confidentiality, integrity, availability, operational continuity, public safety, and regulatory compliance. These ratings are intended as conceptual qualitative assessments derived from the literature review, comparative legal analysis, and representative CPS case studies, rather than statistically derived quantitative measurements.

The Stuxnet malware attack is a well-known example that demonstrated how malware can exploit and manipulate industrial control systems to cause physical damage. The recent Colonial Pipeline ransomware attack is another example of the growing threats to critical infrastructure by cybercriminals. Another example of the types of risks associated with cyber-physical systems is illustrated by the Jeep Cherokee hack; a demonstration of the dangers associated with automobile/cyber-physical systems.

In summary, when security measures fail within a cyber-physical system, multiple issues are created, including: legal liability; monetary losses; regulatory scrutiny; and issues related to public safety. [Table 4](#) presents the qualitative risk heat matrix used for prioritising CPS threats.

Table 4 CPS Risk Heat Matrix

Impact ↓ / Likelihood →	Low	Medium	High
Critical	-	Sensor Spoofing	Ransomware
High	-	Insider Threat AI Manipulation Supply Chain	Malware
Medium	-	-	-
Low	-	-	-

6 Secure and Compliant Cyber-Physical Systems Framework (SCCF)

The SCCF is a multilayered governance system that links Cyber Security, Privacy, Governance of Artificial Intelligence, and Legal Regulation.

The framework consists of five layers:

- (1) Physical Security Layer
- (2) Cyber Security Layer
- (3) Data Governance Layer
- (4) AI Governance and Ethical Layer
- (5) Legal and Regulatory Compliance Layer

As a result, the framework is built to facilitate the implementation of Security by Design, Privacy by Design, Compliance by Design, and Governance by Design principles. [Figure 4](#) presents the proposed Secure and Compliant Cyber-Physical Systems Framework (SCCF). [Figure 5](#) demonstrates how legal obligations are translated into governance policies, security controls, compliance evidence, and audit verification. [Figure 6](#) summarises the fundamental principles of AI governance adopted within the proposed SCCF.

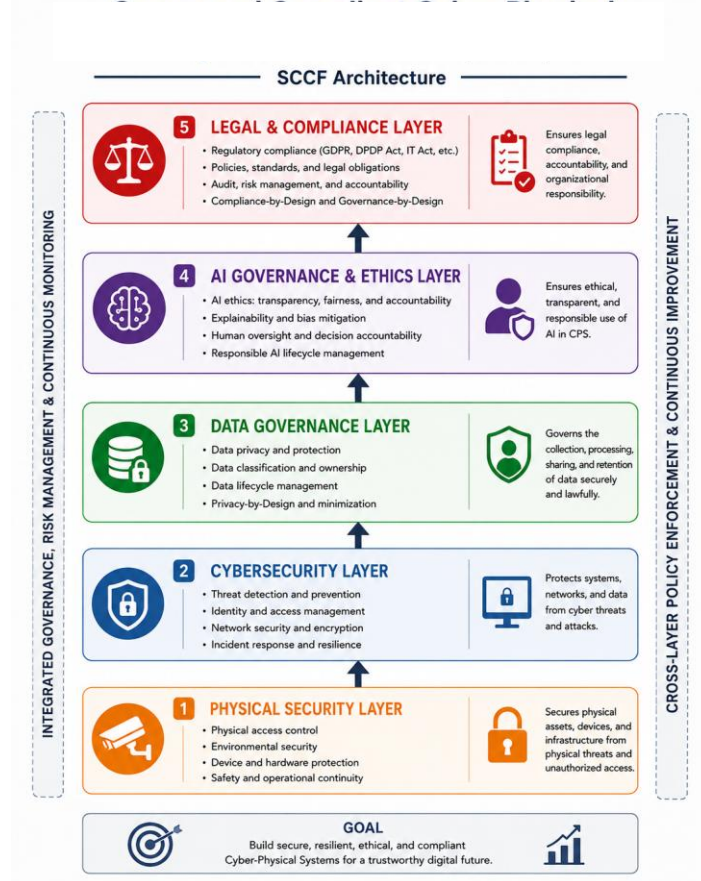


Fig. 4 Secure and Compliant Cyber-Physical Systems Framework (SCCF)



Fig. 5 SCCF Compliance Mapping Model

[Table 5](#) depicts a comparison of SCCF with existing frameworks. The five governance layers operate in a hierarchical yet interdependent manner. Physical Security provides the foundation upon which cybersecurity controls protect digital assets. Data Governance regulates information throughout its lifecycle, while AI Governance ensures ethical, transparent, and accountable decision-making. The Legal and Compliance Layer continuously oversees all operational layers by monitoring regulatory obligations, conducting audits, and ensuring organizational accountability. In practice, trade-offs such as transparency versus operational security are addressed through risk-based governance policies and proportional compliance mechanisms.



Fig. 6 AI Governance and Ethics Model

Table 5 SCCF vs Existing Frameworks

Capability	NIST CSF	ISO 27001	IEC 62443	SCCF
Cybersecurity	✓	✓	✓	✓
Privacy Protection	Partial	Partial	No	✓
AI Governance	No	No	No	✓

Capability	NIST CSF	ISO 27001	IEC 62443	SCCF
Legal Compliance	Partial	Partial	Partial	✓
Unified Governance	No	No	No	✓

7 Illustrative Application of SCCF

Consider a smart hospital deploying interconnected medical devices, AI-assisted diagnostics, and cloud-based patient monitoring. Under the SCCF, physical security safeguards protect medical equipment from unauthorised access, cybersecurity controls secure communications and connected devices, data governance ensures lawful processing of patient information under the DPDP Act and GDPR, AI governance promotes explainability and human oversight of diagnostic recommendations, while the legal and compliance layer continuously monitors regulatory obligations, audit trails, and breach notification requirements. This example demonstrates how SCCF integrates technical, organisational, and legal controls into a unified governance framework for critical healthcare infrastructure.

8 Evaluation and Discussion

SCCF was compared to NIST CSF 2.0, ISO/IEC 27001 and IEC 62443 using a comparative analysis for evaluation. The results of this evaluation showed that the SCCF provides broader governance coverage due to its integration of cyber security, privacy protection, artificial intelligence governance and legal compliance into one overall architecture.

The SCCF Framework has contributed to both theory and practice. It has contributed to theory through the extension of the literature on governance of Cyber-Physical Systems (CPS), and it has provided organisations with a roadmap to implement secure and compliant CPS ecosystems. The SCCF Framework will also provide accountability, transparency and readiness for regulations.

The implementation of SCCF may differ across organizations depending on their size, sector, regulatory obligations, technological maturity, and available resources. Small and medium-sized enterprises may adopt the framework incrementally, whereas critical infrastructure operators may implement all governance layers as part of comprehensive enterprise risk management programs.



[Table 6](#) compares the proposed SCCF with existing governance frameworks. This Table demonstrates that the proposed SCCF provides broader governance coverage than existing cybersecurity frameworks by integrating cybersecurity, privacy protection, AI governance, and legal compliance into a unified governance architecture. Unlike traditional frameworks, SCCF adopts a multidisciplinary approach that simultaneously addresses technical, organizational, ethical, and regulatory requirements for Cyber-Physical Systems.

Table 6 Comparative Evaluation of SCCF and Existing Frameworks

Framework	Cyber	Privacy	AI	Legal
NIST	✓	Partial	✗	Partial
ISO	✓	Partial	✗	Partial
IEC	✓	✗	✗	Partial
SCCF	✓	✓	✓	✓

9 Limitations and Future Work

The proposed SCCF is primarily a conceptual governance framework derived through qualitative analysis of literature, legal instruments, and representative case studies. Although the framework provides a comprehensive governance perspective, it has not yet been empirically validated through industrial deployment, expert evaluation, or quantitative performance assessment. Future work should focus on validating the framework in real-world CPS environments, developing governance maturity models, and integrating automated compliance monitoring mechanisms.

10 Conclusion

Cyber-Physical Systems (CPSs) are an important part of the modern digital infrastructure. With the increase in convergence of Artificial Intelligence (AI) and the Internet of Things (IoT) with other technologies, the growing complexity of these systems has resulted in new challenges associated with security, privacy, governance, and compliance. Presently existing frameworks are a good source of information for organisations but are not sufficiently integrated regarding their treatment of cybersecurity and legal obligations. The purpose of this research is to introduce the Secure and Compliant Cyber-Physical Systems Framework (SCCF), which consists of five layers of governance focused on Physical Security, Cybersecurity, Data Governance, AI Governance, as well as Legal & Regulatory Compliance. The SCCF provides organisations with the ability to better align their technical controls to the

requirements of the law and provides a foundation for Building Resilient and Trustworthy Cyber-Physical Systems (CPS). Overall, the proposed SCCF provides a comprehensive governance roadmap capable of supporting the development of secure, trustworthy, resilient, and legally compliant Cyber-Physical Systems for future intelligent digital infrastructures.

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Contribution of each Author

Sunetra Chatterjee: Conceptualisation, Research design, SCCF framework development, Cybersecurity analysis, Methodology, Manuscript review and editing.

Akash Chatterjee: Legal analysis, Literature review, Comparative legal framework analysis, Case law interpretation, Draft preparation, Reference compilation.

Conflict of Interest Statement

The authors declare no conflict of interest.

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